MB504T- METHODS IN MOLECULAR BIOLOGY

COMPLETE PPT

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MERGE BY BAQIR HUSSAIN BUZDAR

BS ZOOLOGY DATE 04-09-2024

03336361244 8TH SEMESTER

https://chat.whatsapp.com/K2MsHeqhWkWIHjrQhSVvRU

Essentials of Genetics

MODULE# 01

POLYMERASE CHAIN REACTION

Polymerase Chain Reaction

PCR

A technique widely used in Molecular Biology and Biotechnology.

Its name is from one of its key component - DNA polymerase.

Polymerase Chain Reaction

PCR

As PCR progresses, DNA generated is itself used as template for replication.

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Polymerase Chain Reaction

PCR- A CHAIN REACTION

This sets in motion of a chain reaction in which the DNA template is exponentially amplified.

Polymerase Chain Reaction

HEAT STABLE DNA POLYMERASE

During PCR, a heat-stable DNA polymerase, such as Taq polymerase - an enzyme derived from the bacterium Thermus aquaticus.

Polymerase Chain Reaction

CONCLUSION

PCR can be performed to amplify DNA.

It can be extensively modified to perform a wide array of genetic manipulations.

Essentials of Genetics

MODULE# 02

STEPS OF PCR

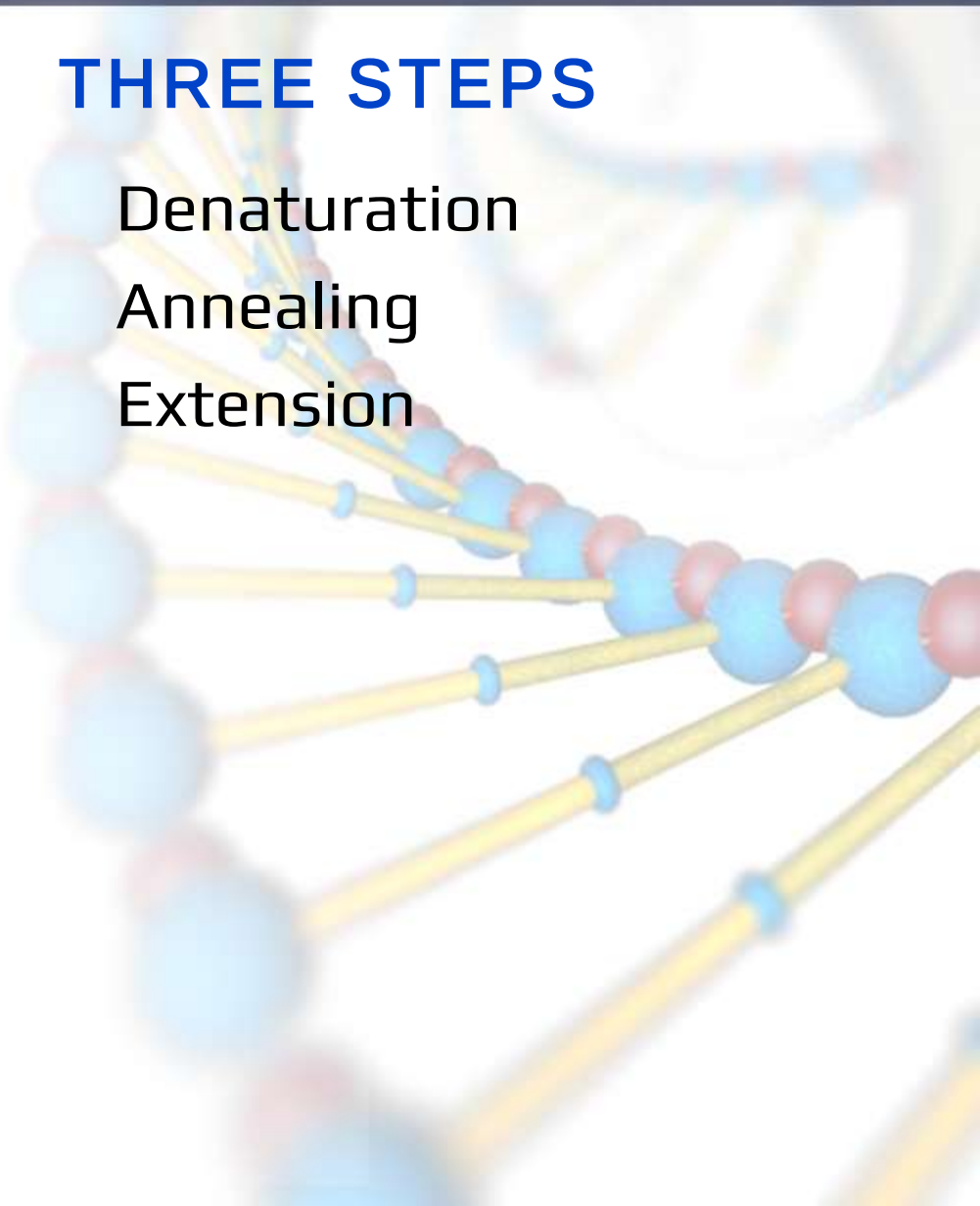
Steps of PCR

THREE STEPS

Denaturation

Annealing

Extension



Steps of PCR

INITIALIZATION STEP

This step consists of heating the reaction to a temperature of 93°C - 96°C, which is held for 1-10 minutes.

Steps of PCR

DENATURATION

First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20-45 seconds.

It causes melting of DNA template yielding single strands of DNA.

Steps of PCR

ANNEALING

The reaction temperature is lowered to 40-65°C for 20-40 seconds.

Annealing of the primers to the single-stranded DNA template.

Steps of PCR

EXTENSION

The temperature at this step depends on the type of DNA polymerase used.

Taq polymerase has its optimum activity temperature at 70-80°C.

Commonly 72°C is used.

Steps of PCR

EXTENSION

At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's.

Steps of PCR

FINAL ELONGATION

Single step is performed at a temperature of 70-74°C for 5-15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully extended.

Steps of PCR

FINAL HOLD TEMPERATURE

This step at 4-15°C for an indefinite time may be employed for short-term storage of the reaction.

MODULE# 03

INGREDIENTS OF PCR

Ingredients of PCR

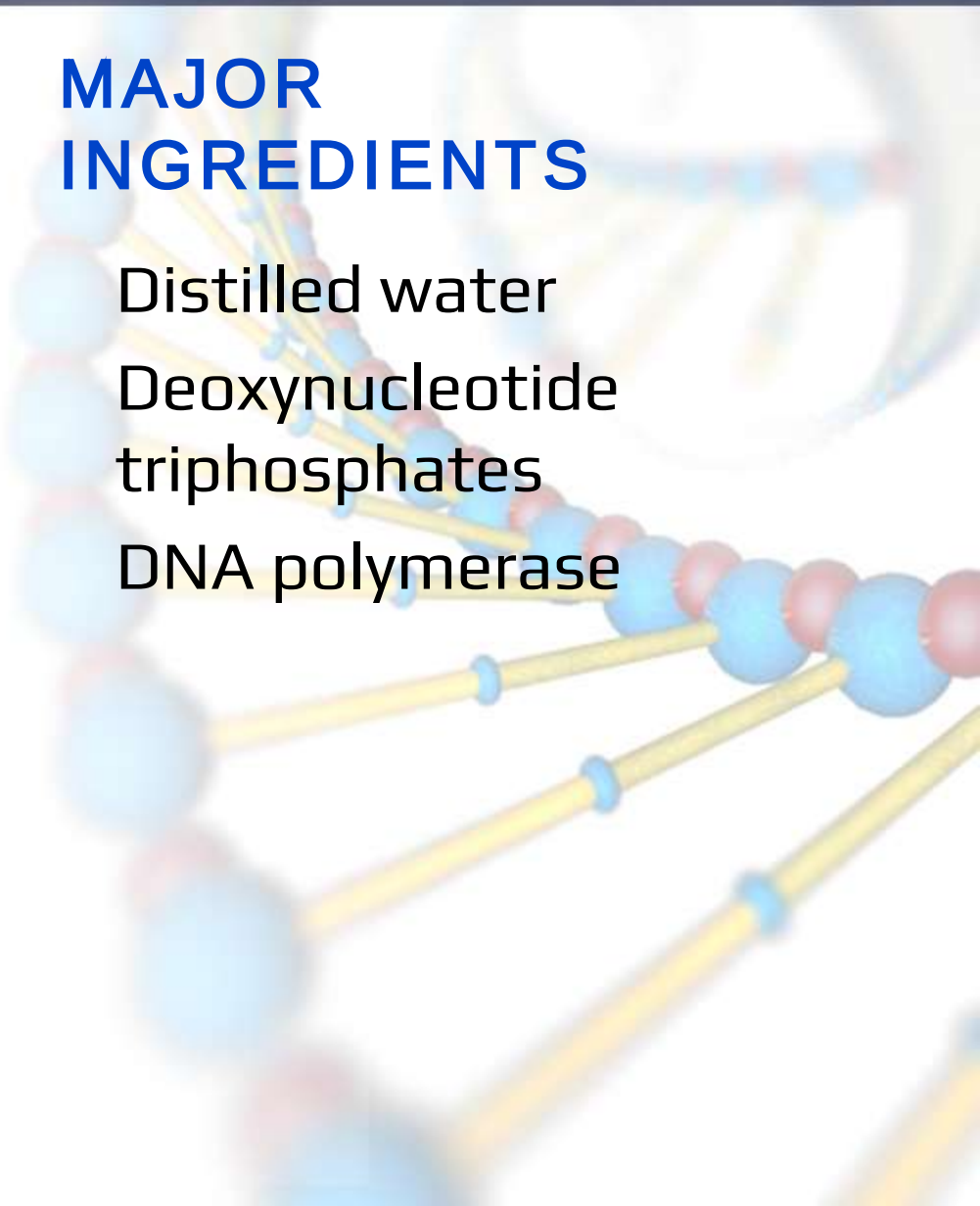
MAJOR INGREDIENTS

- Microfuge tube
- Thermal cycler
- DNA template
- Primers
- Buffer
- $MgCl_2$

Ingredients of PCR

MAJOR INGREDIENTS

- Distilled water
- Deoxynucleotide triphosphates
- DNA polymerase



Ingredients of PCR

MICROFUGE TUBES

These are small cylindrical plastic tubes with conical bottoms.





Ingredients of PCR

THERMOCYCLER

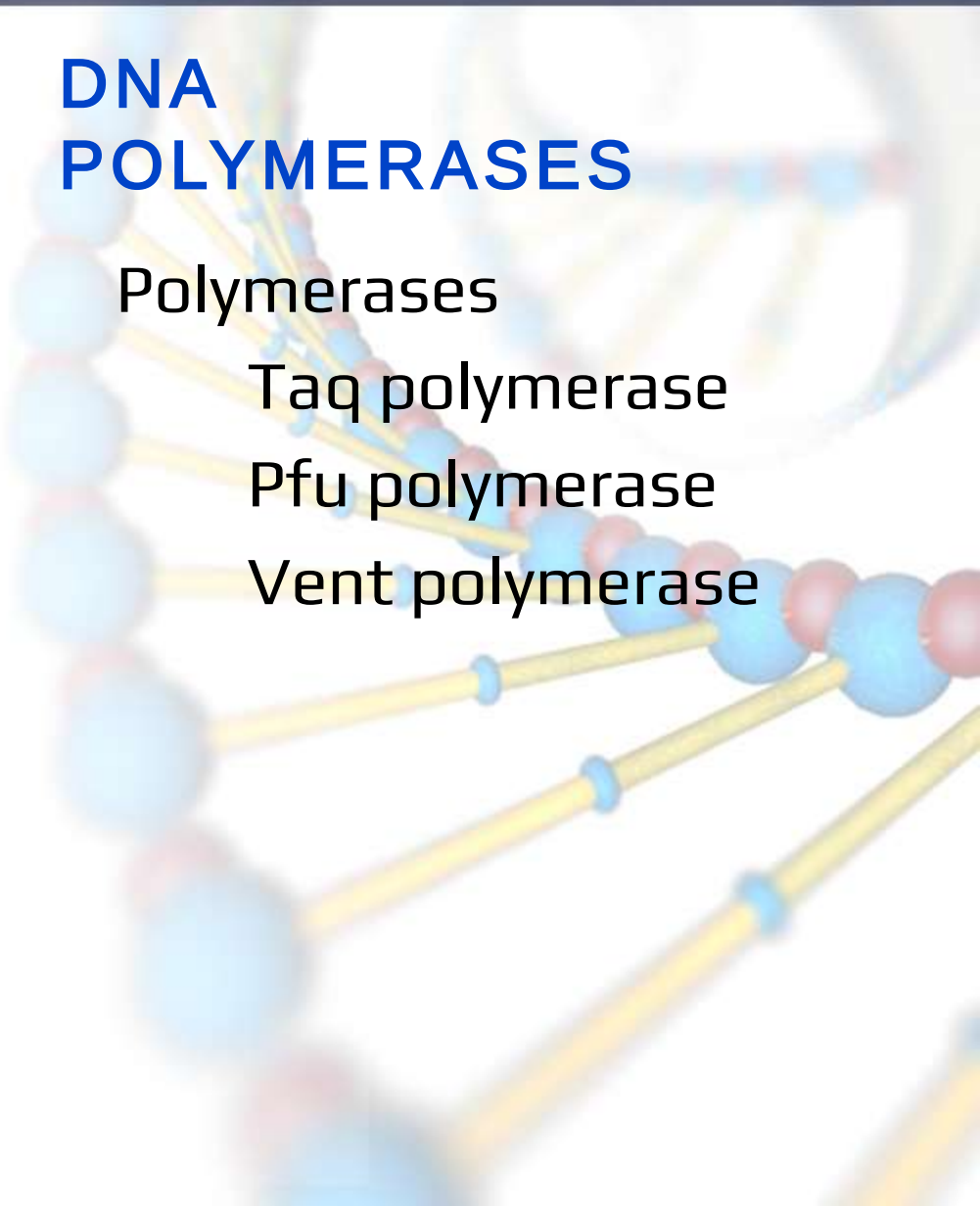
The thermocycler works on the principle of Peltier effect, which raises and lowers the temperature of the block in a pre-programmed manner.



Ingredients of PCR

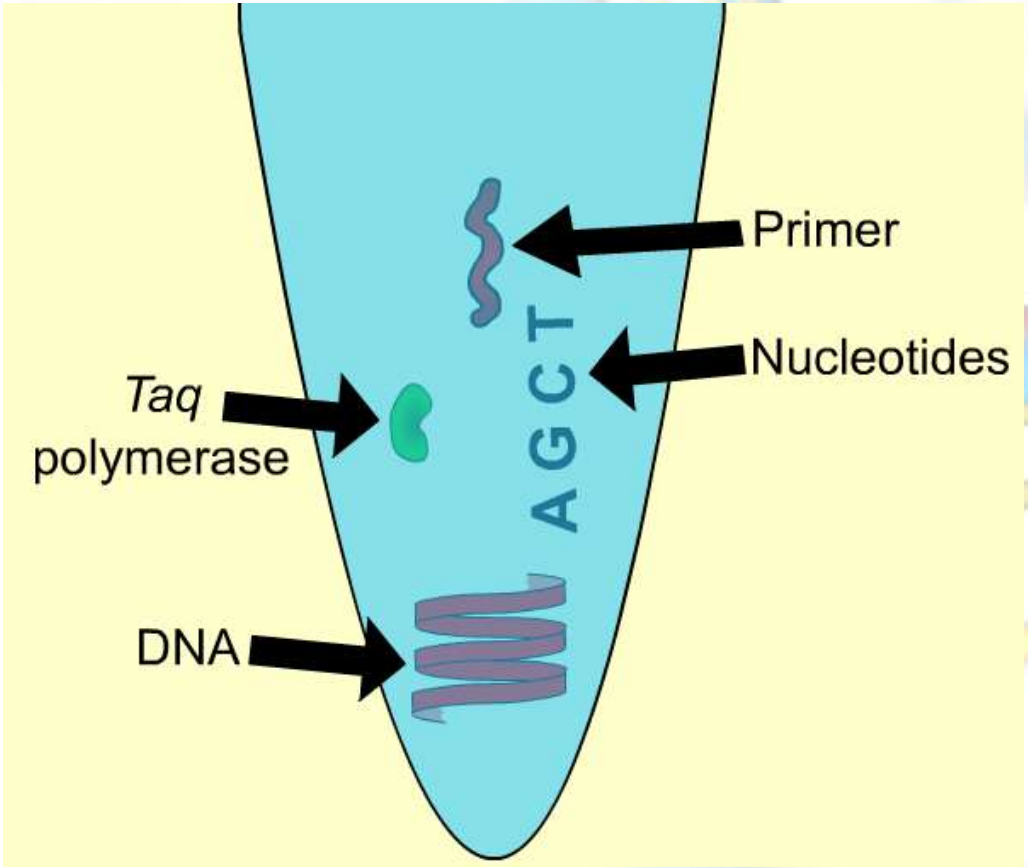
DNA POLYMERASES

- Polymerases
  - Taq polymerase
  - Pfu polymerase
  - Vent polymerase



Ingredients of PCR

INGREDIENTS IN PCR TUBE

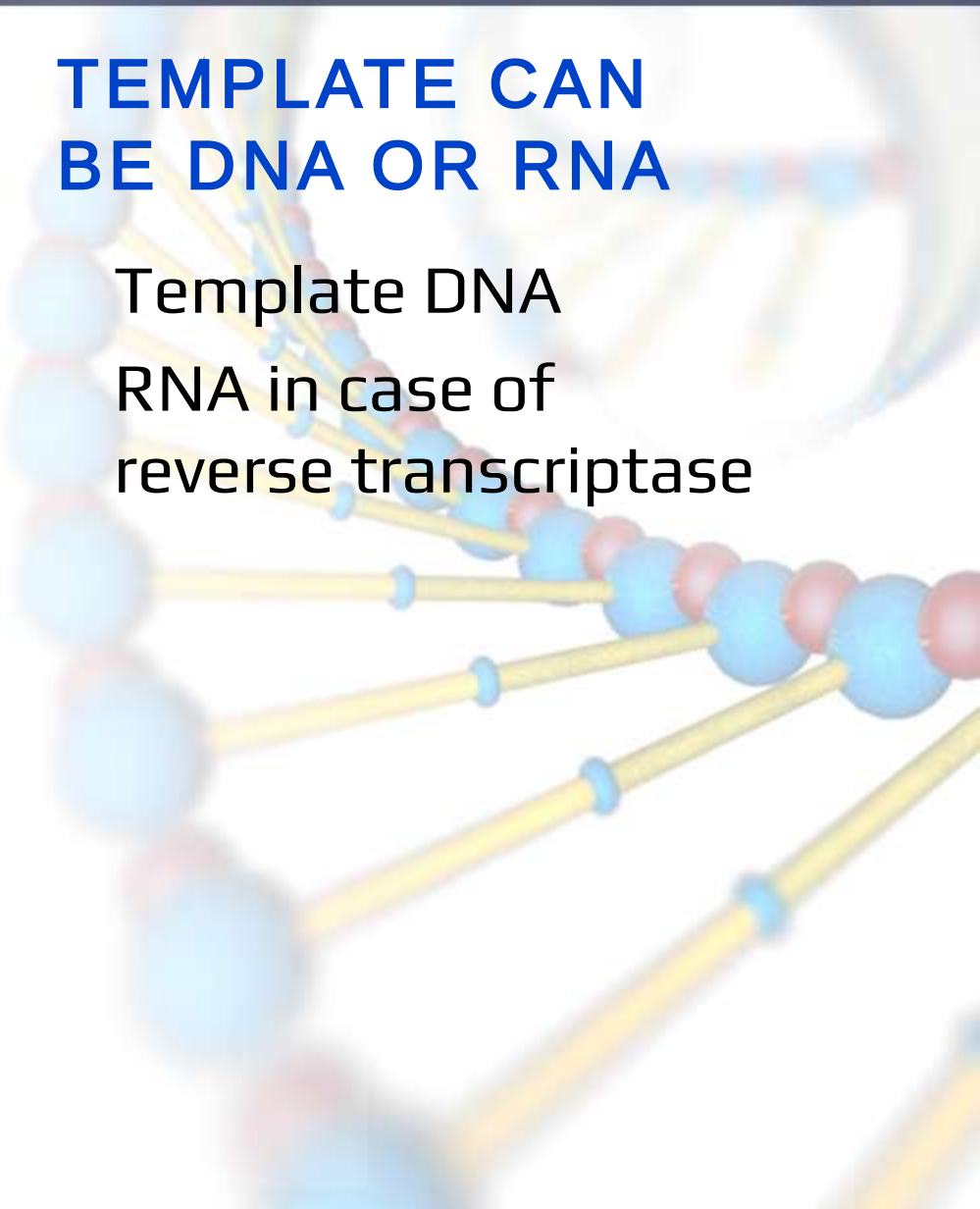


Ingredients of PCR

TEMPLATE CAN BE DNA OR RNA

Template DNA

RNA in case of reverse transcriptase



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MODULE# 04

PRIMERS FOR PCR

PRIMERS FOR PCR

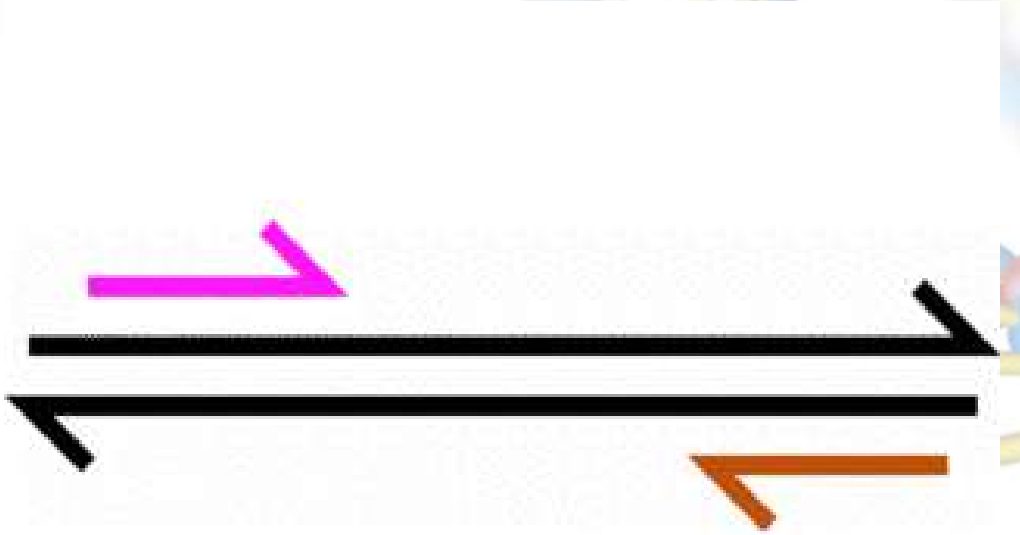
PRIMERS

Primers are single-stranded 18–30 bp long DNA fragments

Complementary to sequences flanking the region to be amplified.

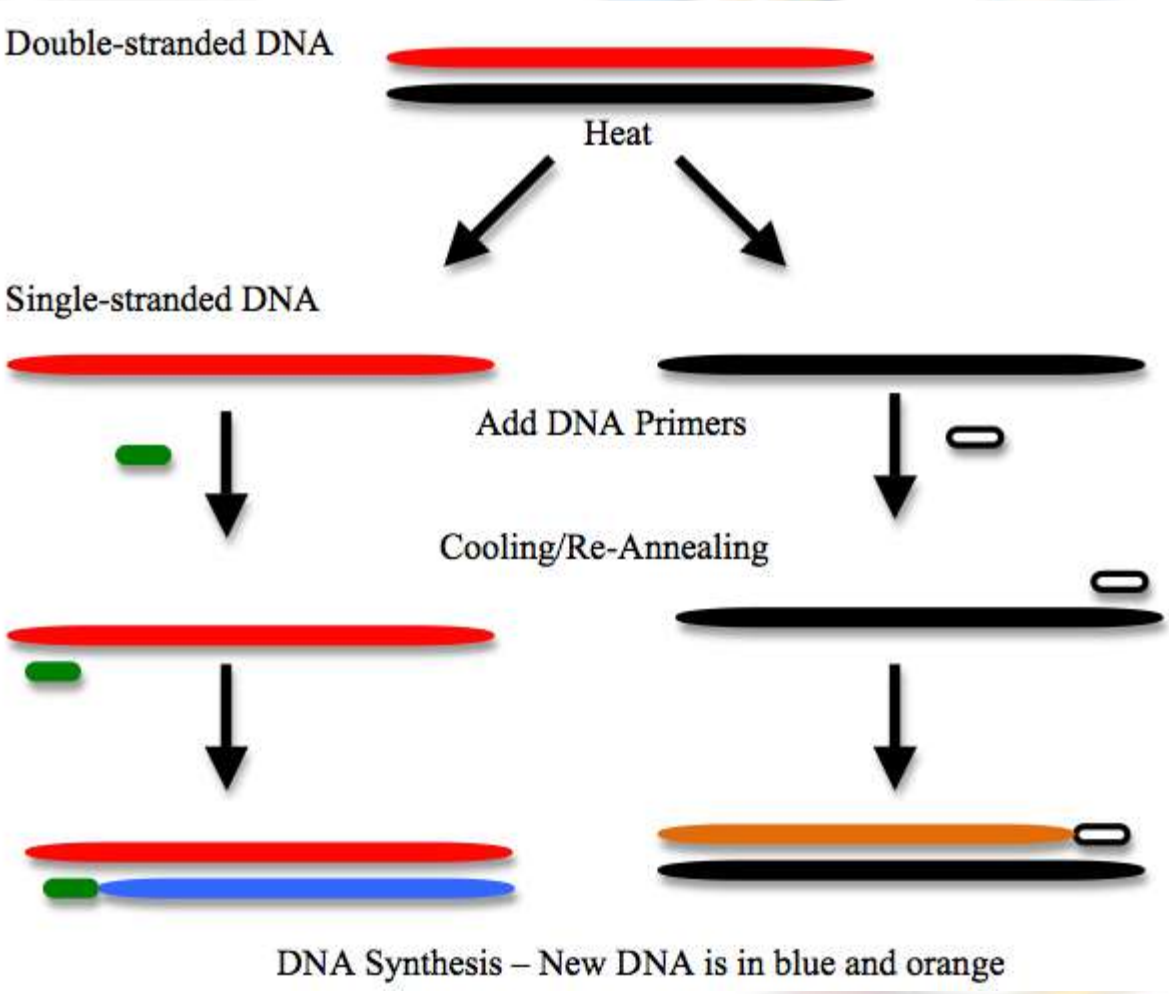
PRIMERS FOR PCR

PRIMERS



PRIMERS FOR PCR

PRIMERS



DNA Synthesis – New DNA is in blue and orange

PRIMERS FOR PCR

PRIMERS CAN BE SPECIFIC OR RANDOM

Primers determine the specificity of the PCR reaction.

Distance between the primers binding sites will determine the size of PCR product.

PRIMERS FOR PCR

FEATURES OF PRIMERS

- Types of primers- random or specific
- Primer length
- Annealing temperature
- Specificity
- Nucleotide composition

PRIMERS FOR PCR

PRIMERS

Avoid inter-strand homologies

Avoid intra-strand homologies

T_m of forward primer = T_m of reverse primer

G/C content of 20–80%; avoid longer than GGGG

Product size (100–700 bp)

Target specificity

PRIMERS FOR PCR

FORMULA FOR CALCULATING MELTING TEMPERATURE OF PRIMERS

$$Tm = 4(G+C) + 2(A+T)$$

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MODULE# 05

DNA POLYMERASES

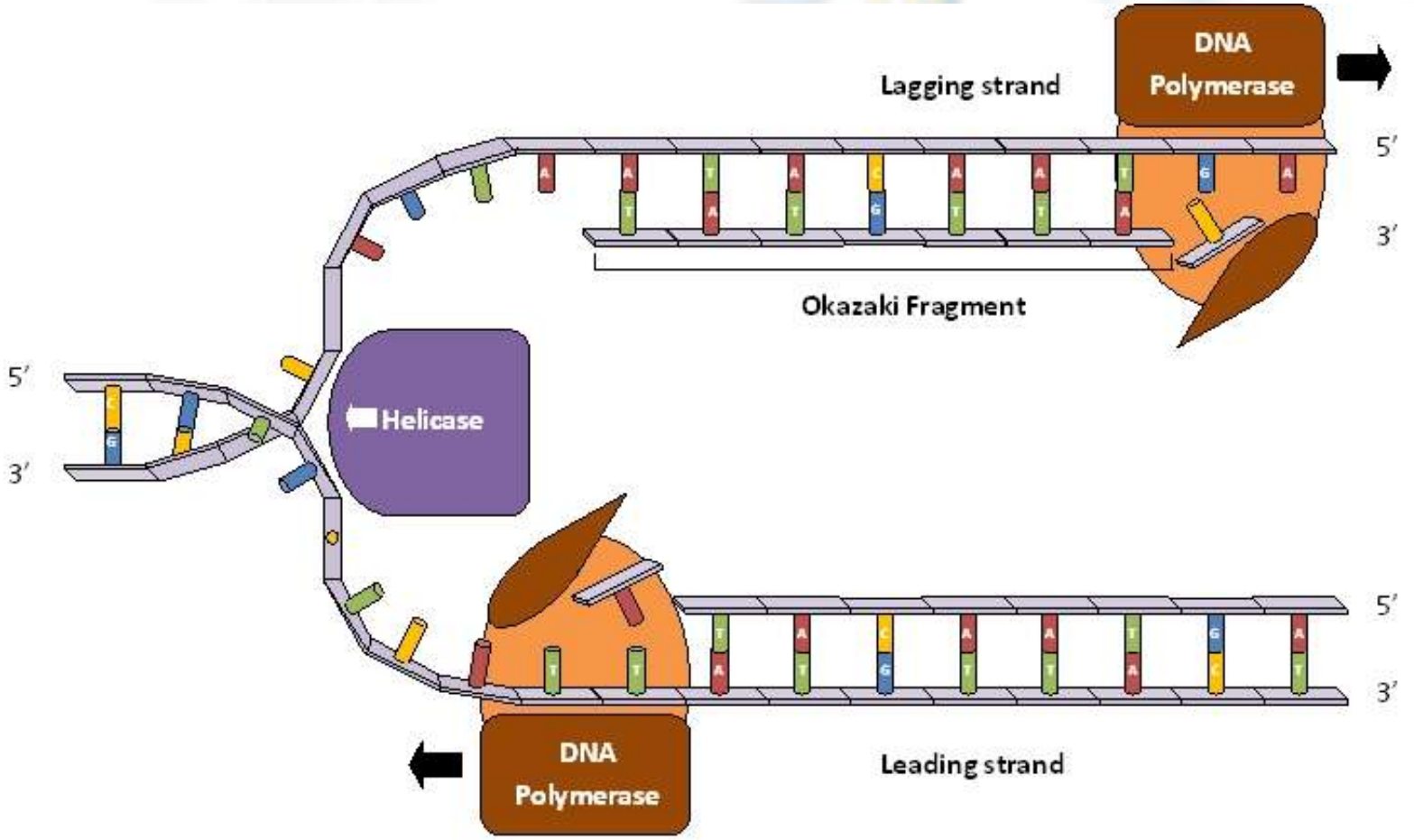
DNA Polymerases

POLYMERASES

Polymerase	Extension Rate (nt/sec)	Source
Taq pol	75	T. aquaticus
Amplitaq (Stoffel fragment)	>50	T. aquaticus
Vent	>80	Thermococcus litoralis
Pfu	60	Pyrococcus furiosus
Tth (RT activity)	>33	T. thermophilus

DNA Polymerases

POLYMERASES



DNA Polymerases

POLYMERASES

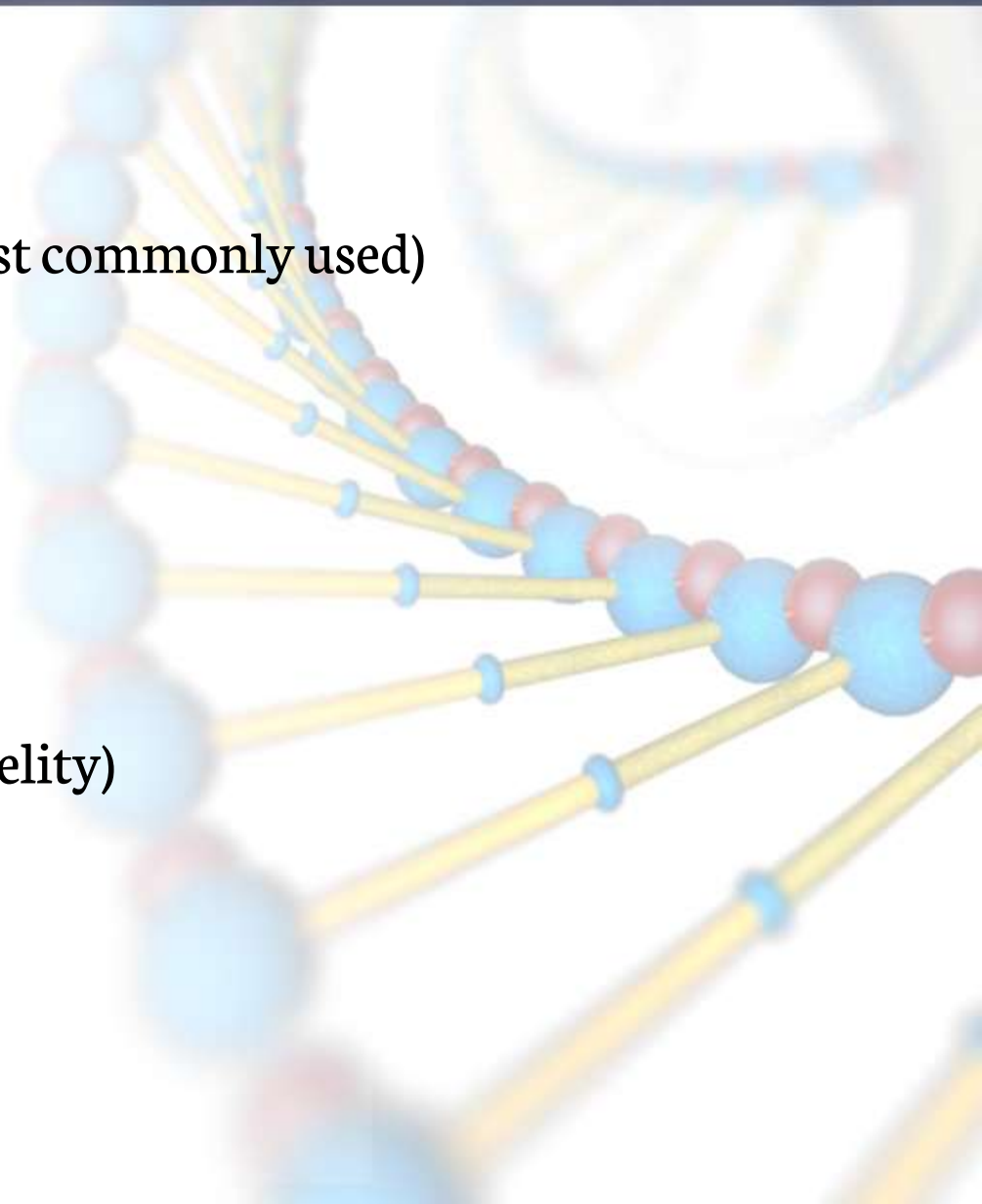
Taq: *Thermus aquaticus* (most commonly used)

Tfl: *T. flavus*

Tth: *T. thermophilus*

Tli: *Thermococcus litoralis*

Pfu: *Pyrococcus furiosus* (fidelity)



DNA Polymerases

DNA POLYMERASES

- Polymerases
  - Taq polymerase
  - Pfu polymerase
  - Vent polymerase

Essentials of Genetics

MODULE# 06

STANDARD PCR REACTION

Standard PCR Reaction

STANDARD REACTION

0.25 mM each primer

0.2 mM each dATP, dCTP, dGTP, dTTP

50 mM KCl

10 mM Tris

1.5 mM $MgCl_2$

2.5 units polymerase

10^2 - 10^5 copies of template

50 ul reaction volume

Standard PCR Reaction

PCR TEMPERATURES

- Denaturation temperature
  - Reduces double stranded molecules to single stranded
  - 90–96°C, 20-45 seconds
- Annealing temperature
  - Controls specificity of hybridization
  - 40–68°C, 20-30 seconds
- Extension temperature
  - Optimized for individual polymerases
  - 70–75°C, 30-45 seconds

Standard PCR Reaction

TEMPERATURES

Temp	For	Comments
94-60-72	Perfect, long primers	Higher temp can be used; maximum annealling temp
94-55-72	Good or perfectly matched primers between 19-24 nt	Standard conditions
94-50-72	Adequate primers	Allows 1-3 mismatches/20 nt
94-48-68	Poorly matched primers	Allows 4-5 mismatches/20 nt
94-45-65	Unknown match, likely poor	Primers of questionable quality

Standard PCR Reaction

TEMPERATURES

- Amplification takes place as the reaction mix is subject to an amplification program and temperatures.
- The amplification program consists of a series of 20–50 PCR cycles.

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MODULE# 07

INTERPRETATION OF RESULTS

Interpretation of Results

INTERPRETATION

- The PCR product should be of the expected size.
- Misprimes may occur due to non-specific hybridization of primers.

Interpretation of Results

PRIMERS DIMERS MAY REDUCE AMPLIFICATION

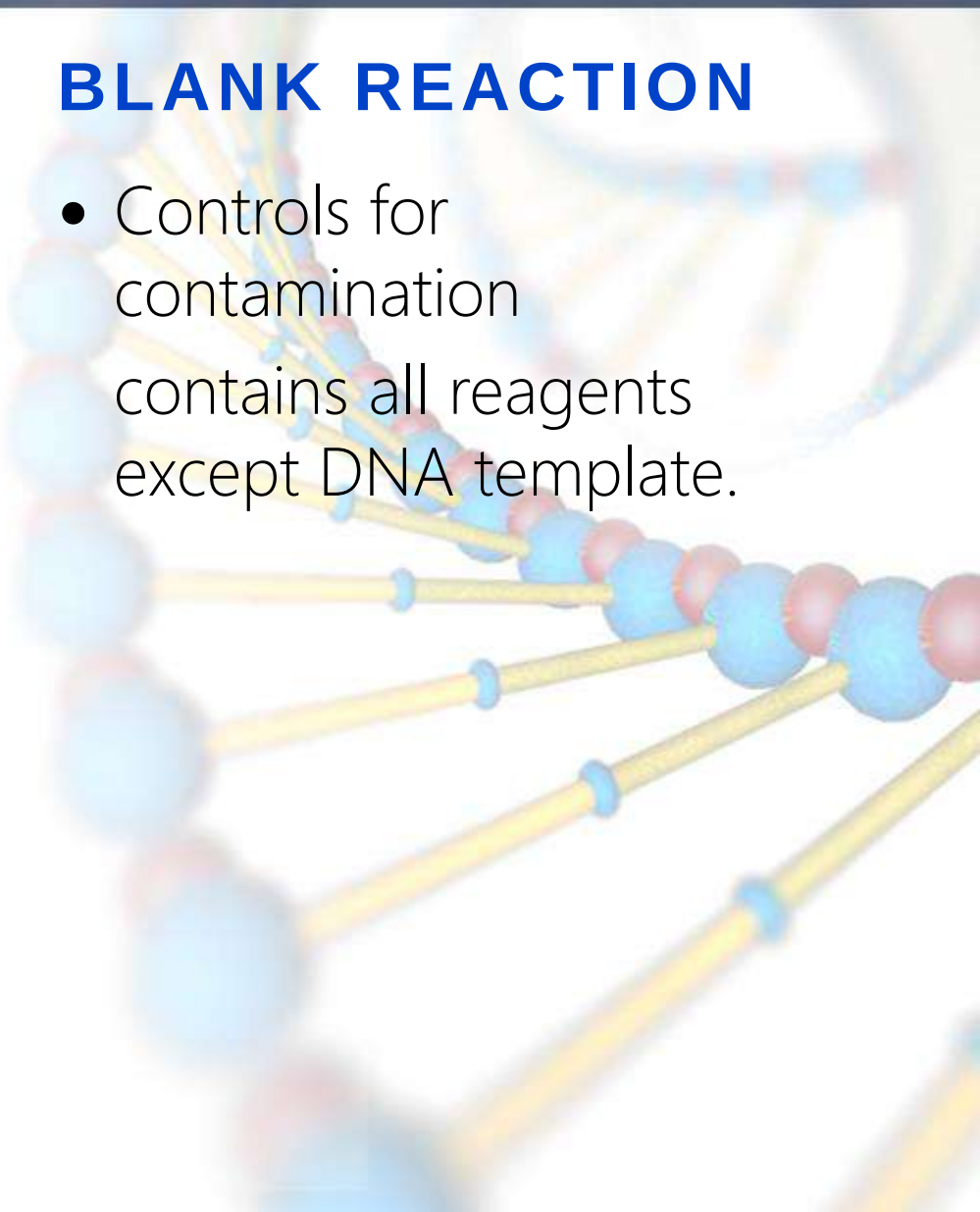
- Primer dimers may occur due to hybridization of primers to each other.

Interpretation of Results

BLANK REACTION

- Controls for contamination

  contains all reagents except DNA template.



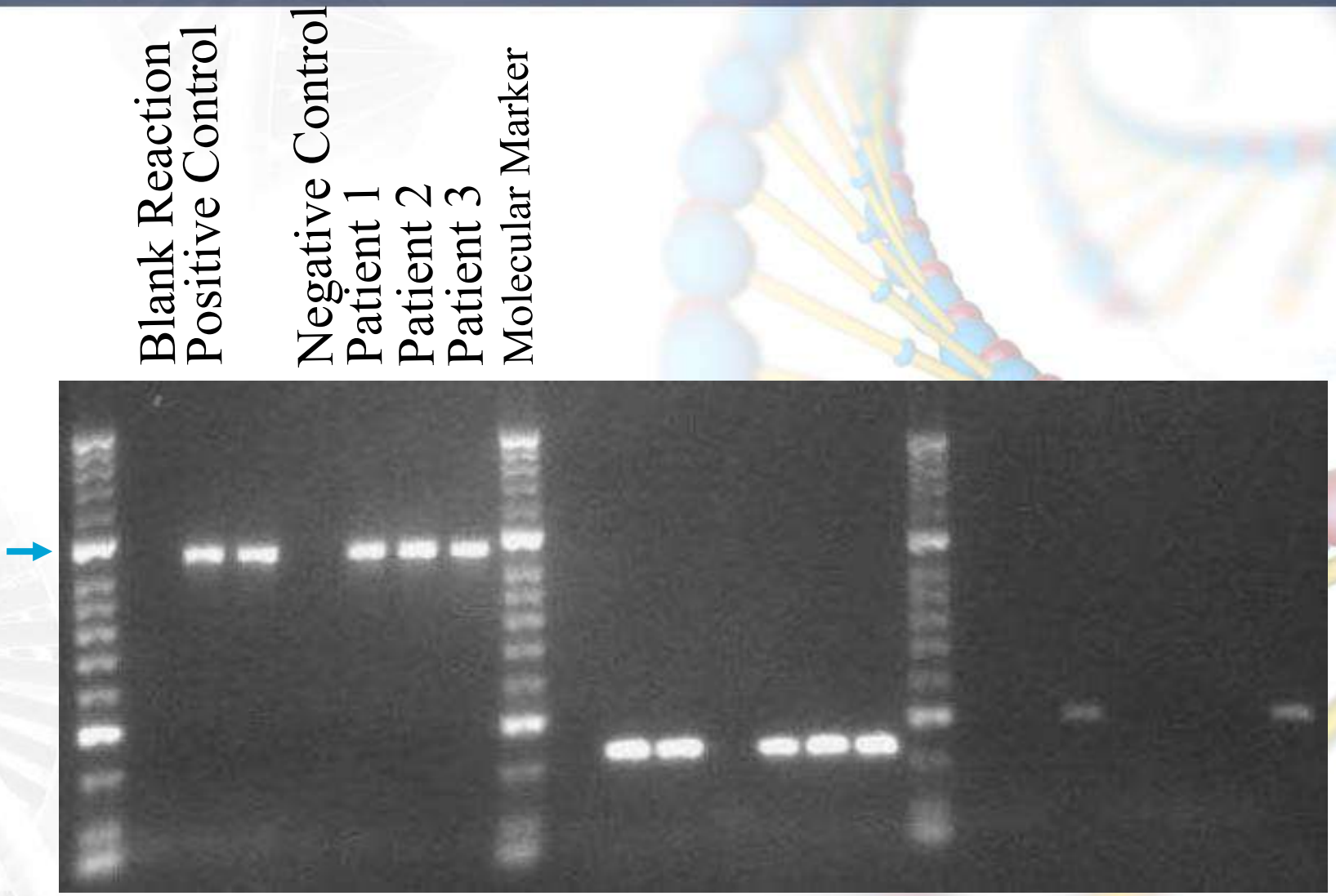
Interpretation of Results

NEGATIVE CONTROL

- Controls for specificity of the amplification reaction

  contains all reagents and a DNA template lacking the target sequence.

Interpretation of Results



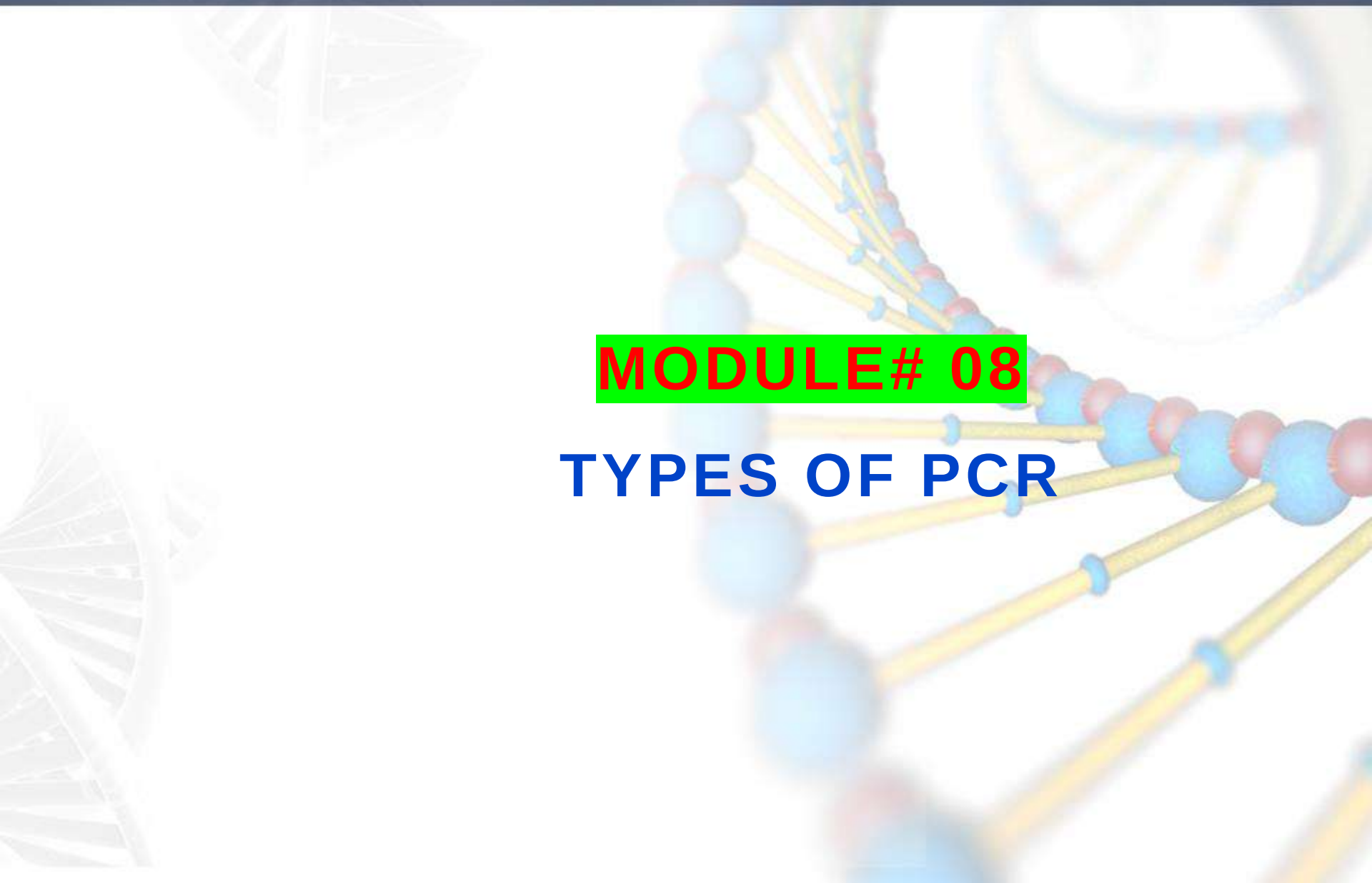
Interpretation of Results

CONCLUSION

- The PCR product should be of the expected size.

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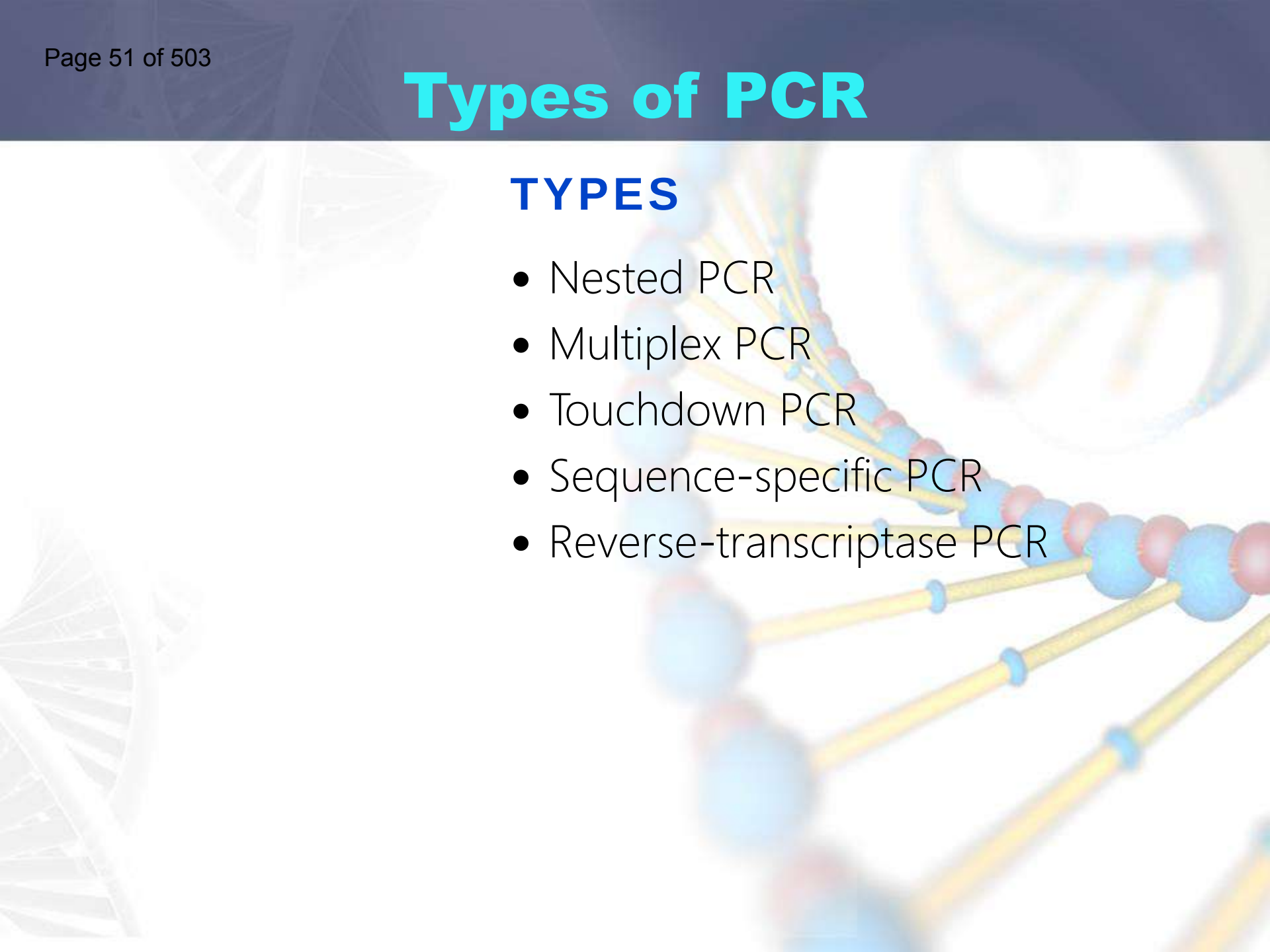
MODULE# 08

TYPES OF PCR

Types of PCR

TYPES

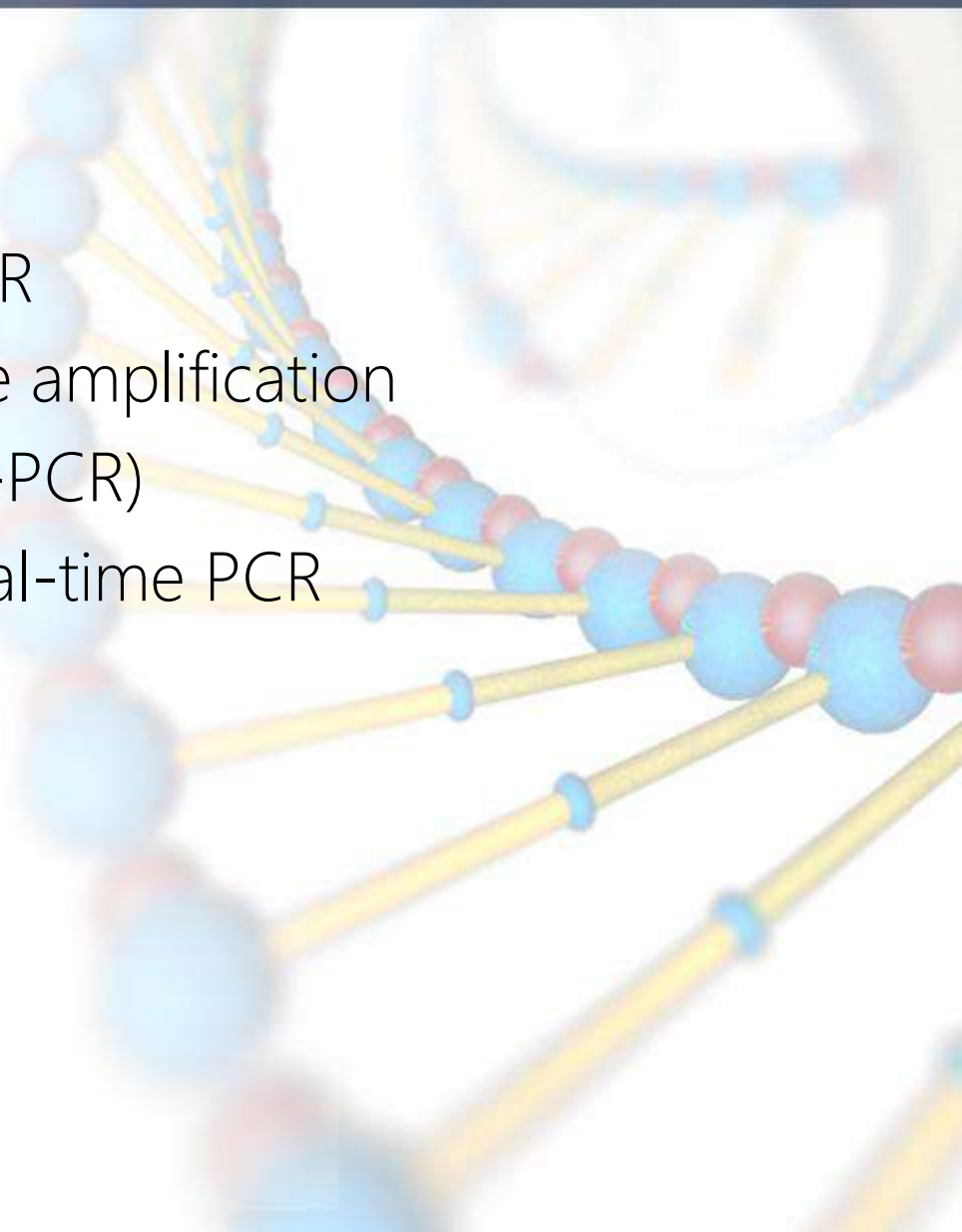
- Nested PCR
- Multiplex PCR
- Touchdown PCR
- Sequence-specific PCR
- Reverse-transcriptase PCR



Types of PCR

TYPES

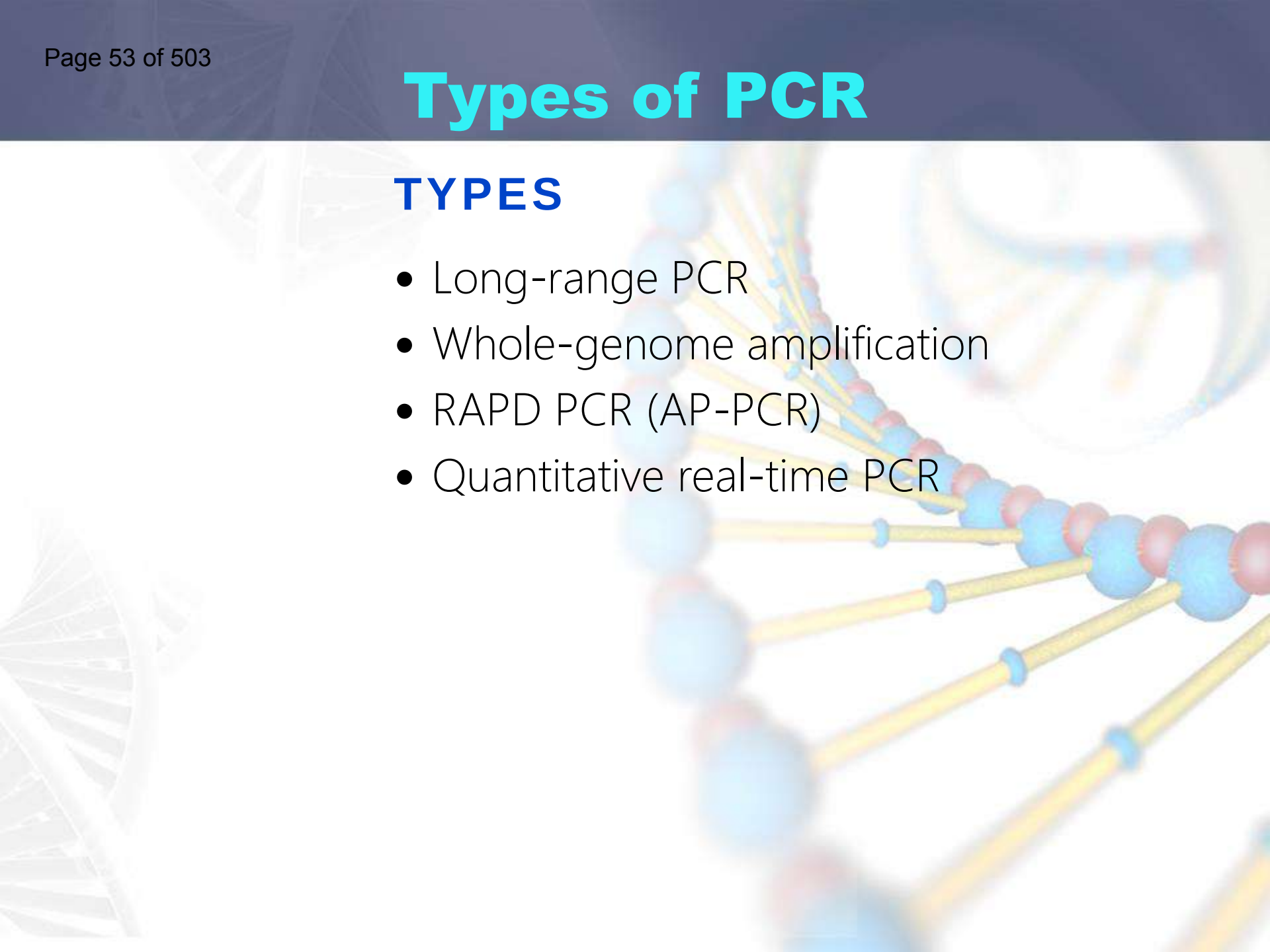
- Long-range PCR
- Whole-genome amplification
- RAPD PCR (AP-PCR)
- Quantitative real-time PCR



Types of PCR

TYPES

- Long-range PCR
- Whole-genome amplification
- RAPD PCR (AP-PCR)
- Quantitative real-time PCR



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MODULE# 09

NESTED PCR

Nested PCR

NESTED PCR

- Two pairs (instead of one pair) of PCR primers are used to amplify a fragment.
- First pair amplify a fragment similar to a standard PCR.

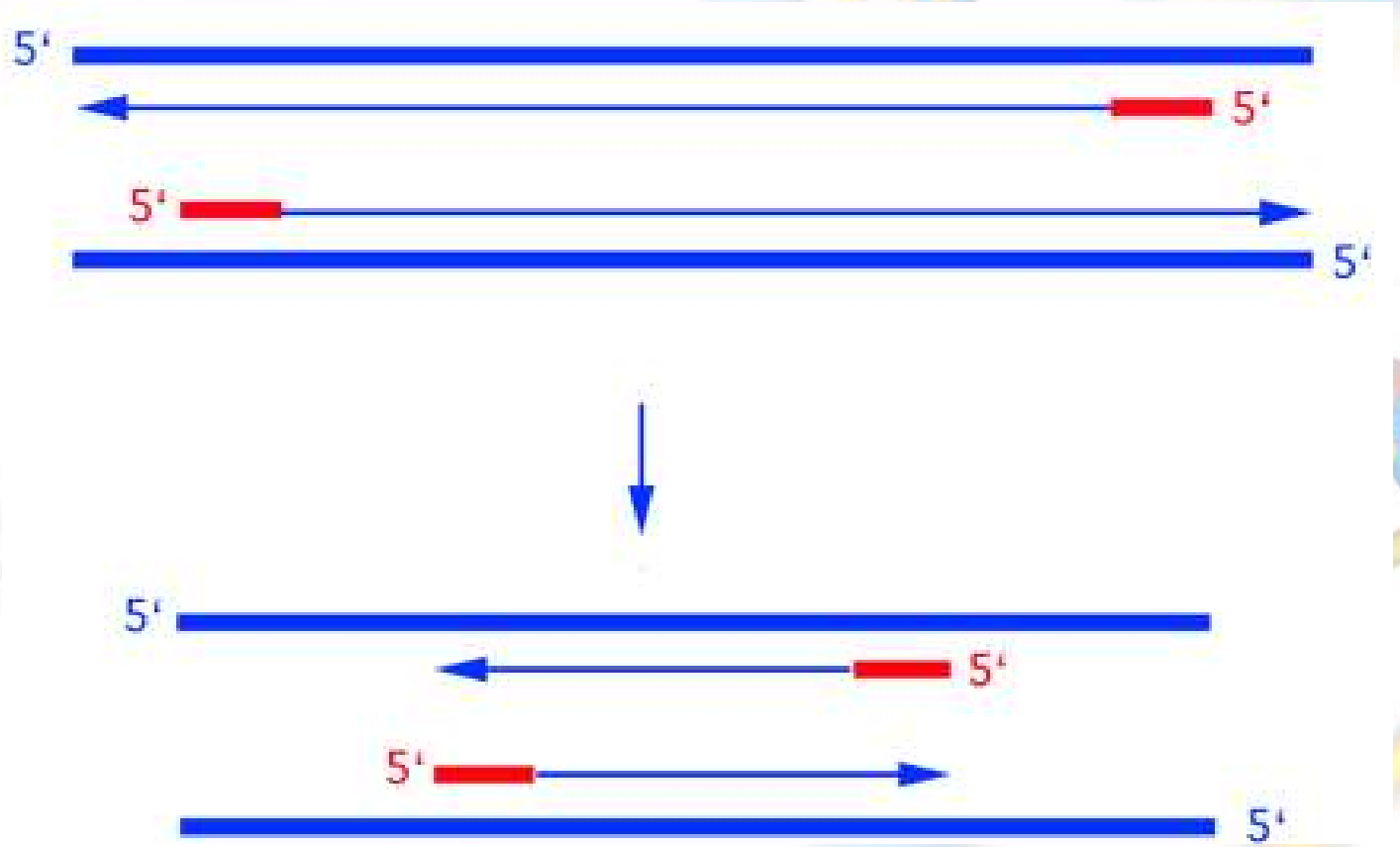
Nested PCR

NESTED PCR – SECOND PAIR OF PRIMERS

- Second pair of primers - nested primers (as they lie / are nested within the first fragment).

Nested PCR

NESTED PCR



Nested PCR

SECOND PCR PRODUCT IS SHORTER THAN FIRST ONE

- Second pair of primers bind inside the first PCR product fragment to allow amplification of a second PCR product.

Nested PCR

SECOND PCR PRODUCT IS SPECIFIC

- increases the specificity of DNA amplification, by reducing background due to non-specific amplification of DNA.

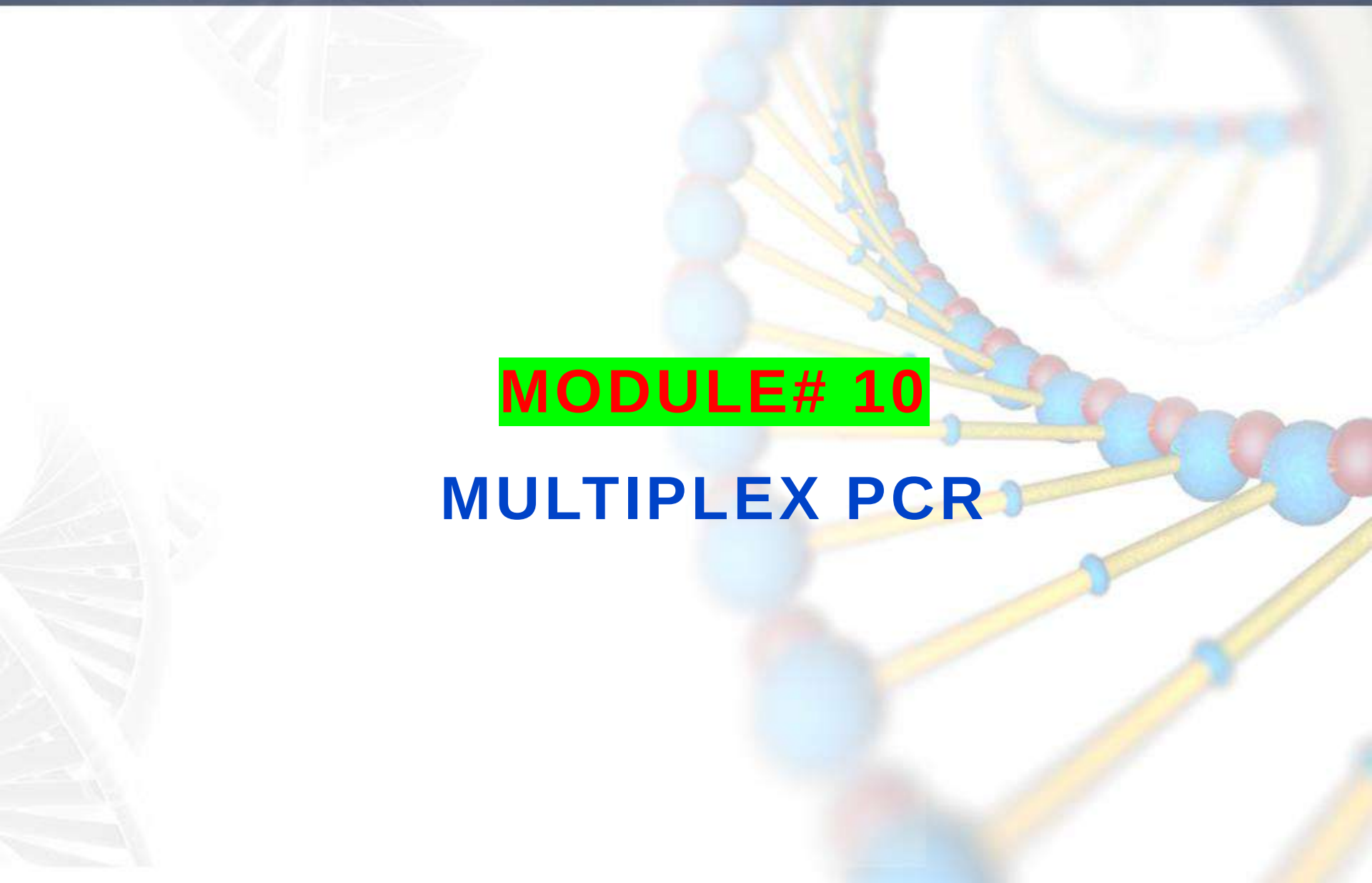
Nested PCR

ADVANTAGE OF NESTED PCR

- Very low probability of nonspecific amplification.

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MODULE# 10

MULTIPLEX PCR

Multiplex PCR

MULTIPLEX PCR

- Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers.

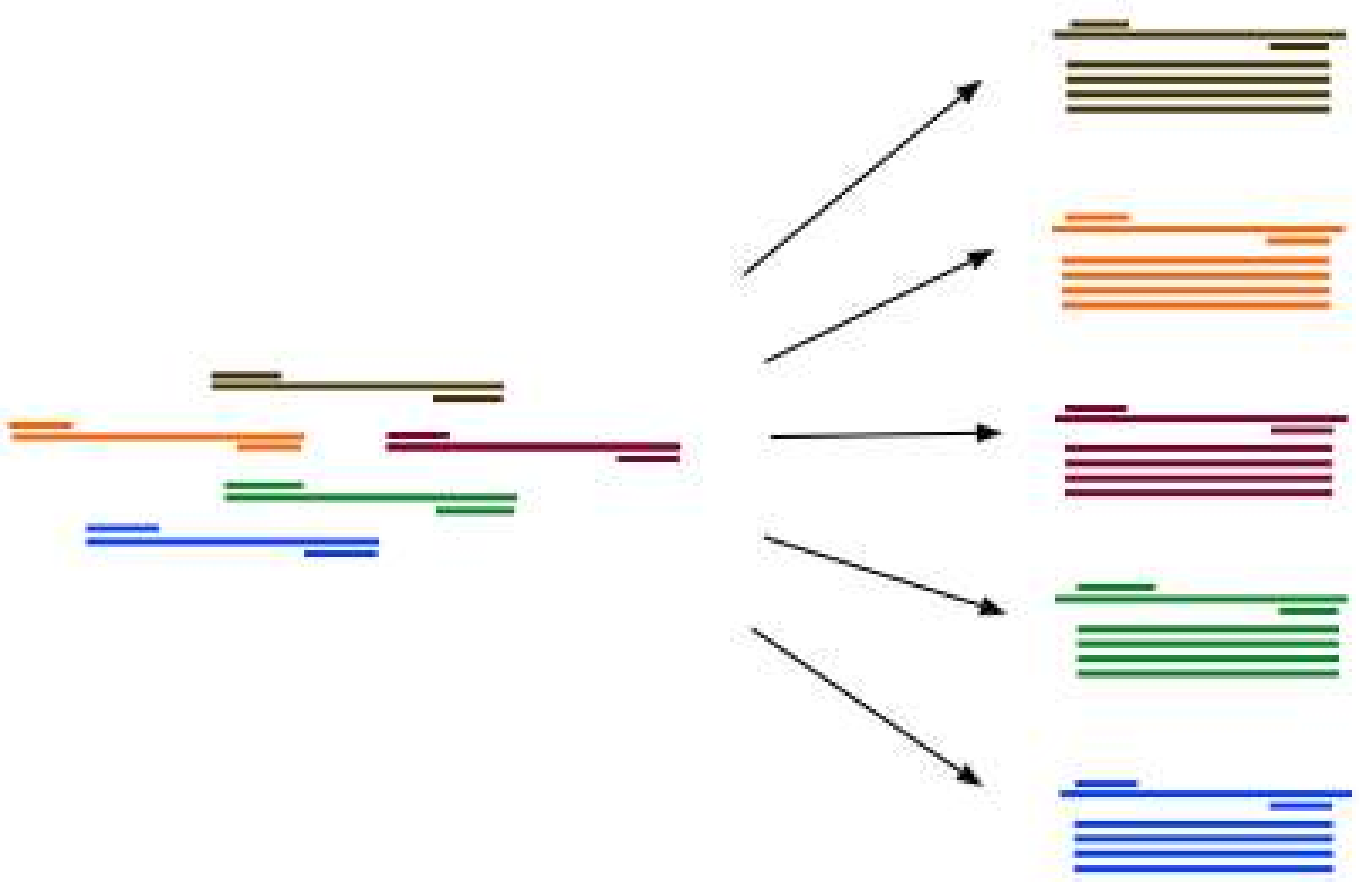
Multiplex PCR

AMPLICONS OF DIFFERENT SIZES ARE PRODUCED

- Multiplex-PCR consists of multiple primer sets within a single PCR mixture to produce amplicons of varying sizes that are specific to different DNA sequences.

Multiplex PCR

MULTIPLEX PCR



Multiplex PCR

SINGLE REACTION, MANY AMPLICONS

- By targeting multiple sequences at once, additional information may be gained from a single test run that otherwise would require several times the reagents and more time to perform.

Multiplex PCR

ANNEALING TEMPERATURES BE OPTIMIZED

- Annealing temperatures for each of the primer sets must be optimized to work correctly within a single reaction.

Multiplex PCR

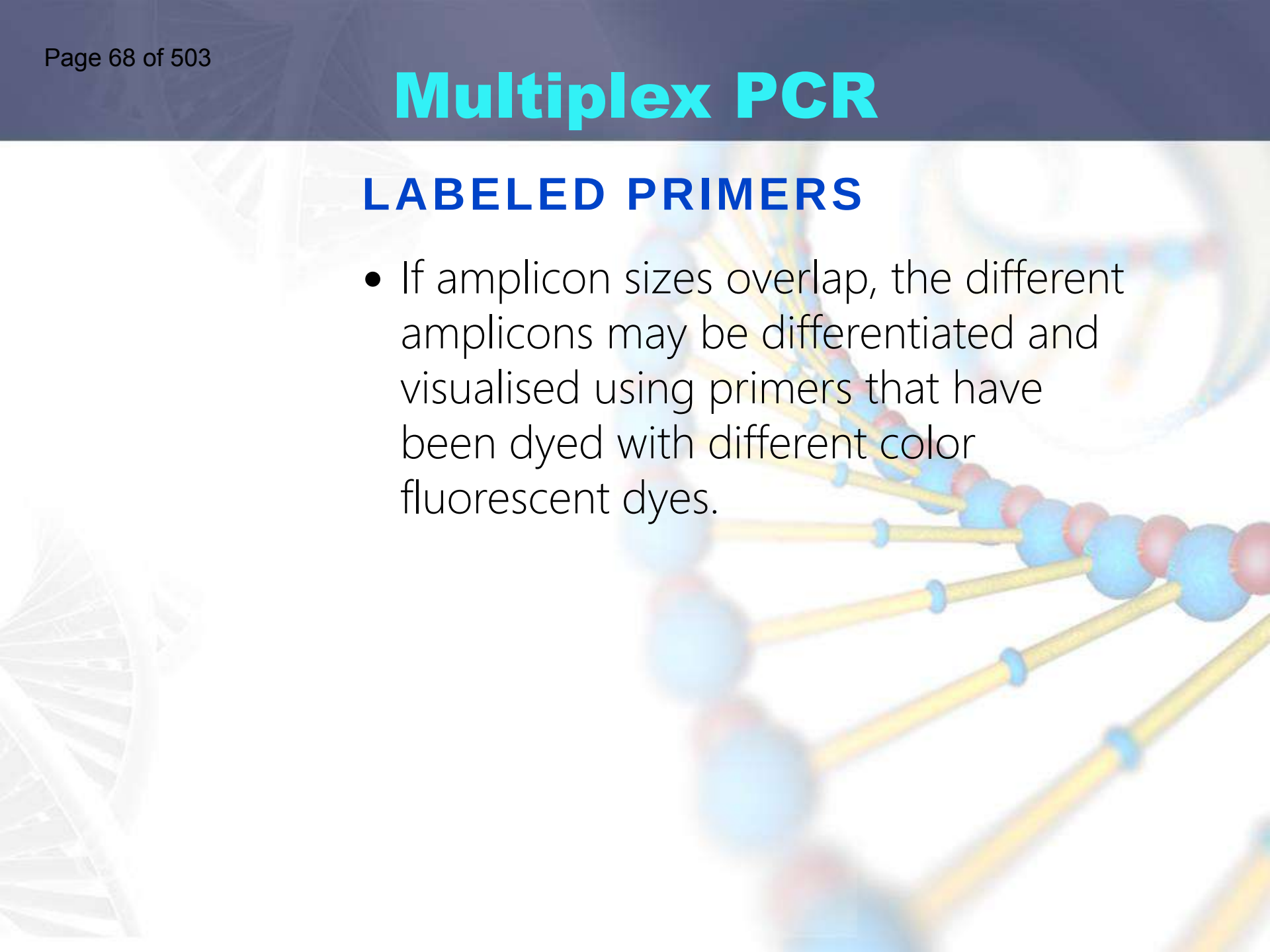
AMPLICON SIZE SHOULD BE DIFFERENT OR BE LABELED

- Amplicon sizes should be different enough to form distinct bands when visualized by gel electrophoresis.

Multiplex PCR

LABELED PRIMERS

- If amplicon sizes overlap, the different amplicons may be differentiated and visualised using primers that have been dyed with different color fluorescent dyes.



Multiplex PCR

PRIMERS PAIRS BE OPTIMIZED

- The primer design for all primers pairs has to be optimized so that all primer pairs can work at the same annealing temperature during PCR.

Multiplex PCR

MULTIPLEX PCR

- The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

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MODULE# 011

REVERSE TRANSCRIPTASE PCR

Reverse Transcriptase PCR

REVERSE TRANSCRIPTASE PCR

- Based on the process of reverse transcription, which reverse transcribes RNA into DNA.

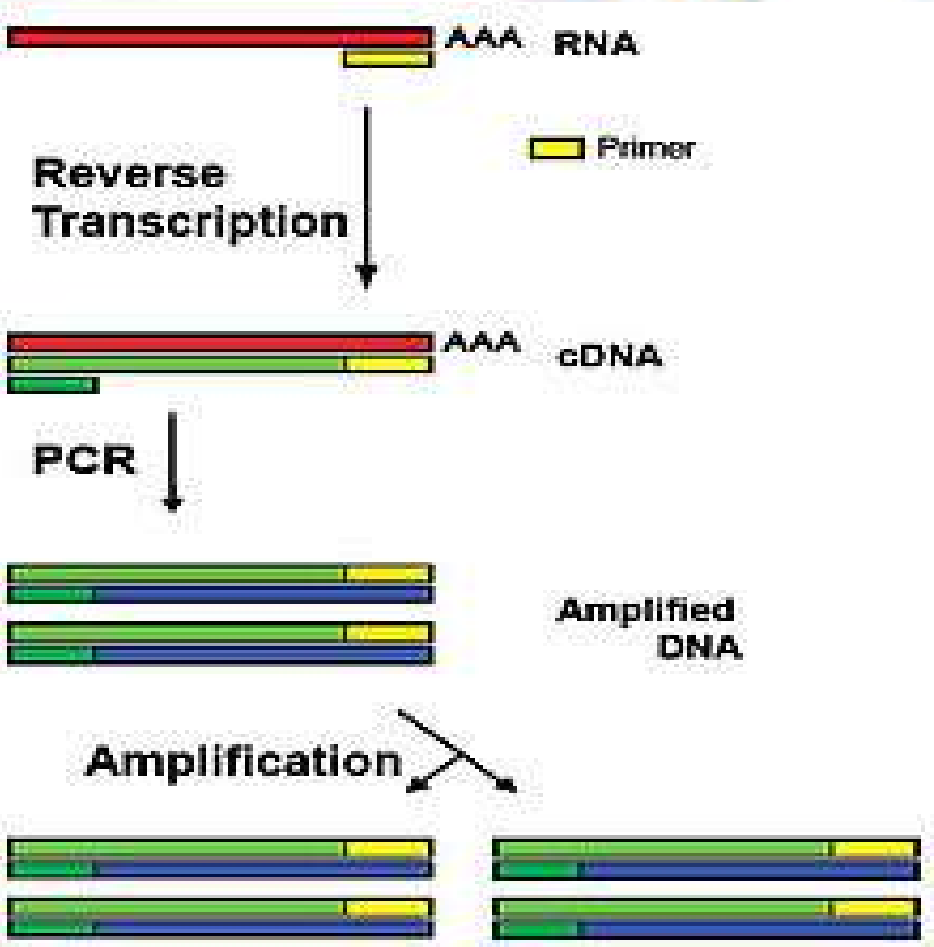
Reverse Transcriptase PCR

REVERSE TRANSCRIPTASE PCR

- First step of RT-PCR - first strand reaction
- Synthesis of cDNA using oligo dT primers (37°C) one hour.
- Second strand reaction - digestion of cDNA:RNA hybrid (RNaseH)-
- Standard PCR with DNA oligo primers .

Reverse Transcriptase PCR

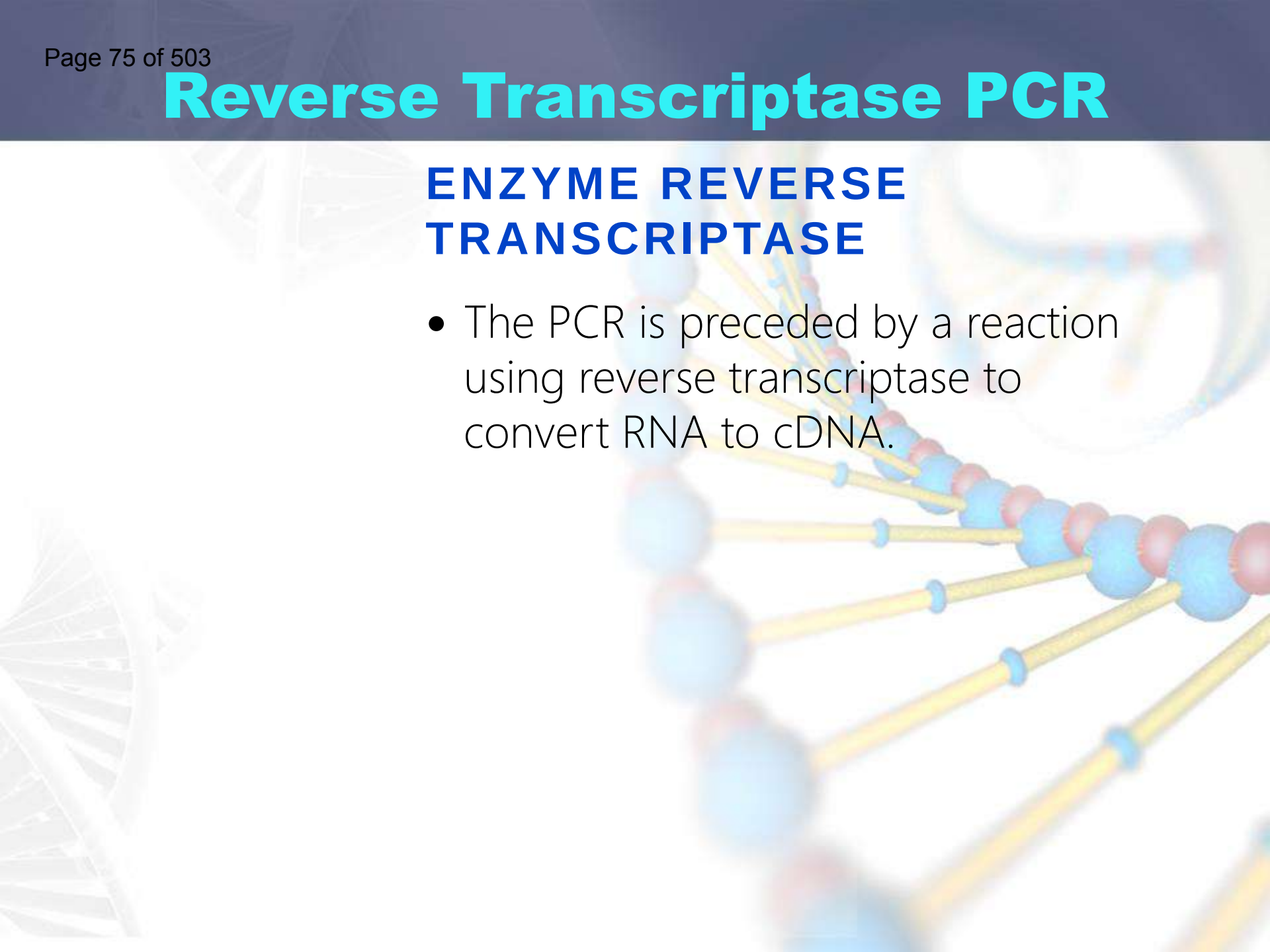
REVERSE TRANSCRIPTASE PCR



Reverse Transcriptase PCR

ENZYME REVERSE TRANSCRIPTASE

- The PCR is preceded by a reaction using reverse transcriptase to convert RNA to cDNA.



Reverse Transcriptase PCR

REVERSE TRANSCRIPTASE PCR

- RT-PCR is widely used in expression profiling, to determine the expression of a gene
- To identify the sequence of an RNA transcript.

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MODULE# 12

REAL TIME PCR

Real Time PCR

REAL TIME PCR

- Real Time PCR is a technique in which fluoroprobes bind to specific target regions of amplicons to produce fluorescence during PCR.

Real Time PCR

REAL TIME PCR

- Real Time PCR is used to measure the quantity of a PCR product.
- The fluorescence, measured in Real Time, is detected in a PCR cycler with an inbuilt filter flurometer.
- It is the method of choice to quantitatively measure starting amounts of DNA, cDNA or RNA.

Real Time PCR

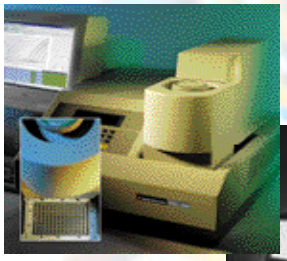
REAL TIME PCR



iCycler
BioRad



LightCycler
Roche



5700
Applied Biosystems



7700
Applied Biosystems



FluorTracker
Stratagene



FluorImager
Molecular Dynamics

Real Time PCR

REAL TIME PCR

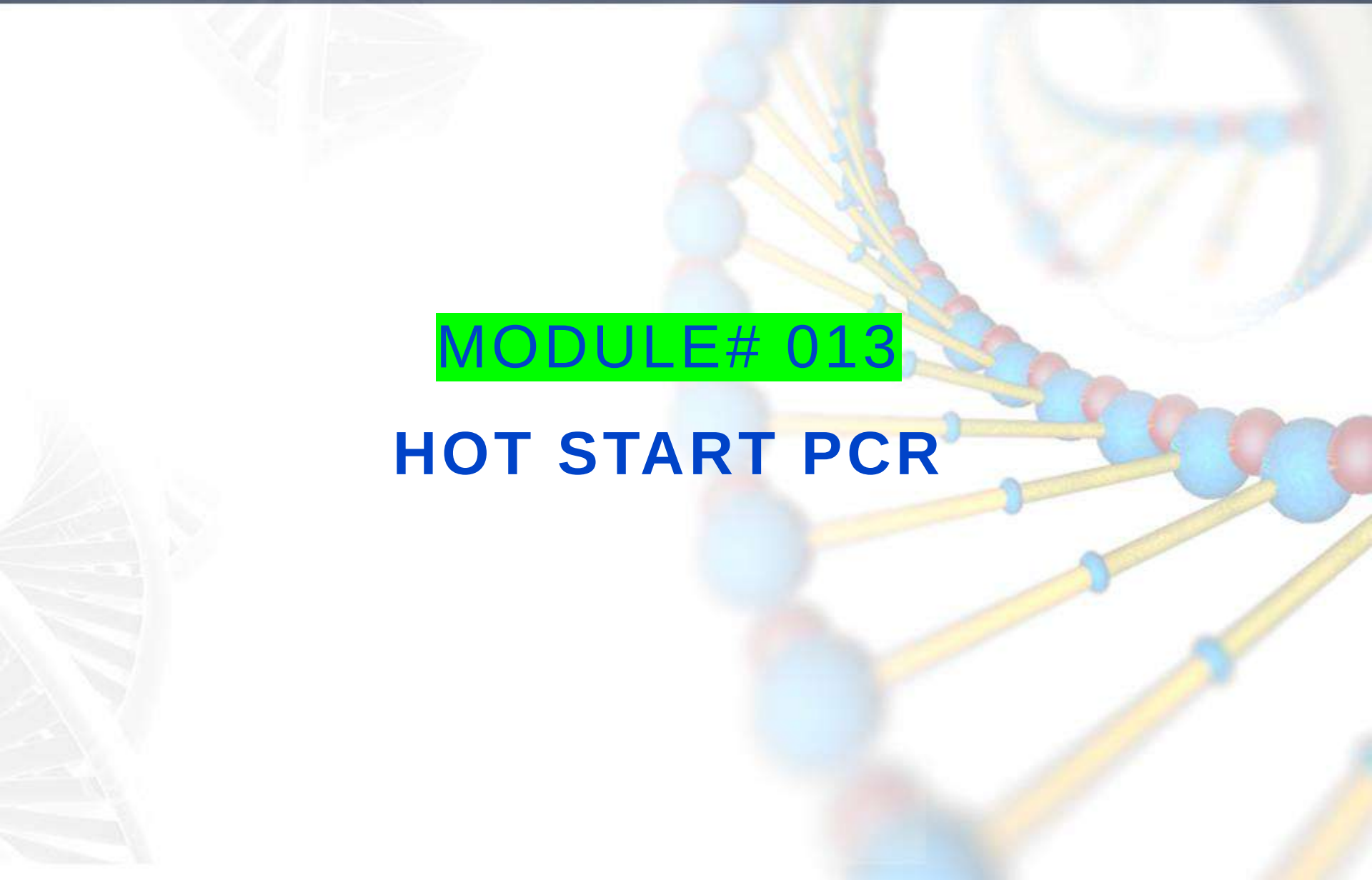
- Quantitative real-time PCR is often confusingly known as RT-PCR (Real Time PCR) or RQ-PCR.
- QRT-PCR or RTQ-PCR are more appropriate contractions.
- QRT-PCR methods use fluorescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.

Real Time PCR

REAL TIME PCR

- Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.

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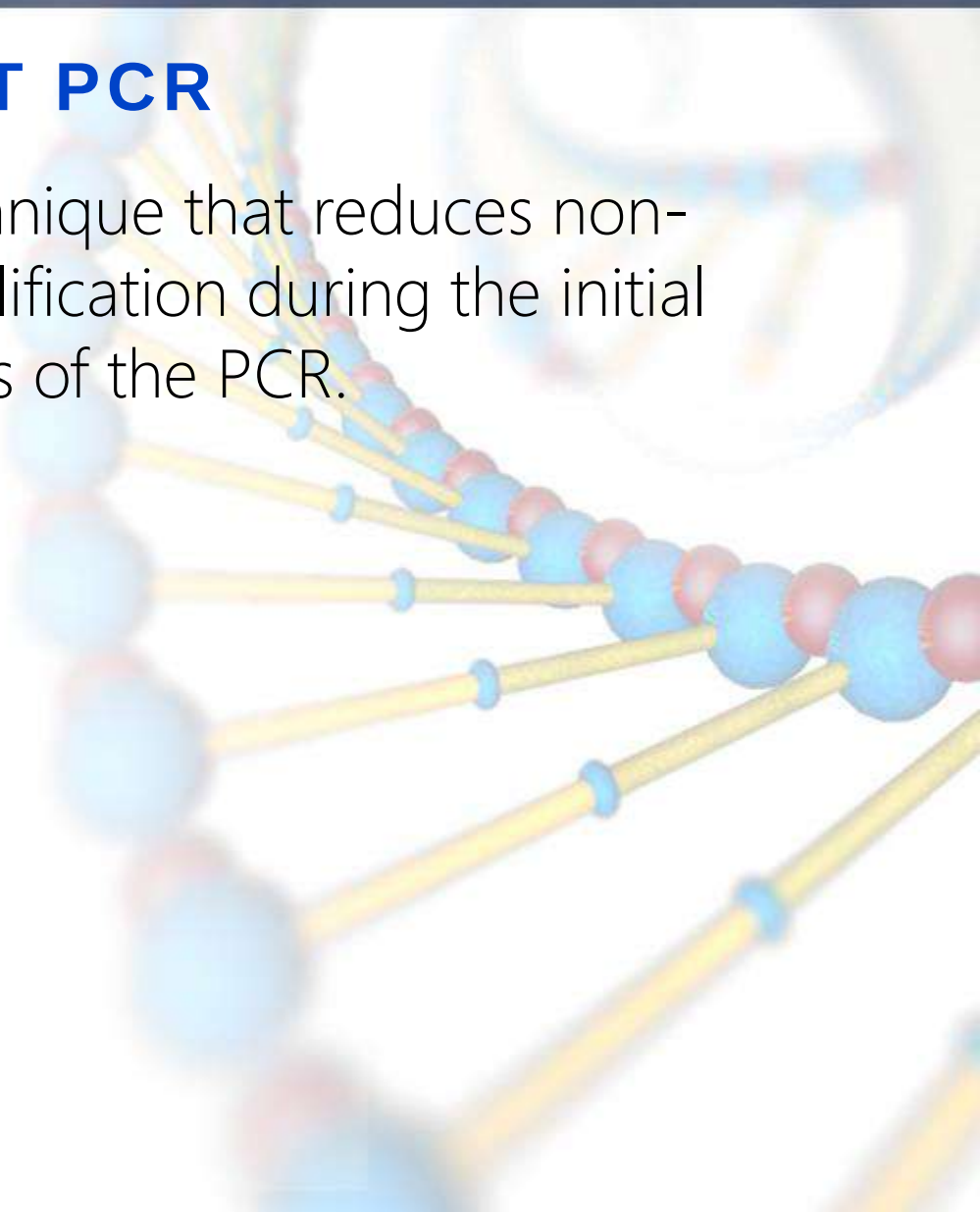
MODULE# 013

HOT START PCR

Hot Start PCR

HOT START PCR

- This is a technique that reduces non-specific amplification during the initial set up stages of the PCR.



Hot Start PCR

HOT START PCR

- The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C).
- Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature.
- This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step.

Hot Start PCR

HOT START PCR

- DNA Polymerase- Eubacterial type I DNA polymerase, Pfu.
- These thermophilic DNA polymerases show a very small polymerase activity at room temperature .

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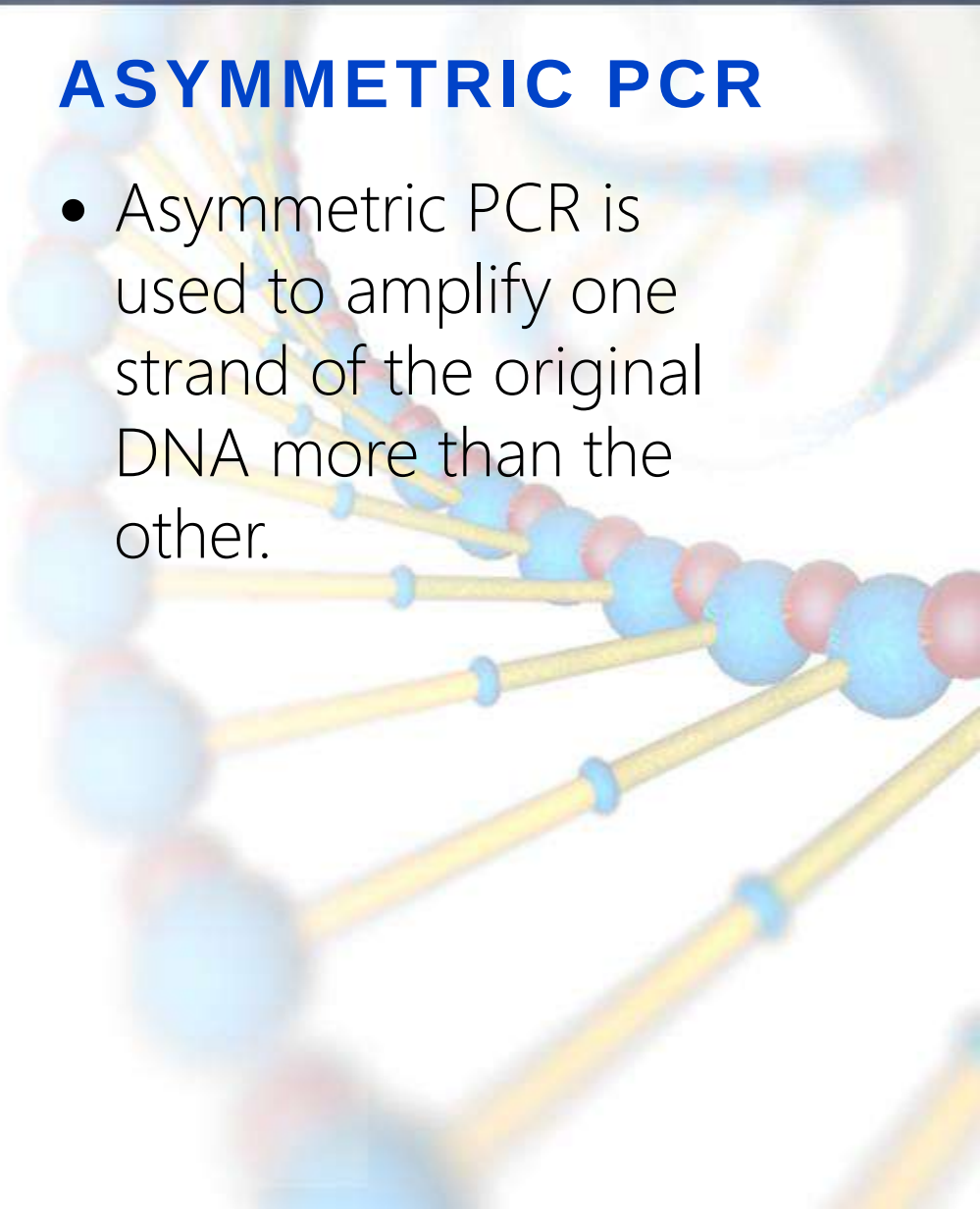
MODULE# 014

ASYMMETRIC PCR

Asymmetric PCR

ASYMMETRIC PCR

- Asymmetric PCR is used to amplify one strand of the original DNA more than the other.



Asymmetric PCR

ASYMMETRIC PCR

- It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.

Asymmetric PCR

ASYMMETRIC PCR

- PCR is carried out as usual, but with a great excess of one primers for the chosen strand.

Asymmetric PCR

ASYMMETRIC PCR

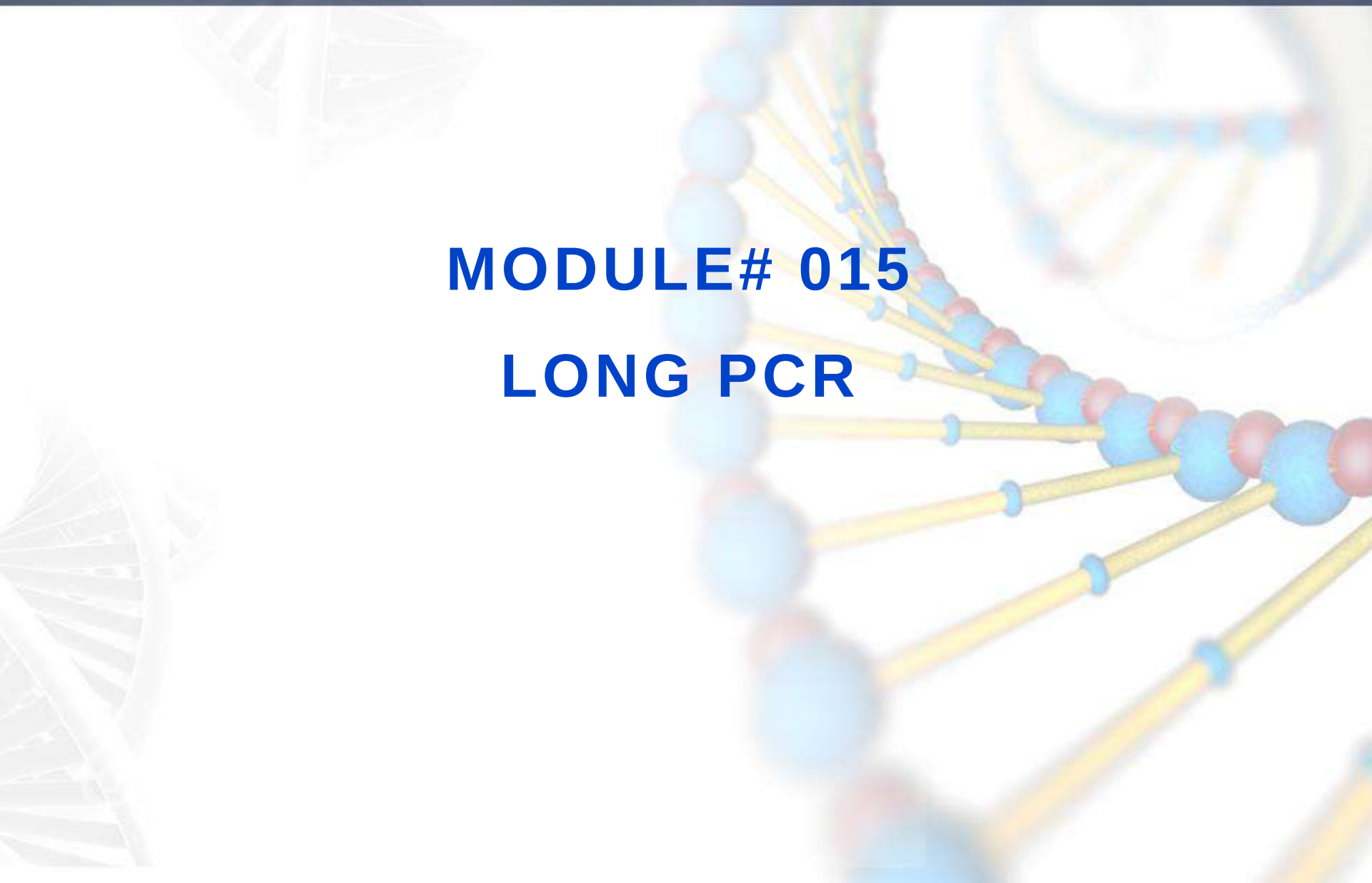
- Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required.

Asymmetric PCR

ASYMMETRIC PCR

- It finds use in some types of sequencing and hybridization probing where having only one of the two complementary strands is required.

Essentials of Genetics



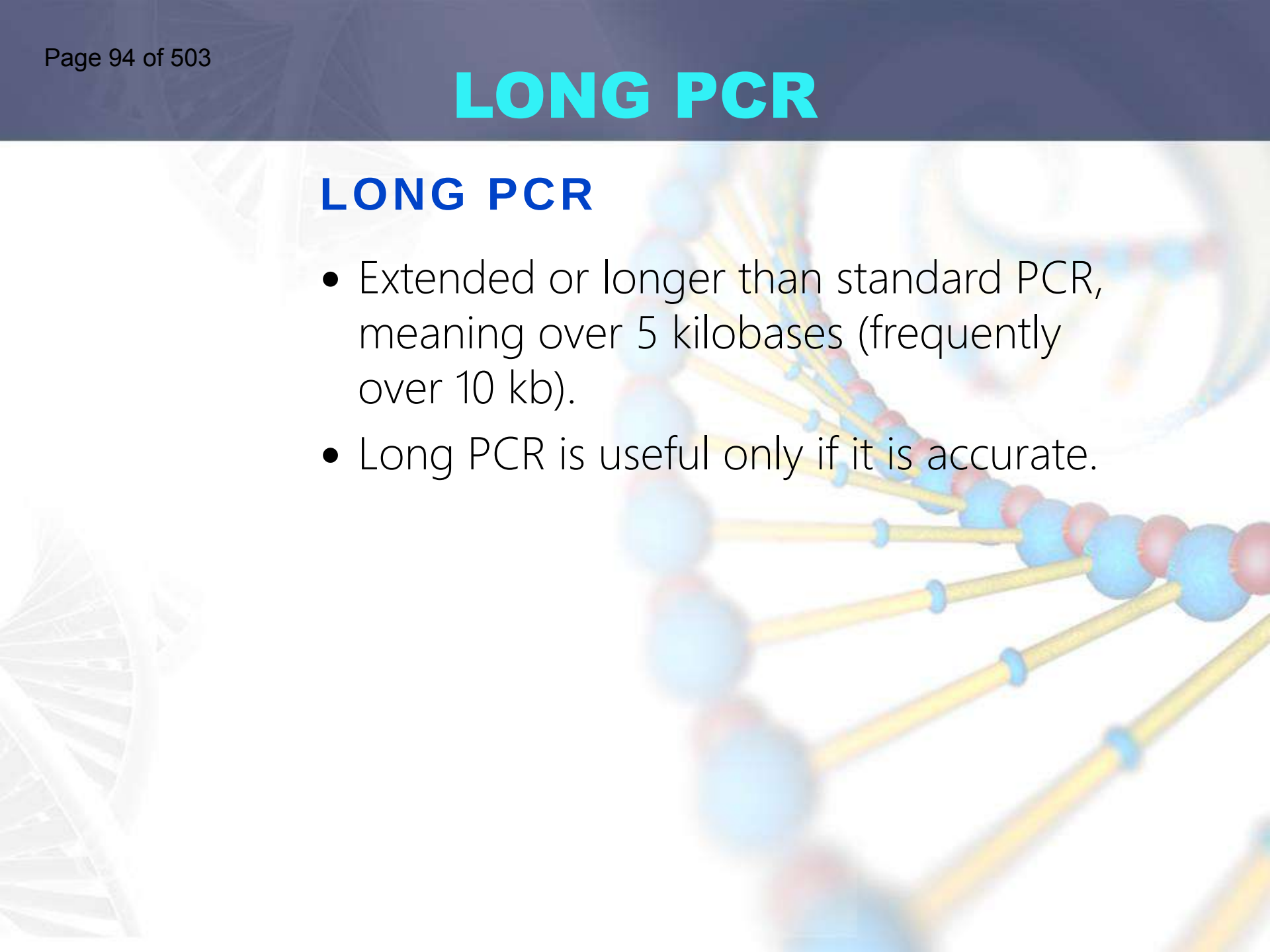
MODULE# 015

LONG PCR

LONG PCR

LONG PCR

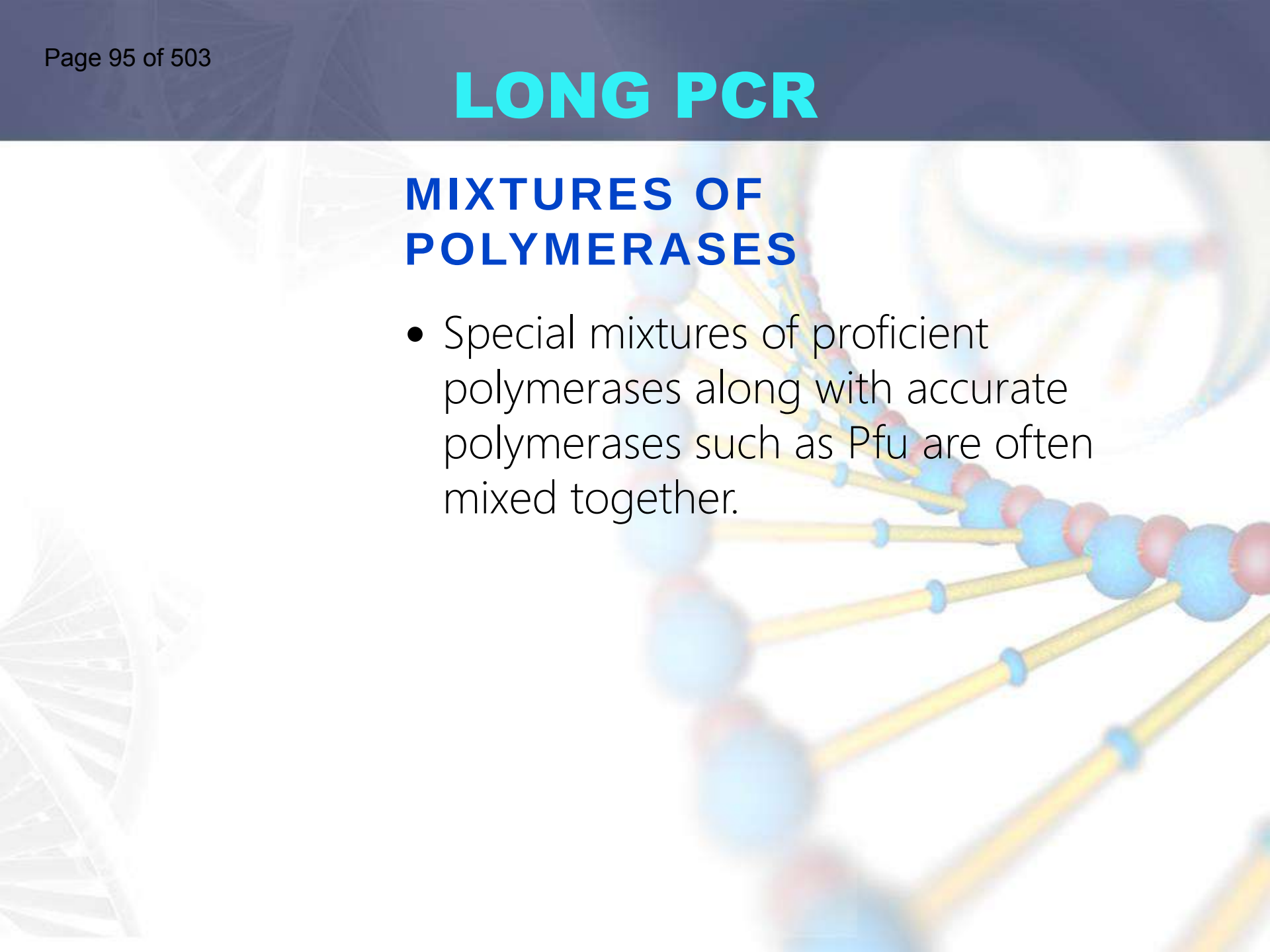
- Extended or longer than standard PCR, meaning over 5 kilobases (frequently over 10 kb).
- Long PCR is useful only if it is accurate.



LONG PCR

MIXTURES OF POLYMERASES

- Special mixtures of proficient polymerases along with accurate polymerases such as Pfu are often mixed together.



LONG PCR

LONG PCR

- Application - to clone large genes not possible with conventional PCR.

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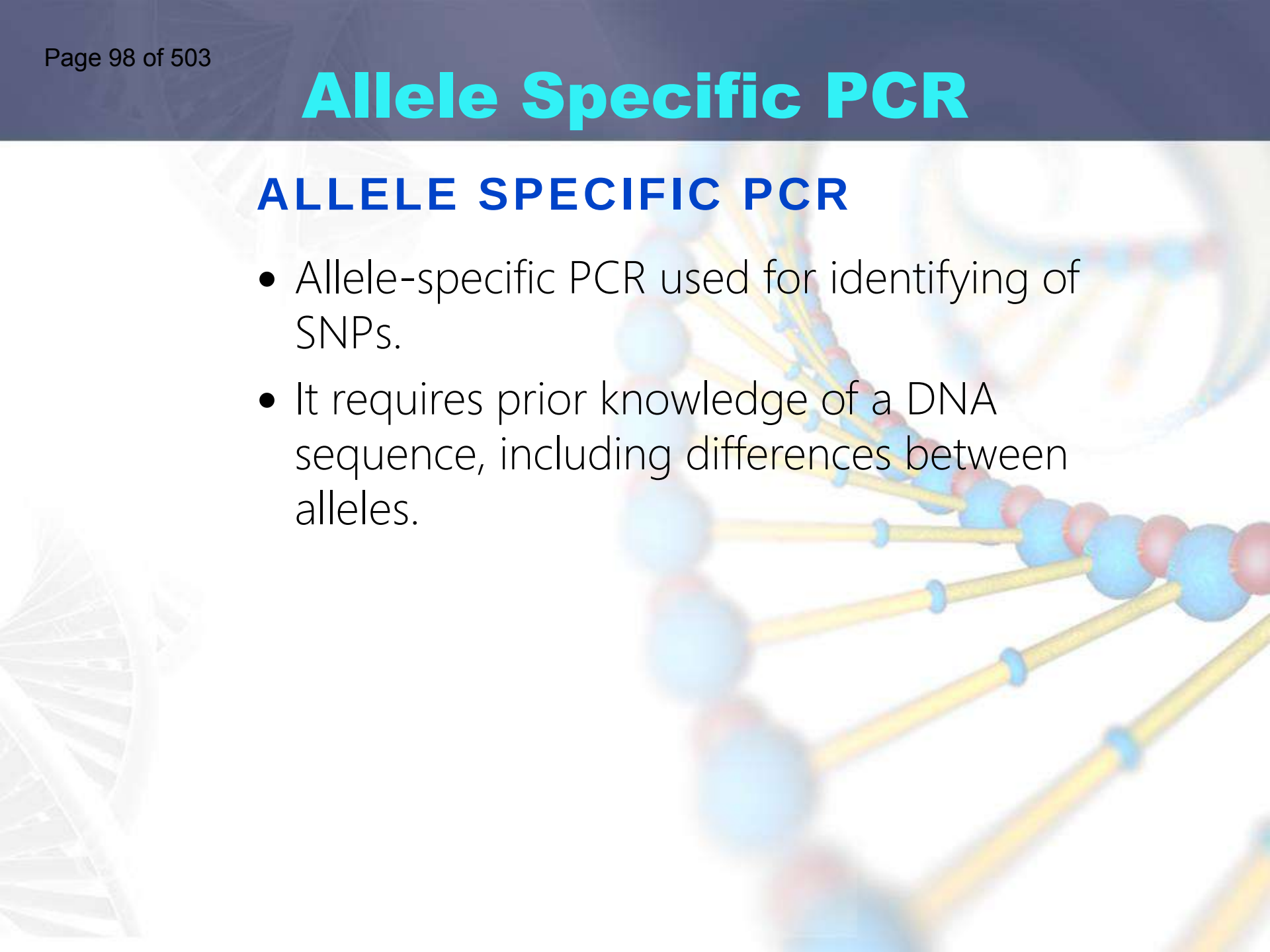
MODULE# 016

ALLELE SPECIFIC PCR

Allele Specific PCR

ALLELE SPECIFIC PCR

- Allele-specific PCR used for identifying of SNPs.
- It requires prior knowledge of a DNA sequence, including differences between alleles.



Allele Specific PCR

ALLELE SPECIFIC PCR

- Uses primers whose 3' ends encompass the SNP.
- PCR amplification under stringent conditions is much less efficient in the presence of a mismatch between template and primer.

Allele Specific PCR

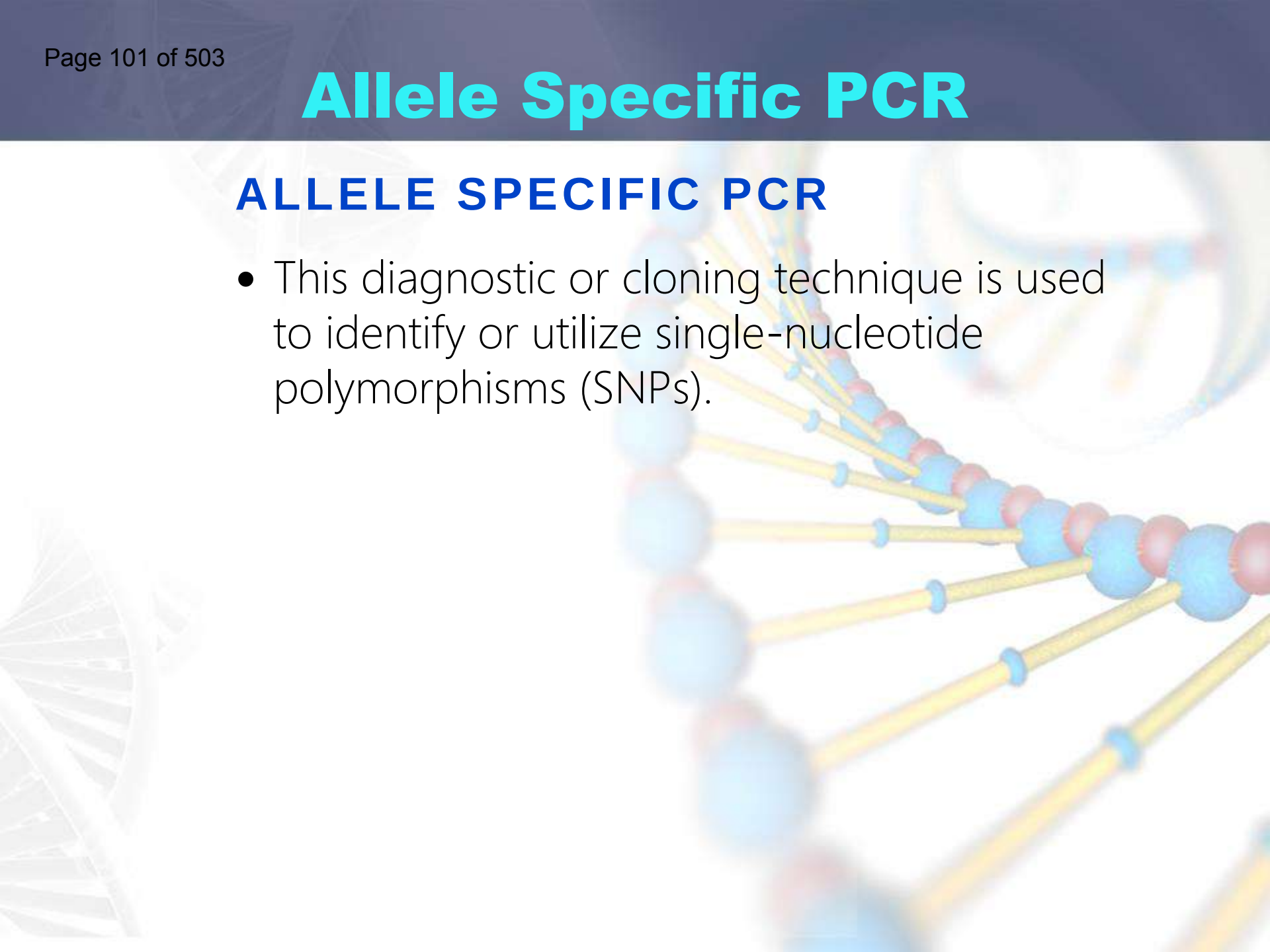
AMPLIFICATION WITH SNP SPECIFIC PRIMER

- Successful amplification with an SNP-specific primer signals presence of the specific SNP in a sequence.

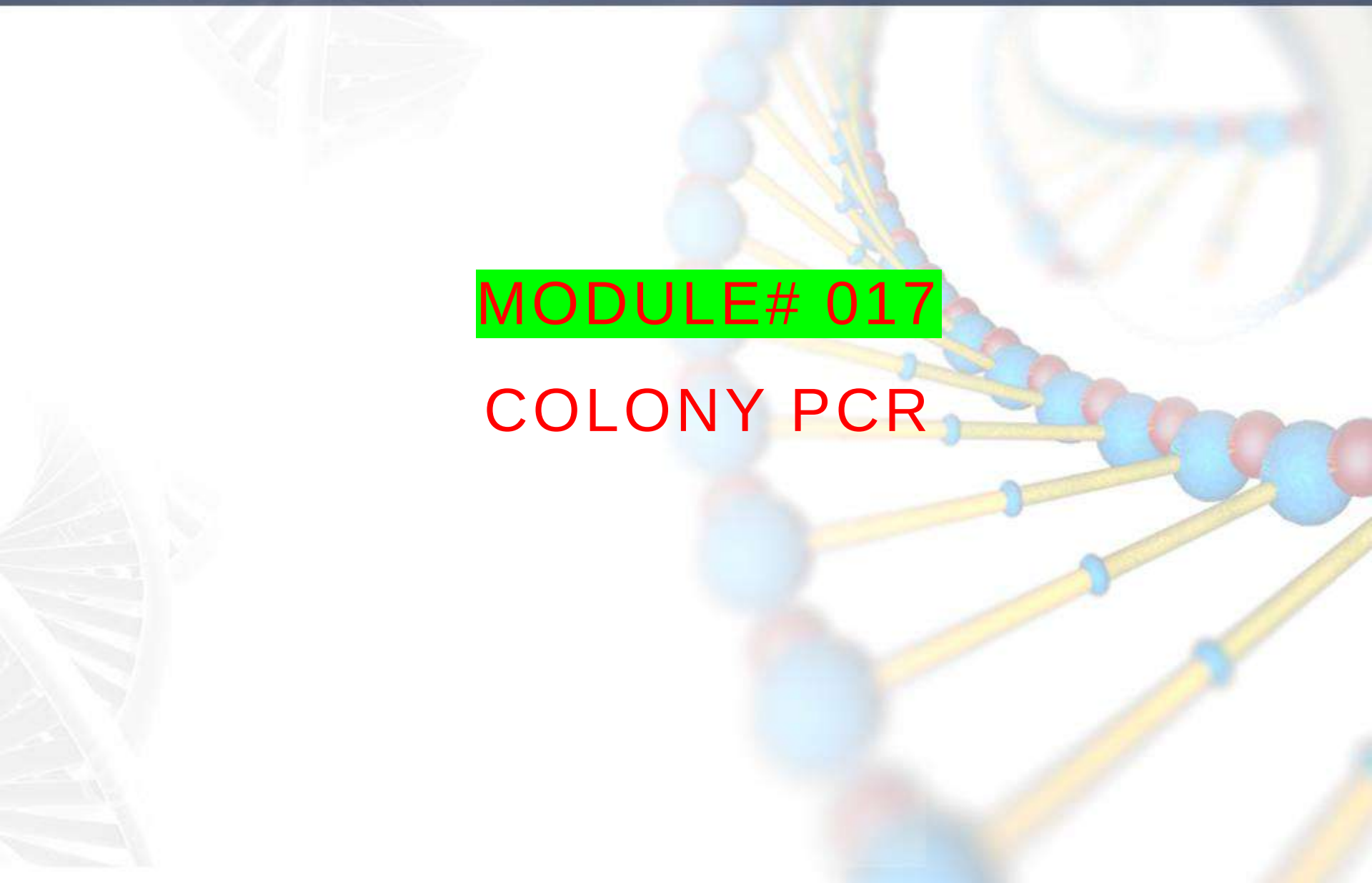
Allele Specific PCR

ALLELE SPECIFIC PCR

- This diagnostic or cloning technique is used to identify or utilize single-nucleotide polymorphisms (SNPs).



Essentials of Genetics



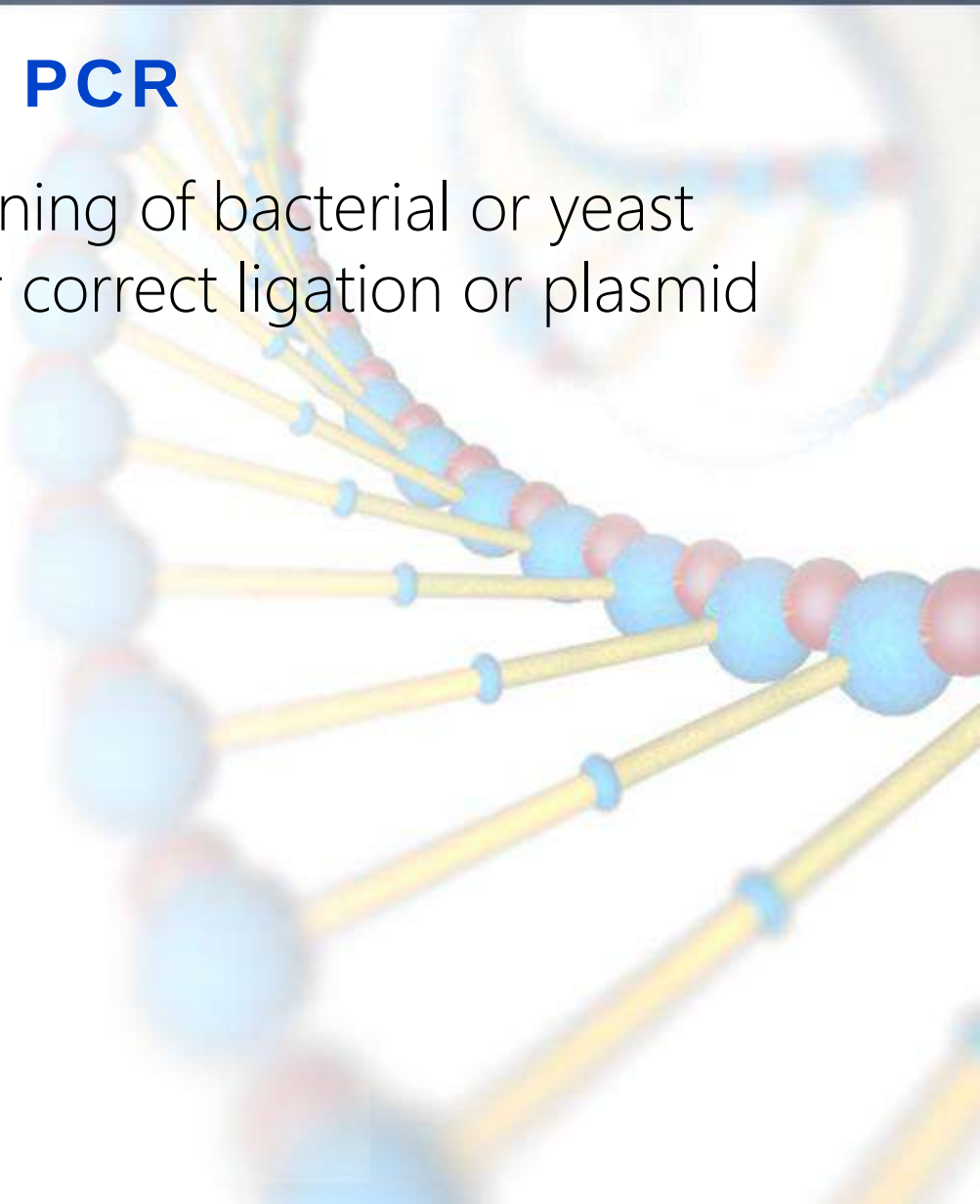
MODULE# 017

COLONY PCR

Colony PCR

COLONY PCR

- The screening of bacterial or yeast clones for correct ligation or plasmid products.



Colony PCR

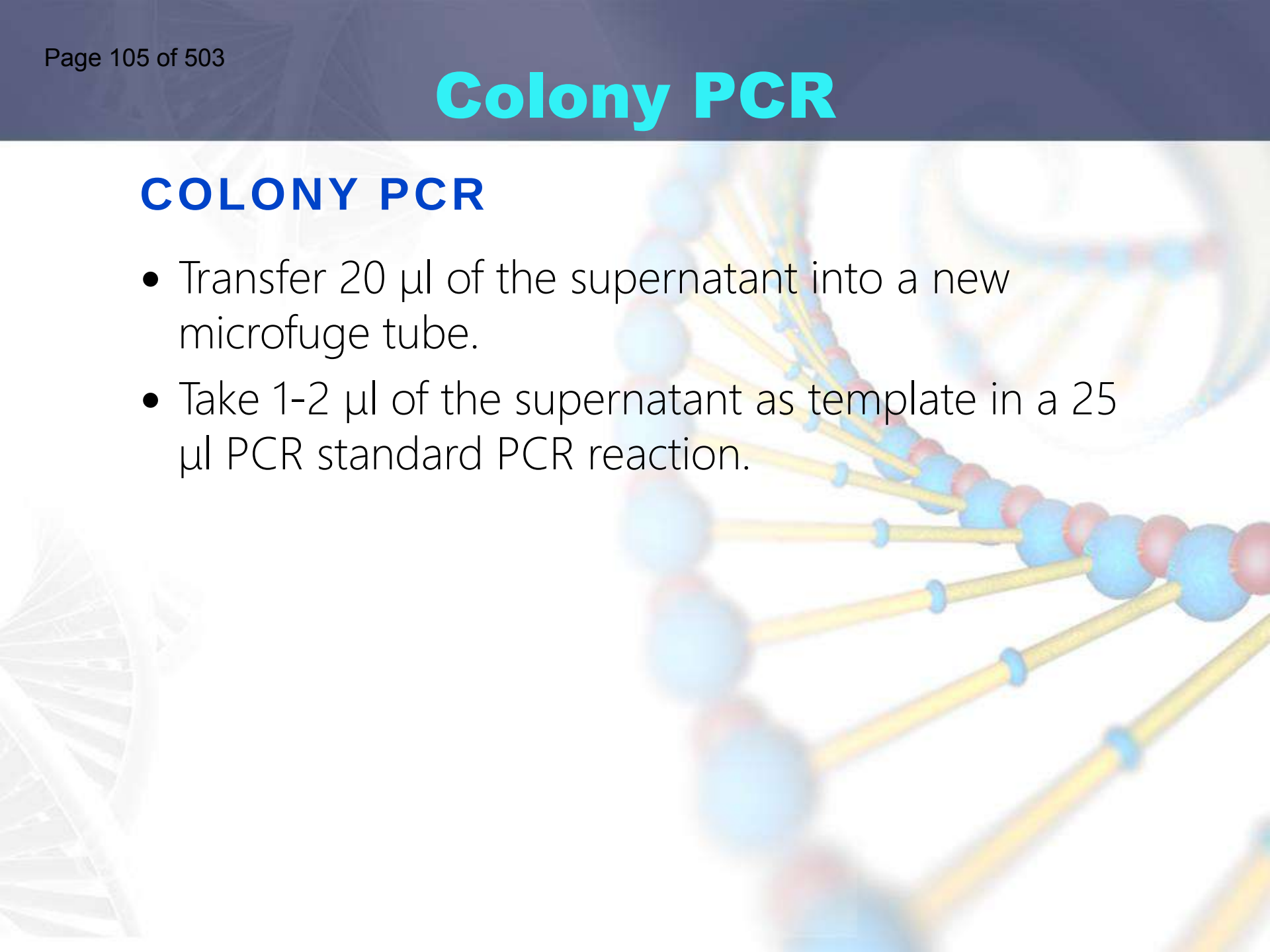
COLONY PCR METHODOLOGY

- The screening of bacterial or yeast clones for correct ligation or plasmid products.
- Pick a bacterial colony with an autoclaved toothpick, swirl it into 25 µl of TE autoclaved dH2O in an microfuge tube.
- Heat the mix in a boiling water bath (90-100^{0}C) for 2 minutes
- Spin sample for 2 minutes high speed in centrifuge.

Colony PCR

COLONY PCR

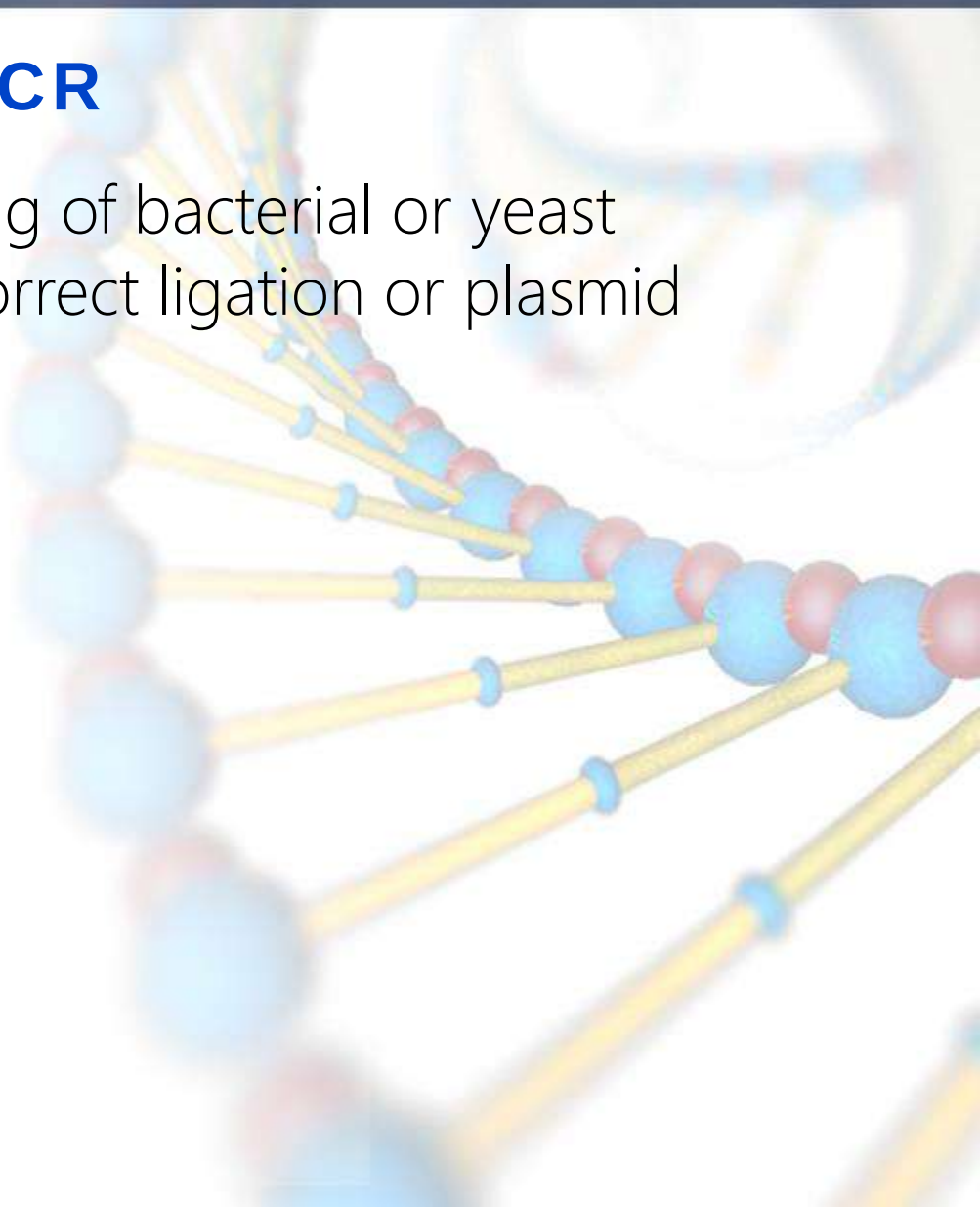
- Transfer 20 µl of the supernatant into a new microfuge tube.
- Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction.



Colony PCR

COLONY PCR

- The screening of bacterial or yeast clones for correct ligation or plasmid products.



Essentials of Genetics

MODULE#018

IN SITU PCR

In Situ PCR

IN SITU PCR

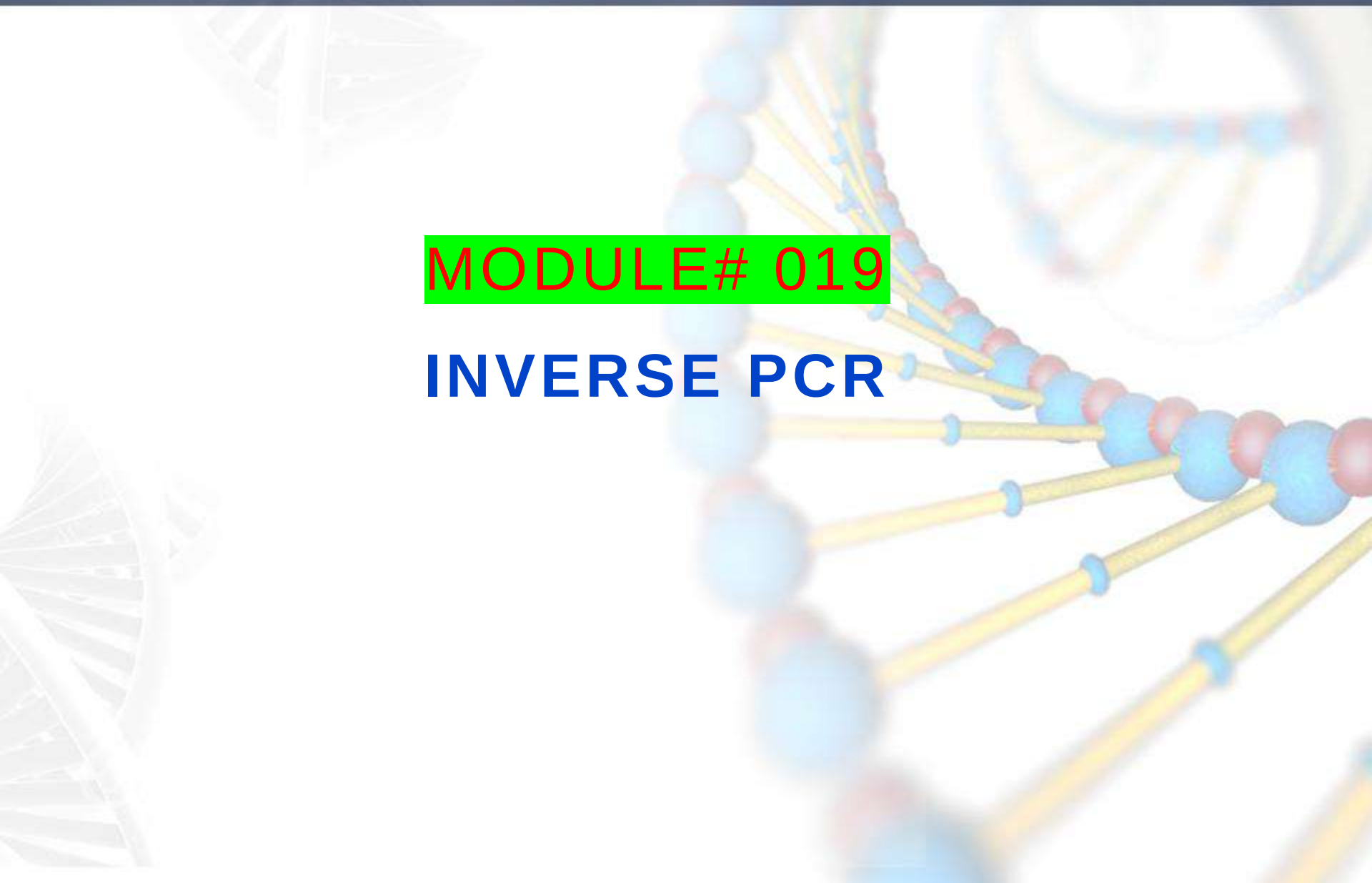
- In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide.
- In situ PCR amplification can be performed on fixed tissue or cells .

In Situ PCR

IN SITU PCR

- Applies the methodology of hybridization of the nucleic acids.
- Allows identification of cellular markers
- Limited to detection of non-genomic material such as RNA.

Essentials of Genetics



MODULE# 019

INVERSE PCR

Inverse PCR

INVERSE PCR

- Inverse PCR uses standard PCR primers oriented in the reverse direction of the usual orientation.

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- **BS ZOOLOGY 03336361244**

Inverse PCR

INVERSE PCR

- The template for the reverse primers is a restriction fragment that has been self-ligated
- Inverse PCR functions to clone sequences flanking a known sequence.
- Flanking DNA sequences are digested and then ligated to generate circular DNA.

Inverse PCR

IDENTIFICATION OF SEQUENCES FLANKING TRANSPOSABLE ELEMENTS

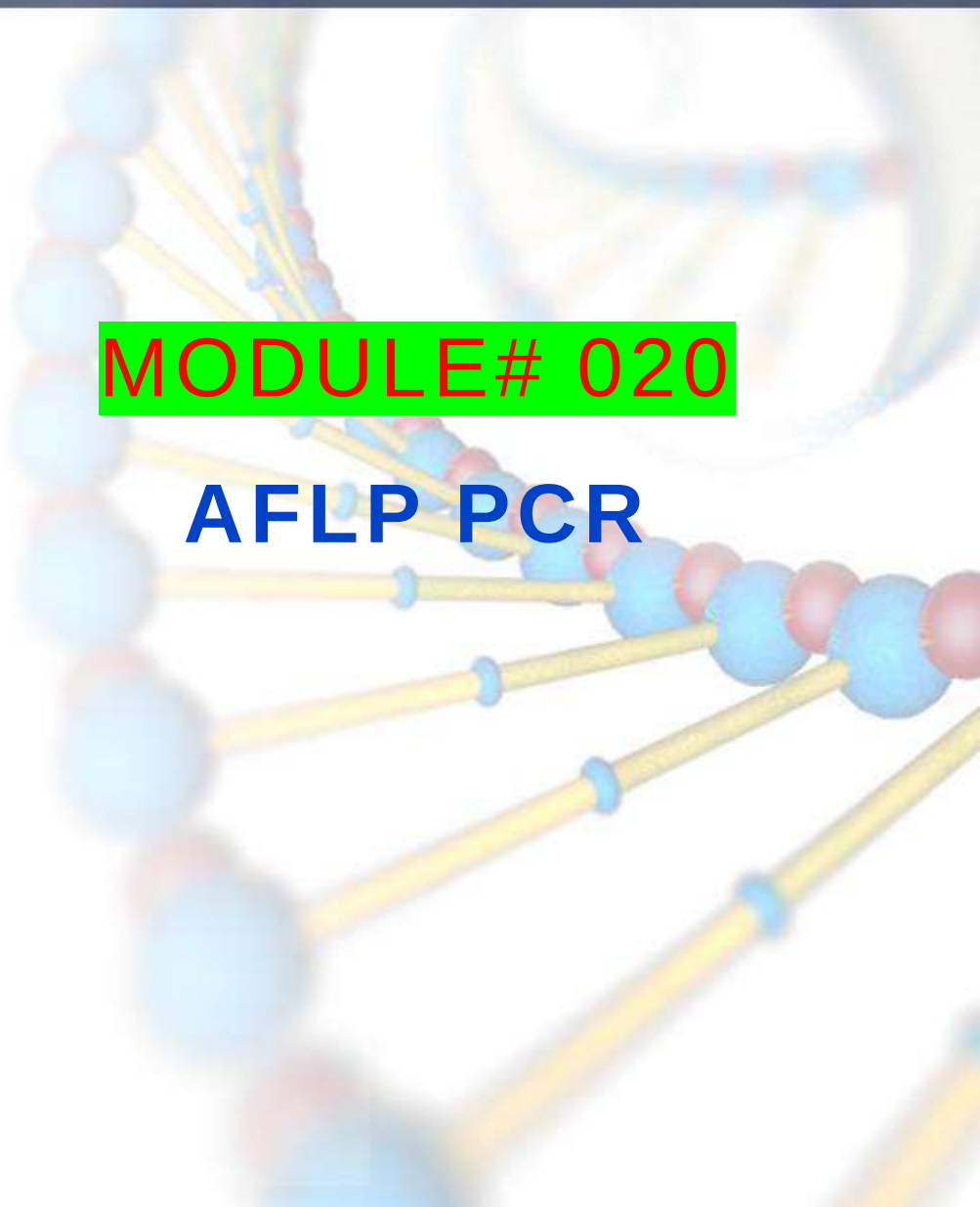
- Amplification and identification of sequences flanking transposable elements, and the identification of genomic inserts.

Inverse PCR

INVERSE PCR

- A method used to allow PCR when only one internal sequence is known.
- Especially useful in identifying flanking sequences to various genomic inserts.

Essentials of Genetics



MODULE# 020

AFLP PCR

AFLP PCR

AFLP PCR - METHODOLOGY

- Genomic DNA is digested with one or more restriction enzymes. tetracutter (MseI) and a hexacutter (EcoRI).
- Ligation of linkers to all restriction fragments. Pre-selective PCR is performed using primers which match the linkers and restriction site specific sequences.
- Electrophoretic separation and amplicons on a gel matrix, followed by visualization of the band pattern.

AFLP PCR

AFLP PCR

- AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA.
- AFLP can be also used for genotyping individuals for a large number of loci.

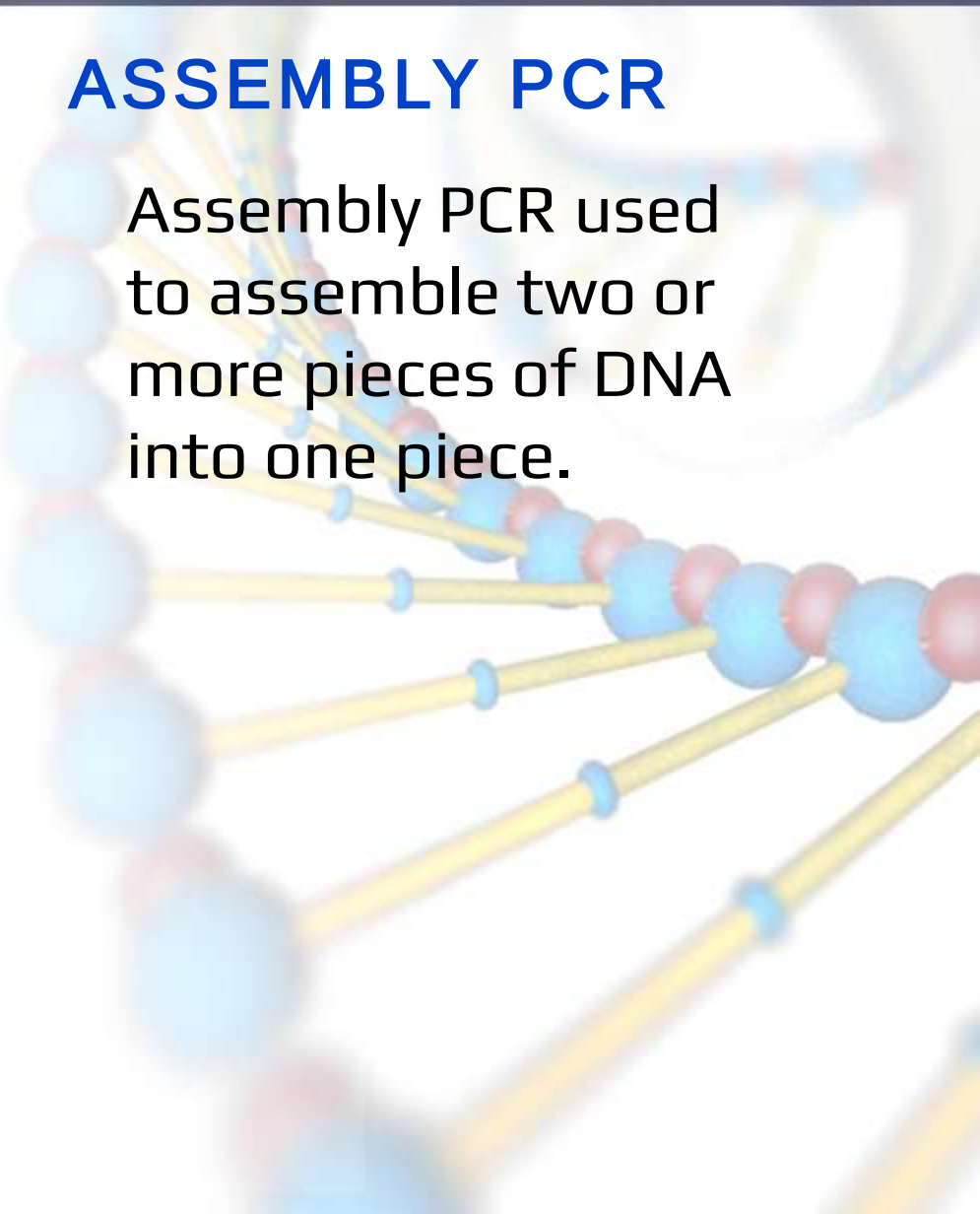
MODULE# 021

ASSEMBLY PCR

Assembly PCR

ASSEMBLY PCR

Assembly PCR used to assemble two or more pieces of DNA into one piece.



Assembly PCR

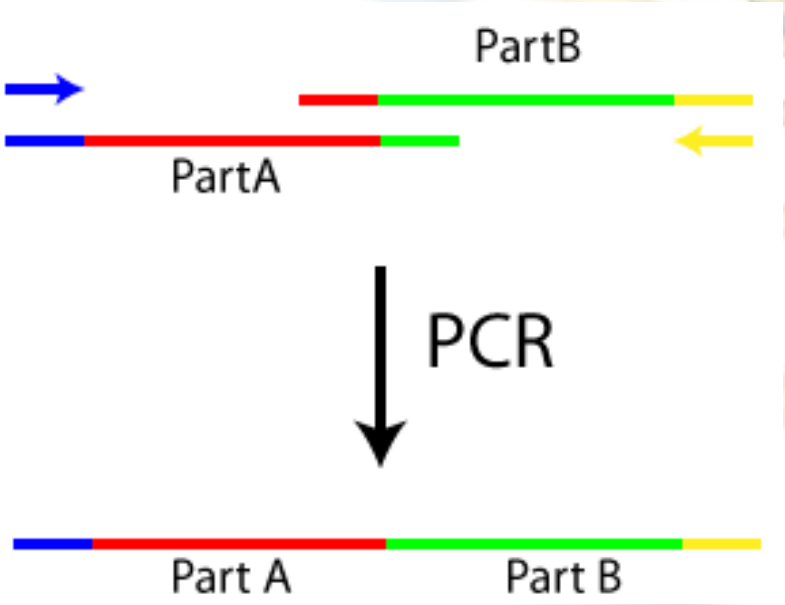
ASSEMBLY PCR - PRINCIPLE

Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

Assembly PCR

ASSEMBLY PCR - METHODOLOGY

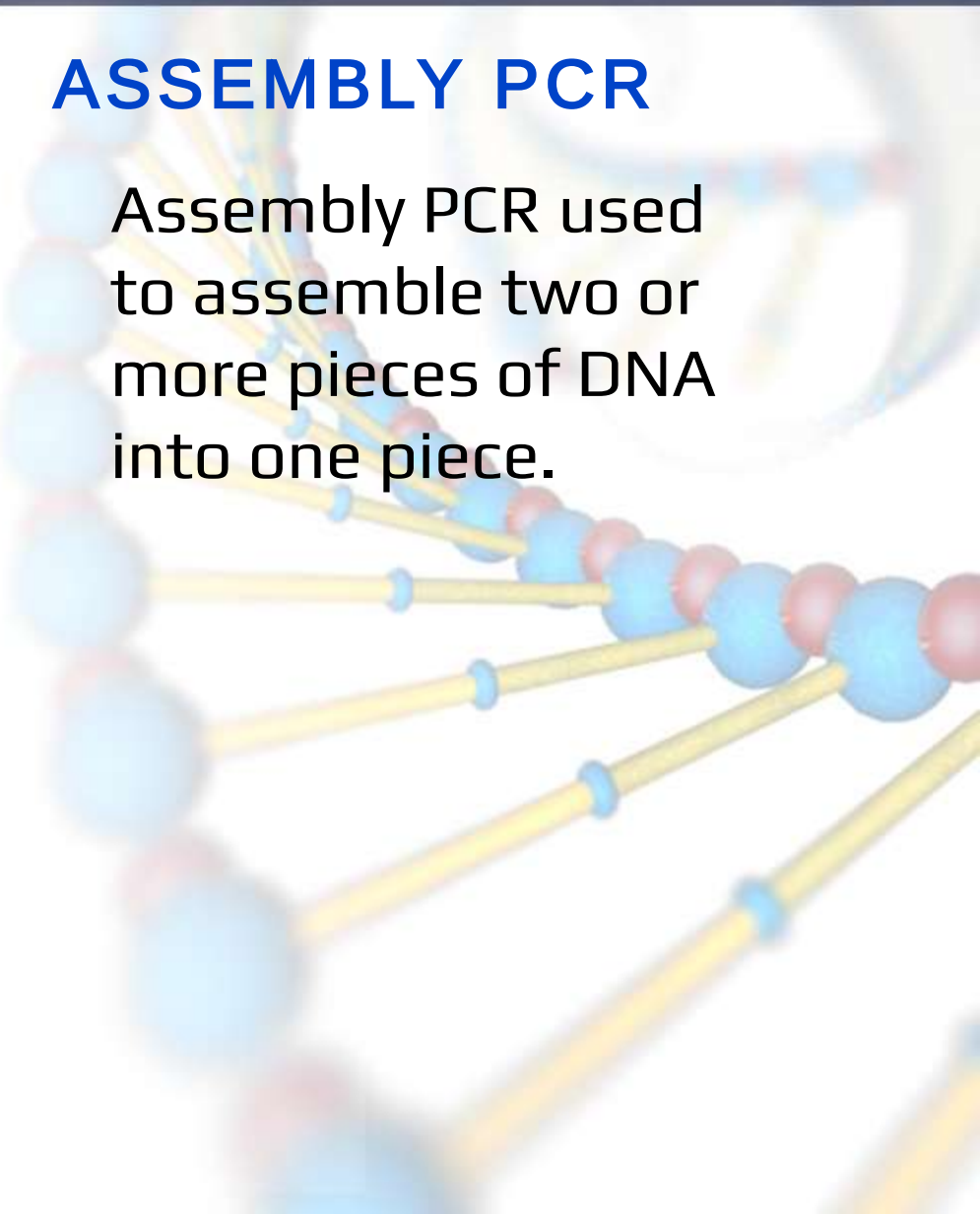
It involves an initial PCR with primers that have an overlap and a second PCR using the products as the template that generates the final full-length product.



Assembly PCR

ASSEMBLY PCR

Assembly PCR used to assemble two or more pieces of DNA into one piece.



Essentials of Genetics

MODULE# 22

SUICIDE PCR

Suicide PCR

SUICIDE PCR - PRINCIPLE

Suicide PCR is typically used in paleogenetics or other studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority.

The method prescribes the use of any primer combination only once in a PCR, which should never have been used in any positive control PCR reaction.

Suicide PCR

SUICIDE PCR

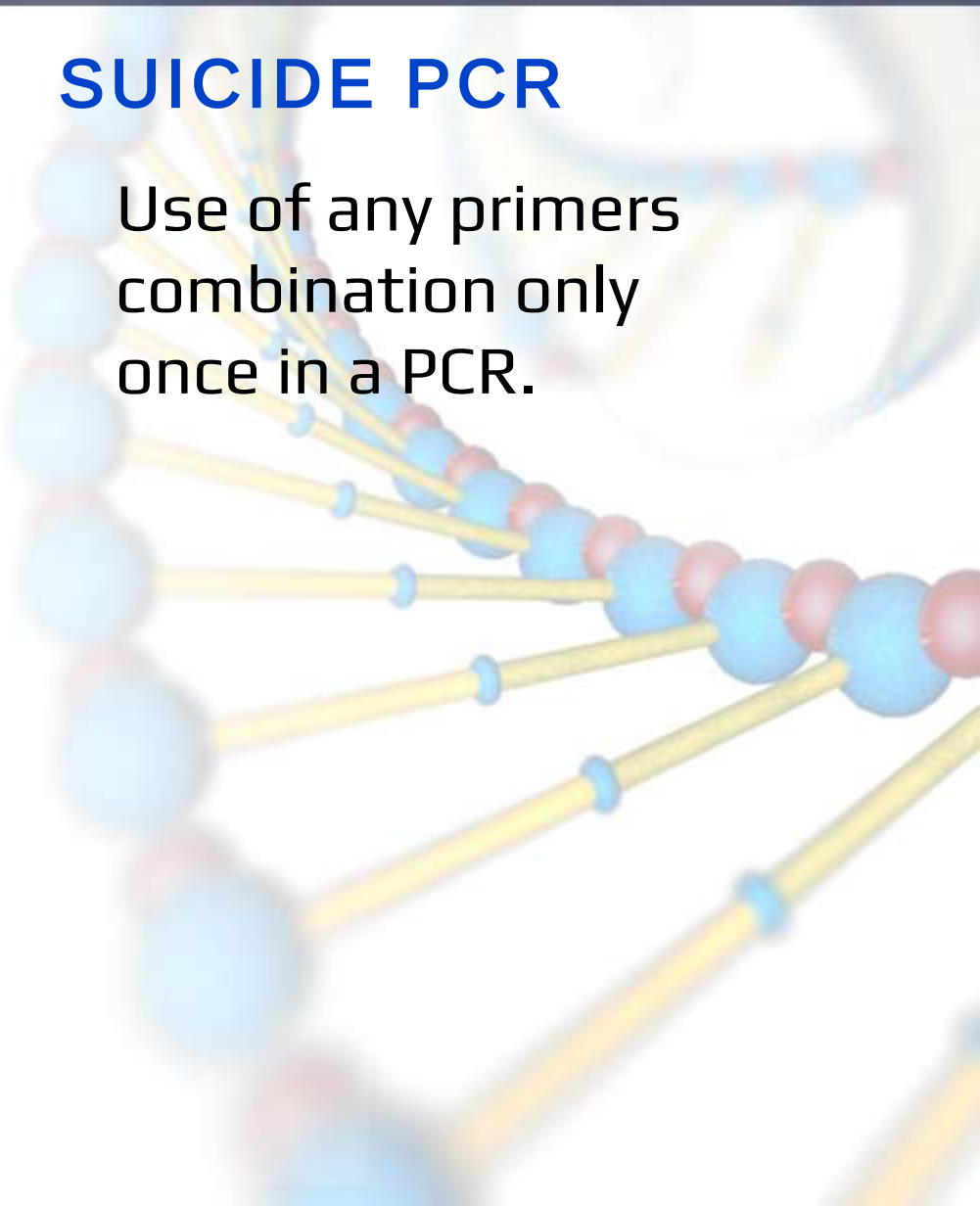
Primers should always target a genomic region never amplified before in the lab using this or any other set of primers.

This ensures that no contaminating DNA from previous PCR reactions is present in the lab, which could otherwise generate false positives.

Suicide PCR

SUICIDE PCR

Use of any primers combination only once in a PCR.



Essentials of Genetics

MODULE# 023

METHYLATION SPECIFIC PCR

Methylation Specific PCR

METHYLATION SPECIFIC PCR

Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA.

Methylation Specific PCR

METHYLATION SPECIFIC PCR

Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

Methylation Specific PCR

MSP

MSP used in quantitative PCR provides information about methylation state of a given CpG island.

Essentials of Genetics

MODULE# 24

INTER SEQUENCE SPECIFIC PCR

InterSequence Specific PCR

INTER SEQUENCE SPECIFIC PCR

A method for DNA fingerprinting that uses primers selected from segments repeated throughout a genome to produce a unique fingerprint of amplified product lengths.

The use of primers from a commonly repeated segment is called Alu-PCR, and can help to amplify sequences adjacent (or between) these repeats.

InterSequence Specific PCR

PCR

PCR used to produce a unique fingerprints of amplified product lengths.

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MODULE# 25

LIGATION MEDIATED PCR

Ligation-mediated PCR

METHODOLOGY

Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA.

PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

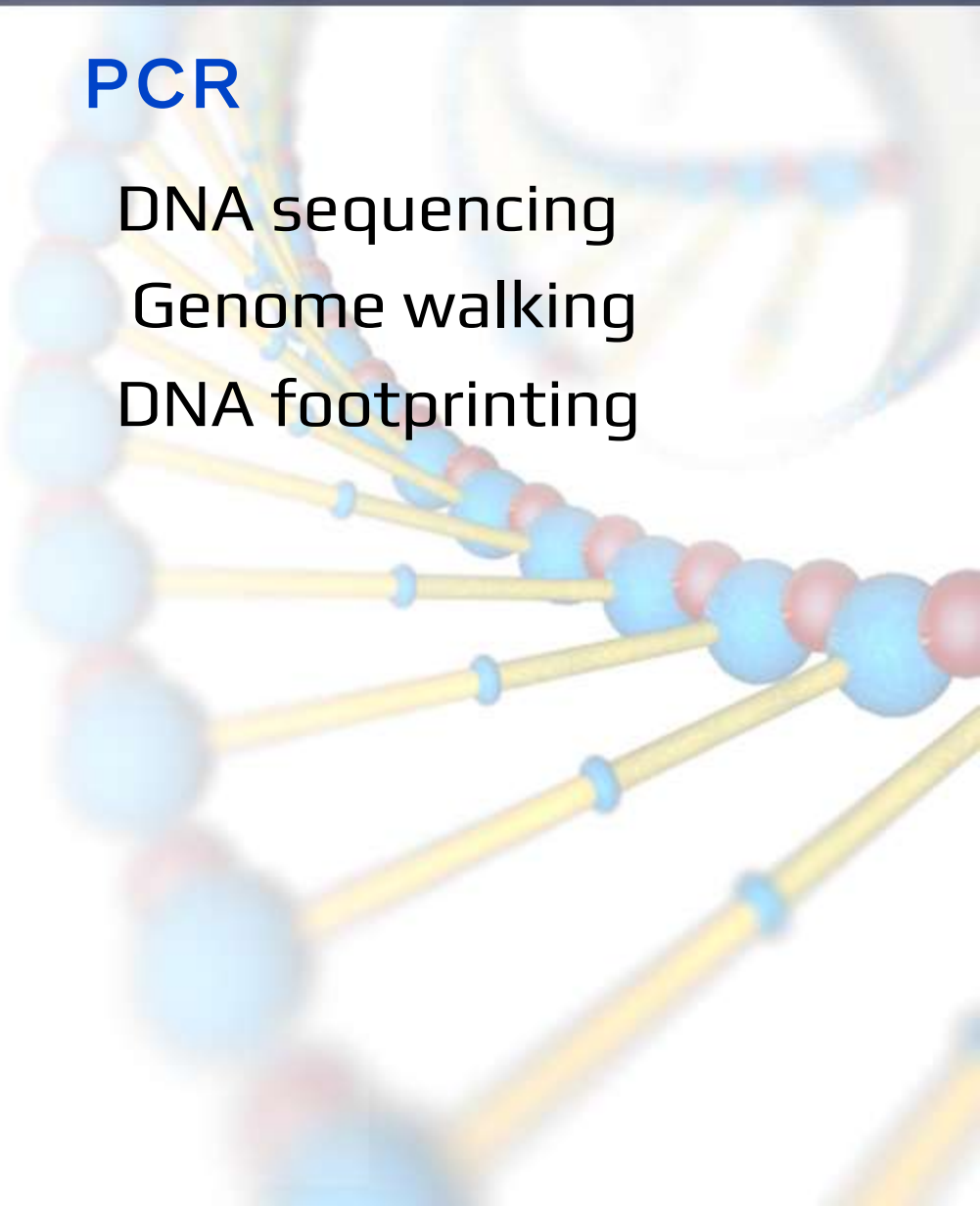
Ligation-mediated PCR

PCR

DNA sequencing

Genome walking

DNA footprinting



Essentials of Genetics

MODULE# 26

WHOLE GENOME AMPLIFICATION PCR

WHOLE GENOME AMPLIFICATION PCR

Primers can be designed to be 'degenerate'

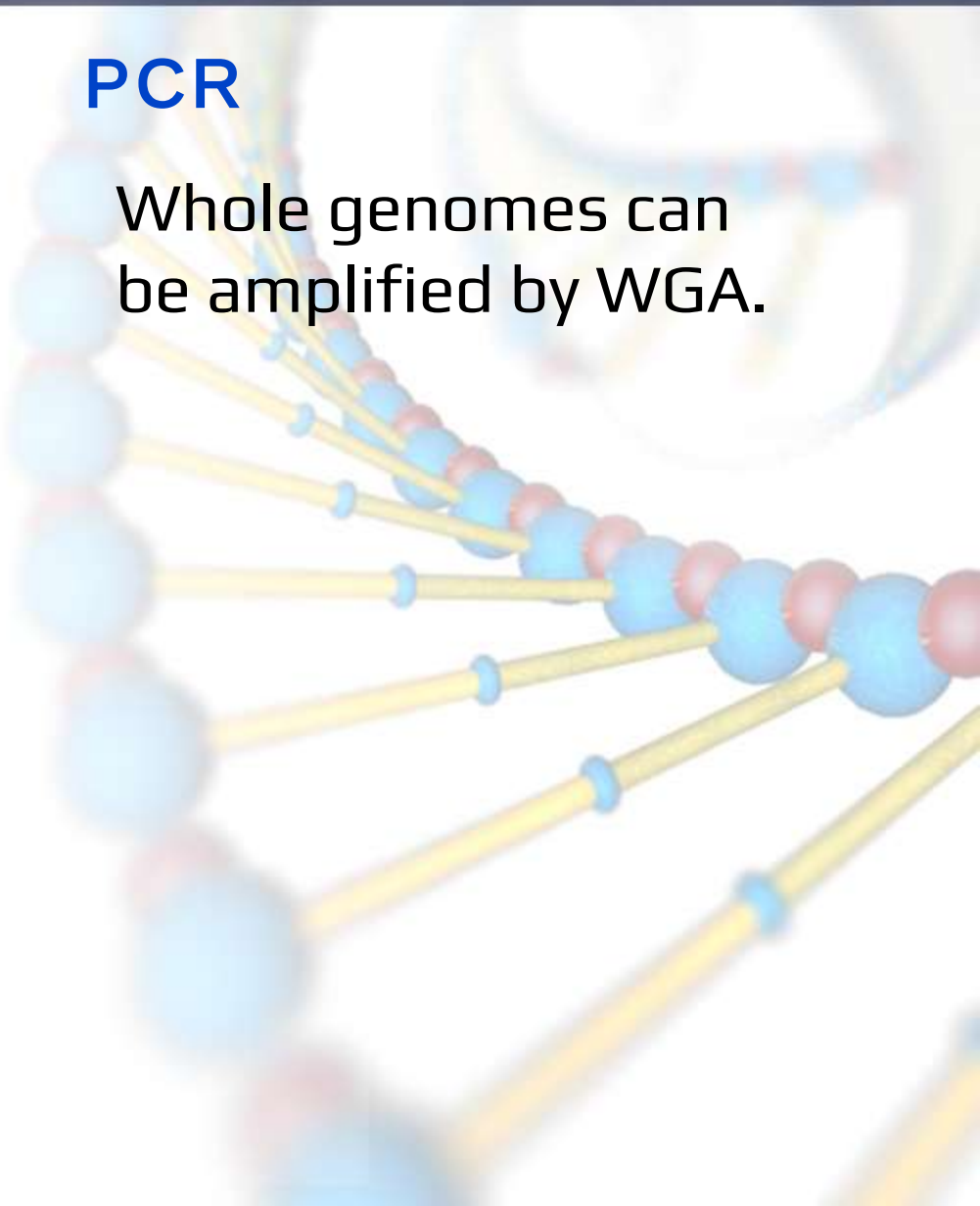
- able to initiate replication from a large number of target locations.

Whole genome amplification (WGA) is a group of procedures that allow amplification to occur at many locations in a genome.

Whole Genome Amplification PCR

PCR

Whole genomes can be amplified by WGA.



Essentials of Genetics

MODULE# 27

MINI-PRIMER PCR

Mini Primer PCR

MINI PRIMER PCR

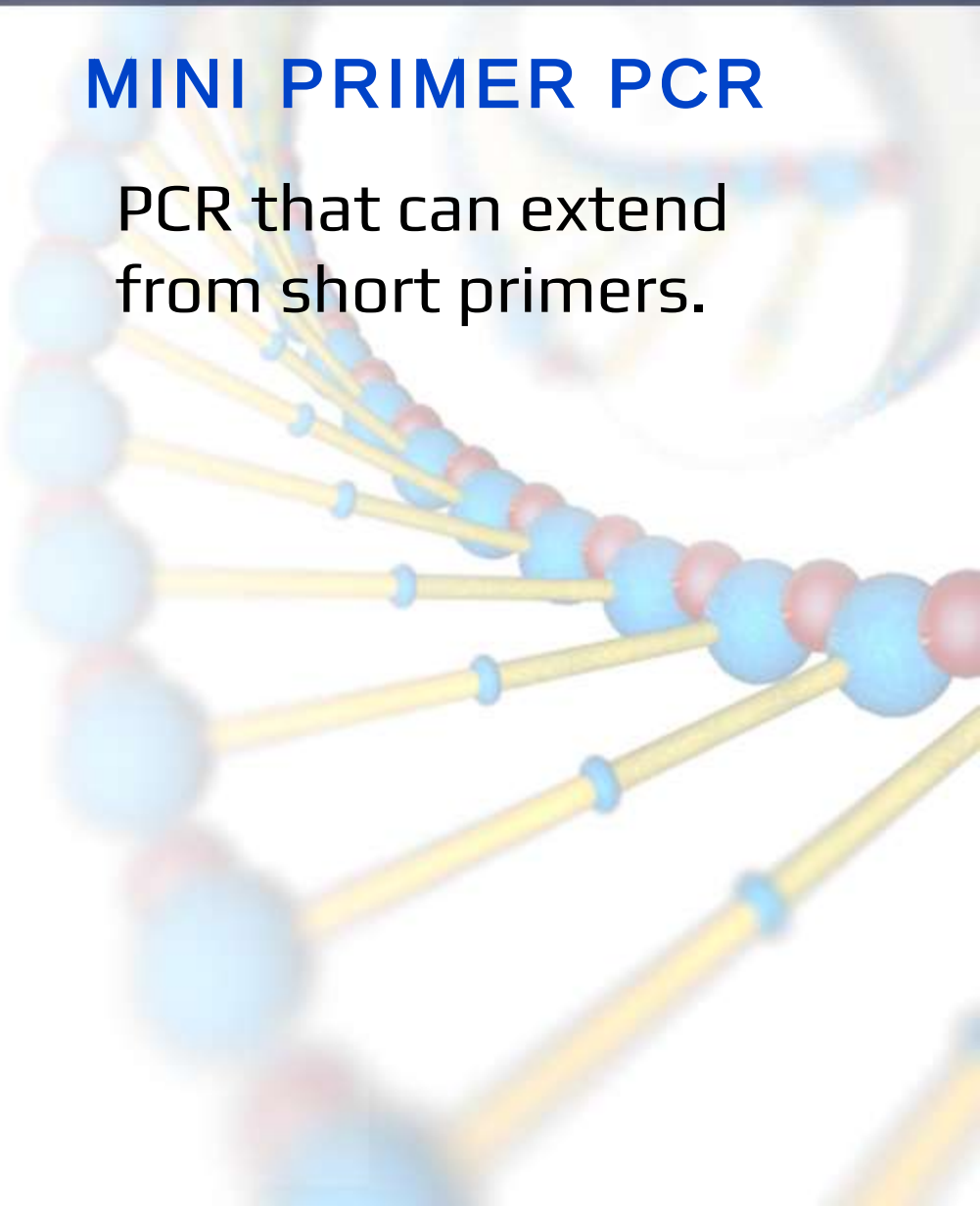
Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.

This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene.

Mini Primer PCR

MINI PRIMER PCR

PCR that can extend from short primers.



MODULE# 28

ADVANTAGES AND LIMITATIONS OF PCR

Advantages & Limitations

ADVANTAGES

- Specific
- Simple, rapid, relatively inexpensive
- Amplifies from low quantities
- Works on damaged DNA
- Sensitive
- Flexible

Advantages & Limitations

LIMITATIONS

- Contamination risk
- Primer complexities
- Primer-binding site complexities
- Amplifies rare species
- Detection methods

MODULE# 29

APPLICATIONS OF PCR

Applications of PCR

APPLICATIONS

- Detection of Infectious diseases: AIDS, TB, CMV, H1N1, etc Viral, Bacterial and fungal infections,
- Diagnosis of latent viruses.
- Forensic applications: DNA finger printing

Applications of PCR

APPLICATIONS

- Detection of Mutations: Inherited disorders & carriers
- Track DNA abnormalities
- Prenatal diagnosis of genetic disorders.

Applications of PCR

APPLICATIONS

- Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening
- Drug discovery
- Classification of organisms Genotyping

Applications of PCR

APPLICATIONS

- Molecular Archaeology
- Molecular Epidemiology
- Molecular Ecology
- Bioinformatics
- Genomic cloning
- Site-directed mutagenesis
- Gene expression studies

Applications of PCR

CLINICAL MICROBIOLY

- Identification of bacteria:
- Slow growing bacteria
- Bacterial antibiotic resistance genes.
- VIRUS: almost all viruses identification and viral load.

Applications of PCR

CLINICAL MICROBIOLY

FUNGUS:

Aspergillus spp

Candida spp

Plasmodium spp

Trypanosoma spp

Leishmania spp

Babesia spp

Polymerase chain reaction (PCR)

Module# 030

Applications of PCR

§ PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

Polymerase chain reaction (PCR)

PCR-Gene cloning and expression

§ PCR has been used in gene cloning and screening of genomic libraries

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Polymerase chain reaction (PCR)

Module# 31

PCR-Medicine

§ PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis

§ Molecular tools have also allowed to perform prenatal genetic diagnosis



Polymerase chain reaction (PCR)

Module# 32

PCR-Forensic sciences

§ Forensic science is the application of scientific procedures to solve criminal and legal matters

§ Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

Polymerase chain reaction (PCR)

PCR-DNA profiling

§ DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA





Polymerase chain reaction (PCR)

Module# 33

PCR-Agricultural sciences and environment

§ PCR has also facilitated research in detection of pathogens in plants, animals and environment

Polymerase chain reaction (PCR)

Module# 34

PCR-Molecular paleontology

§ Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains

Cutting DNA molecules

Module# 35

Types of restriction and modification (R-M) system

- At least four R-M systems are known
- Type I
- Type II
- Type III
- Type IIs

Cutting DNA molecules

Type I

§ Type I systems were the first to be characterized from *E. coli* K12

§ The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit

§ Type I systems are of little value for gene manipulation

Cutting DNA molecules

Type II

§ Most of the useful R-M system is Type II

§ Type II enzymes recognize a defined sequence and cut within it

Cutting DNA molecules

Type III

§ Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value



Cutting DNA molecules

Type IIs

§ Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

Cutting DNA molecules

Module# 36

Nomenclature

§ A suitable system was proposed by Smith and Nathans (1973)

§ The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet

§ *E. coli* = Eco

§ *H. influenzae* = Hin

Cutting DNA molecules

Nomenclature

§ Strain identification is written as *Eco*K

§ In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of *H. influenzae*

§ *HindI, HindII, HindIII* etc

Cutting DNA molecules

Nomenclature

§ All restriction enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of *H. influenzae*

§ R. *HindIII* or M. *HindIII*

Examples of restriction endonuclease nomenclature

Enzyme	Enzyme source	Recognition sequence
SmaI	*Serratia marcescens*, 1st enzyme	CCCGGG
HaeIII	*Haemophilus aegyptius*, 3rd enzyme	GGCC
HindII	*H. influenzae*, strain d, 2nd enzyme	GTPyPuAC
HindIII	*H. influenzae*, strain d, 3rd enzyme	AAGCTT
HamHI	*Bacillus amyloliquefaciens*, strain H, 1st enzyme	GGATCC

Cutting DNA molecules

Module# 37

Target sites

§ Type II endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of rotational symmetry i.e. referred as *palindromes*

5'-GAATTC-3'

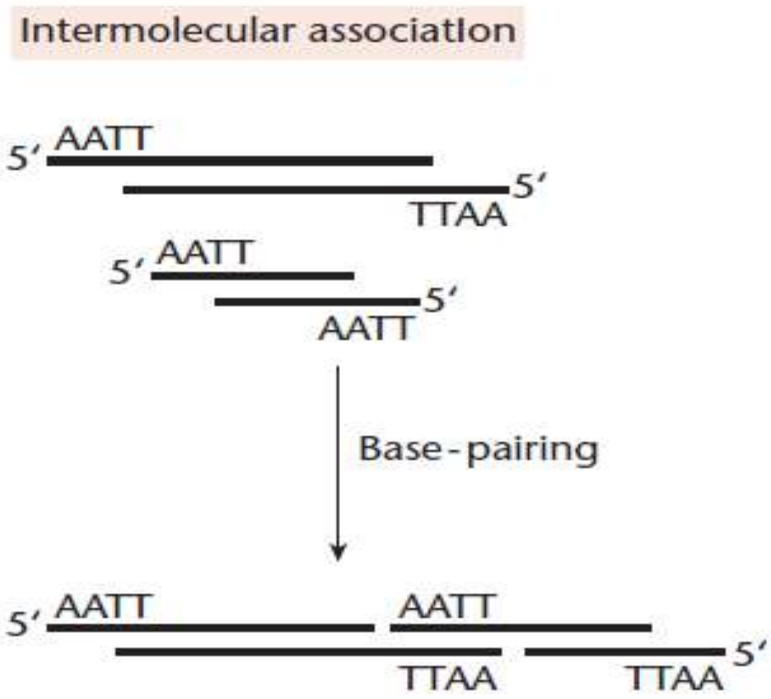
5'-CTTAAG-3'

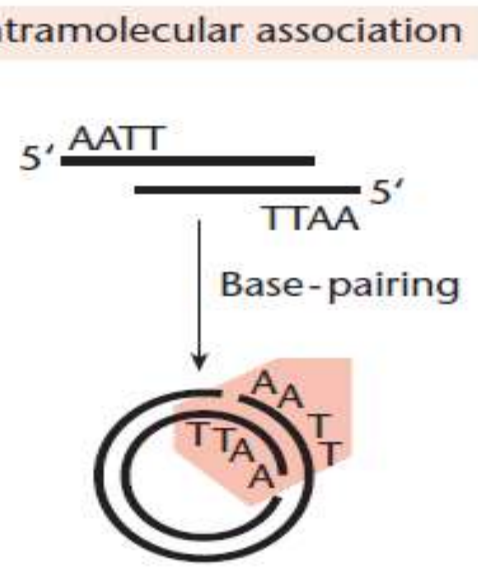
5’-G A A T T C-3’
3’-C T T A A G-5’

5’-G/ A A* T T C-3’
3’-C T T A* A/ G-5’

5’-G 5’- A A T T C-3’
3’-C T T A A-5’ G-5’

Single stranded breaks by *Eco*R1





Cohesive fragments of DNA produced by digestion with EcoR1

Cutting DNA molecules

Module# 38

Number and size of restriction fragments

§ The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut

Cutting DNA molecules

Number and size of restriction fragments

- Four base recognition site occurs every 4^4 (256) bp
- Six base recognition site occurs every 4^6 (4096) bp
- Eight base recognition site occurs 4^8 (65,536) bp

Average fragment size (bp) produced by different enzymes

Enzyme	Target	Arabidopsis	E. Coli	Human
*Apa*I	GGGCCC	25000	15000	2000
*Bam*HI	GGATCC	6000	5000	5000
*Spe*I	ACTAGT	8000	60000	10000

Cutting DNA molecules

Module# 39

Summary of restriction endonucleases

§ Type II restriction enzymes are heavily responsible for the current explosion in the field of gene manipulation in that they are essential in forming recombinant DNA molecules

Joining DNA molecules

Module# 40

DNA modifying enzymes

- Nucleases
- DNA Polymerase
- Reverse transcriptases
- DNA ligases

Joining DNA molecules

Nucleases

§ Nucleases or DNases are the enzymes that degrade DNA

§ Two broad classes of nucleases

i). Exonucleases

ii). Endonucleases

Joining DNA molecules

DNA Polymerase

§ An enzyme that catalyzes template-dependent synthesis of DNA



Joining DNA molecules

Reverse transcriptases

§ An RNA-directed DNA polymerase in retroviruses; capable of making DNA complementary to an RNA



Joining DNA molecules

DNA ligase

§ An enzyme that creates a phosphodiester bond between the 3’ end of one DNA segment and the 5’ of another





Joining DNA molecules

Module# 41

Methods of joining DNA fragments

§ Mainly three methods are used for joining DNA in vitro

i). Joining covalently annealed cohesive ends by DNA ligase

ii). Joining blunt-ended fragments by DNA ligase from phage T4 infected *E. coli*

Joining DNA molecules

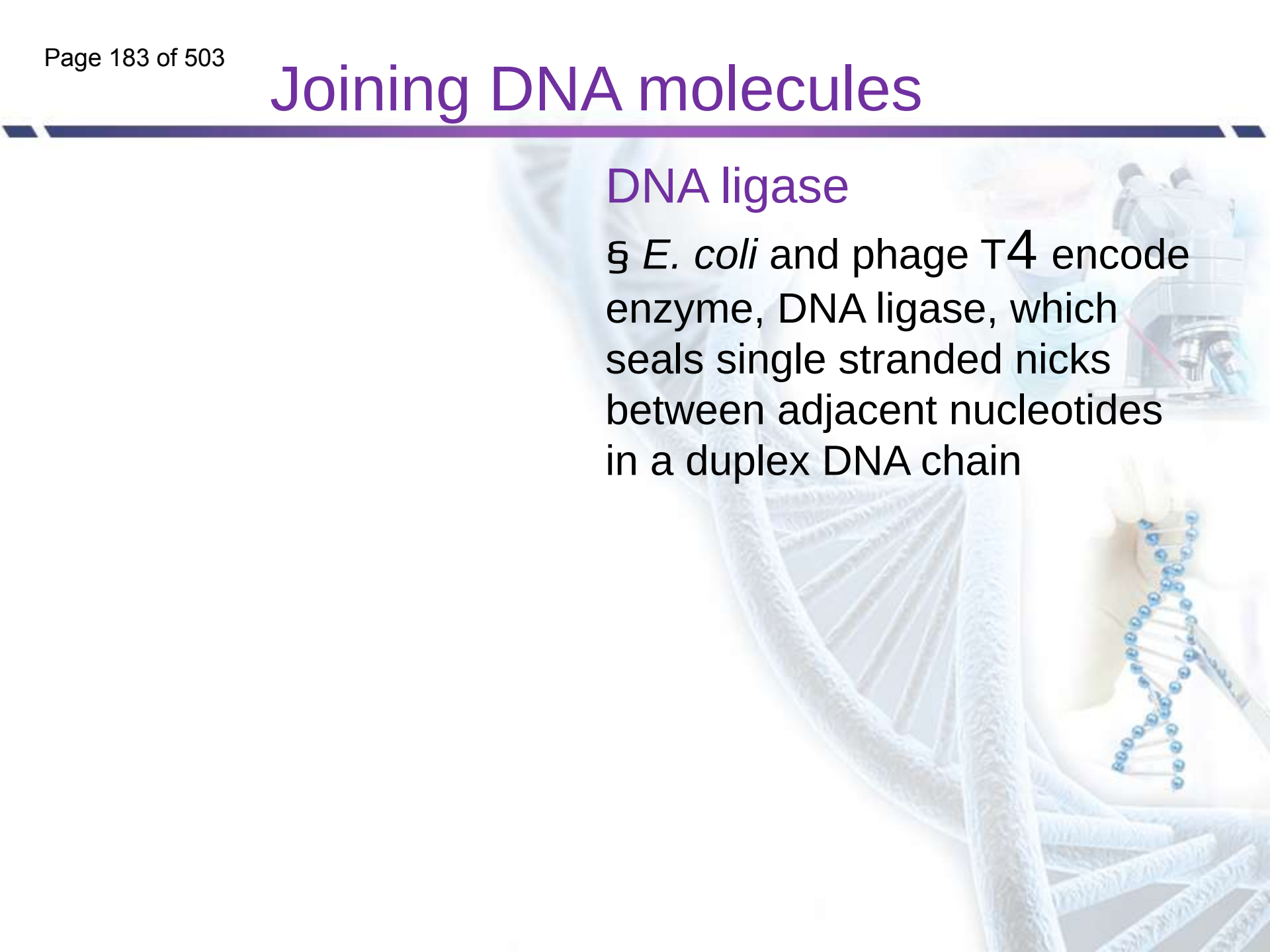
Methods of joining DNA fragments

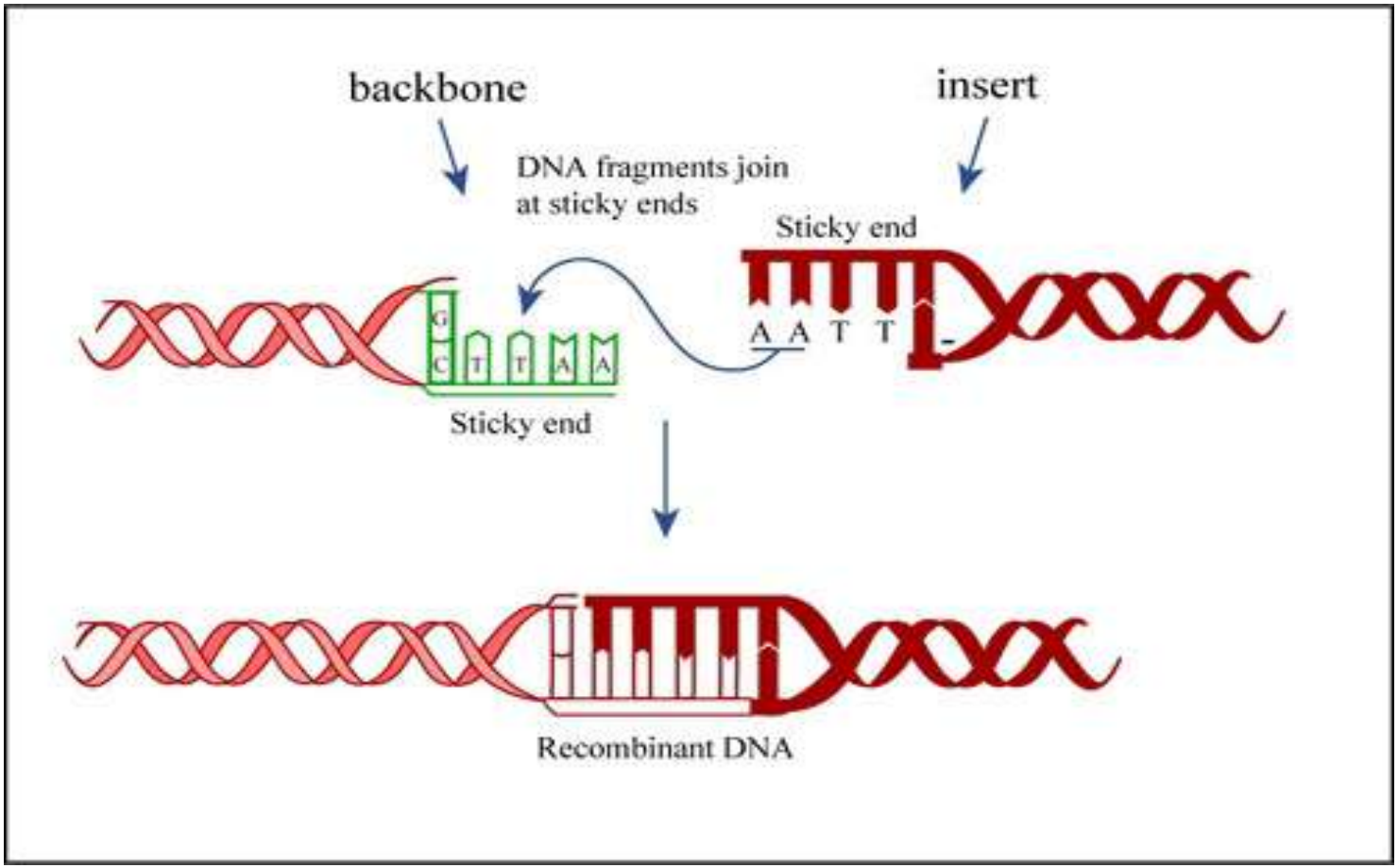
iii). The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric 3' single stranded tails at the ends of fragments

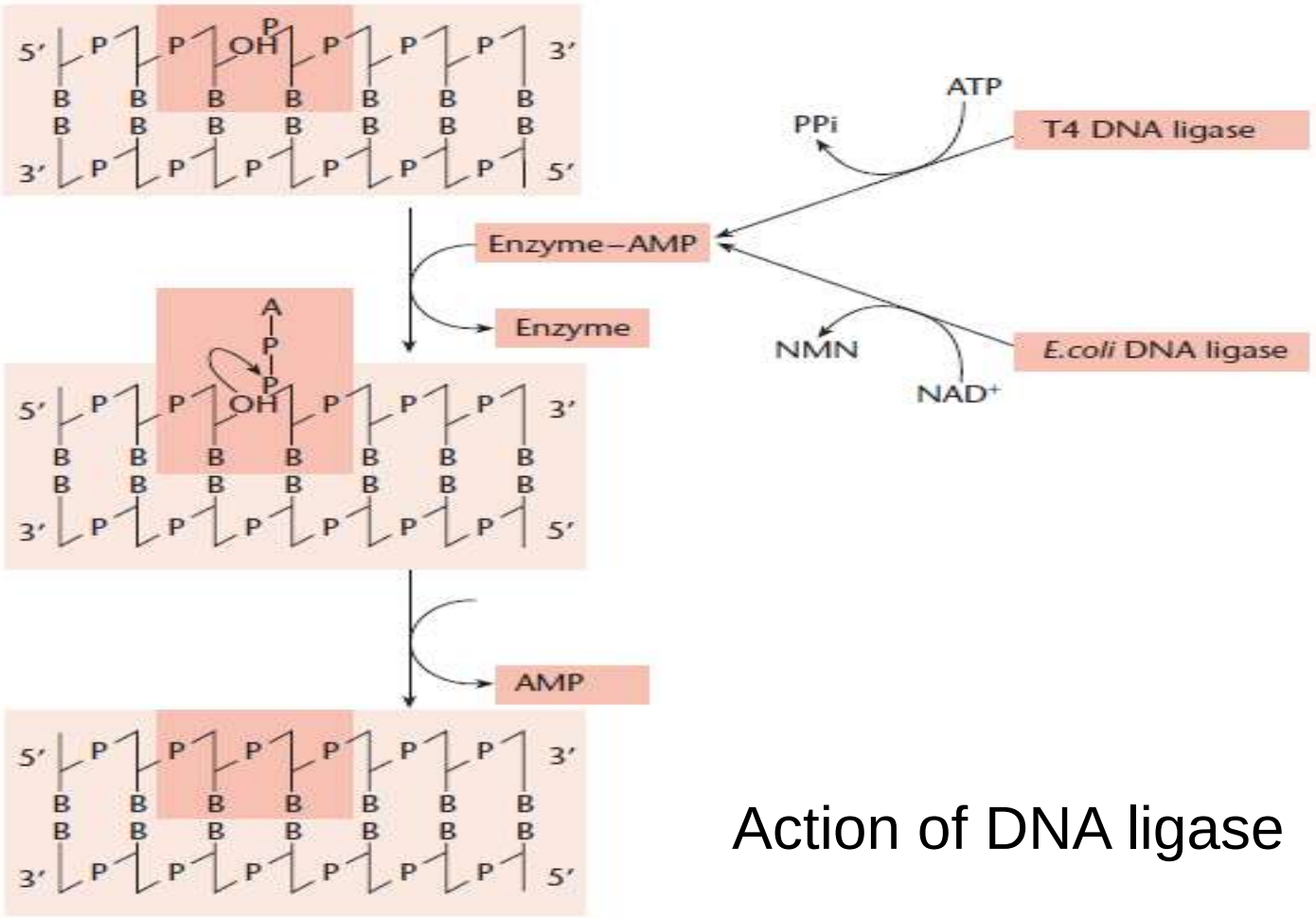
Joining DNA molecules

DNA ligase

§ *E. coli* and phage T4 encode enzyme, DNA ligase, which seals single stranded nicks between adjacent nucleotides in a duplex DNA chain




backbone
insert
DNA fragments join
at sticky ends
Sticky end
G
C T T A A
Sticky end
A A T T
Recombinant DNA



Action of DNA ligase

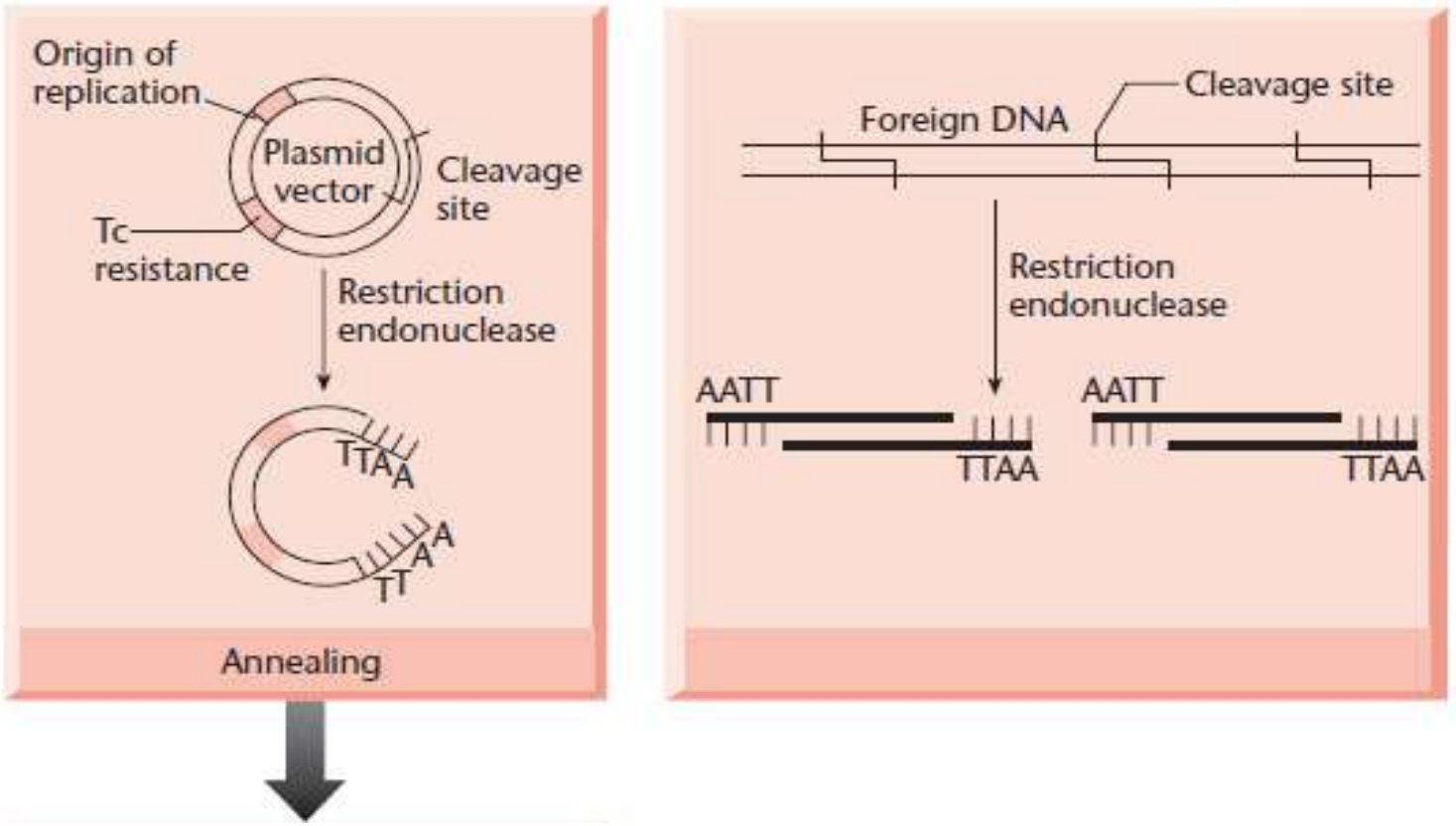
Joining DNA molecules

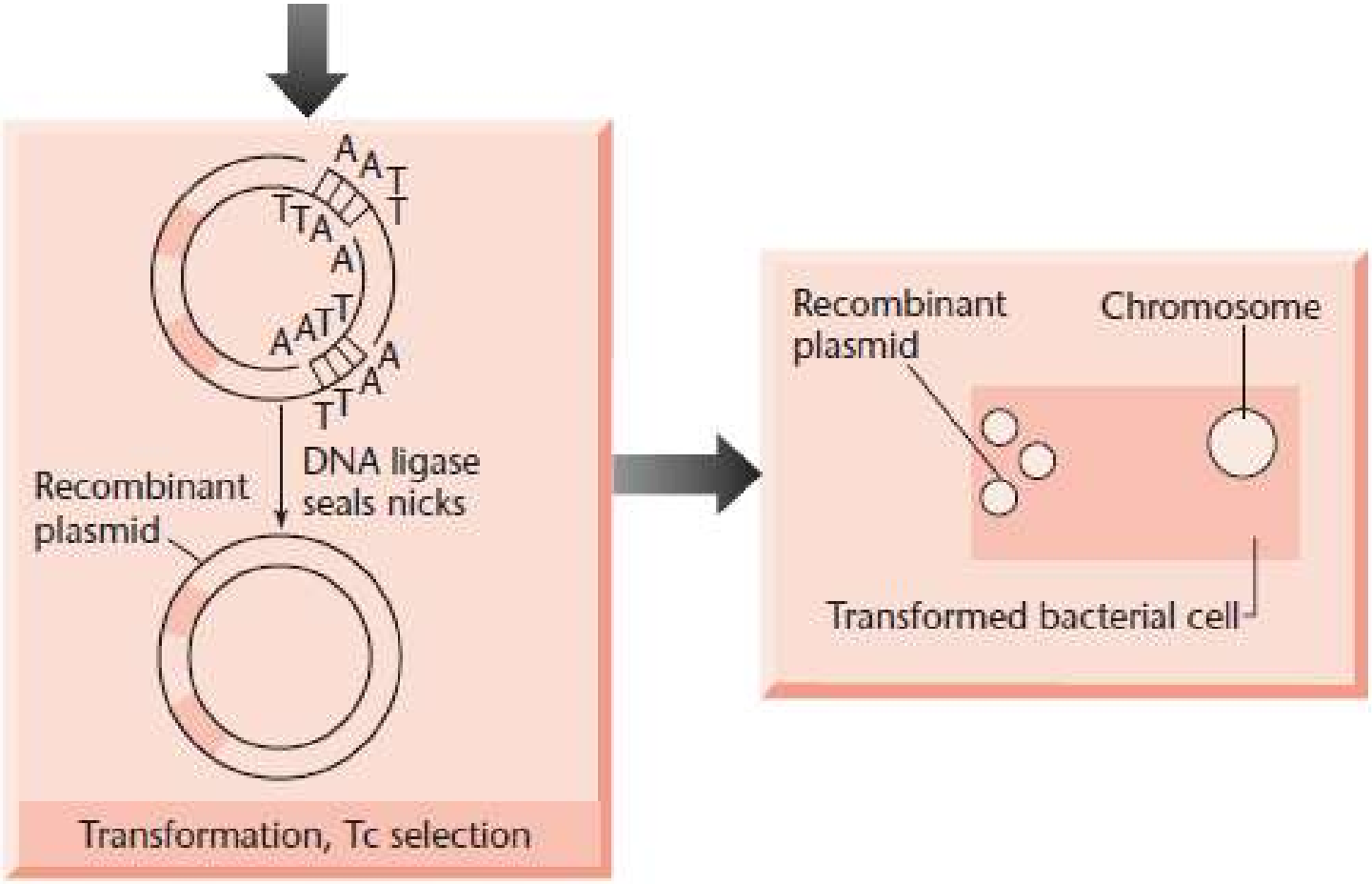
Module# 42

DNA ligase to create covalent recombinant DNA

§ DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands

Use of DNA ligase to create a covalent DNA recombinant




DNA ligase
seals nicks
Recombinant
plasmid
Transformation, Tc selection
Recombinant
plasmid
Chromosome
Transformed bacterial cell

Joining DNA molecules

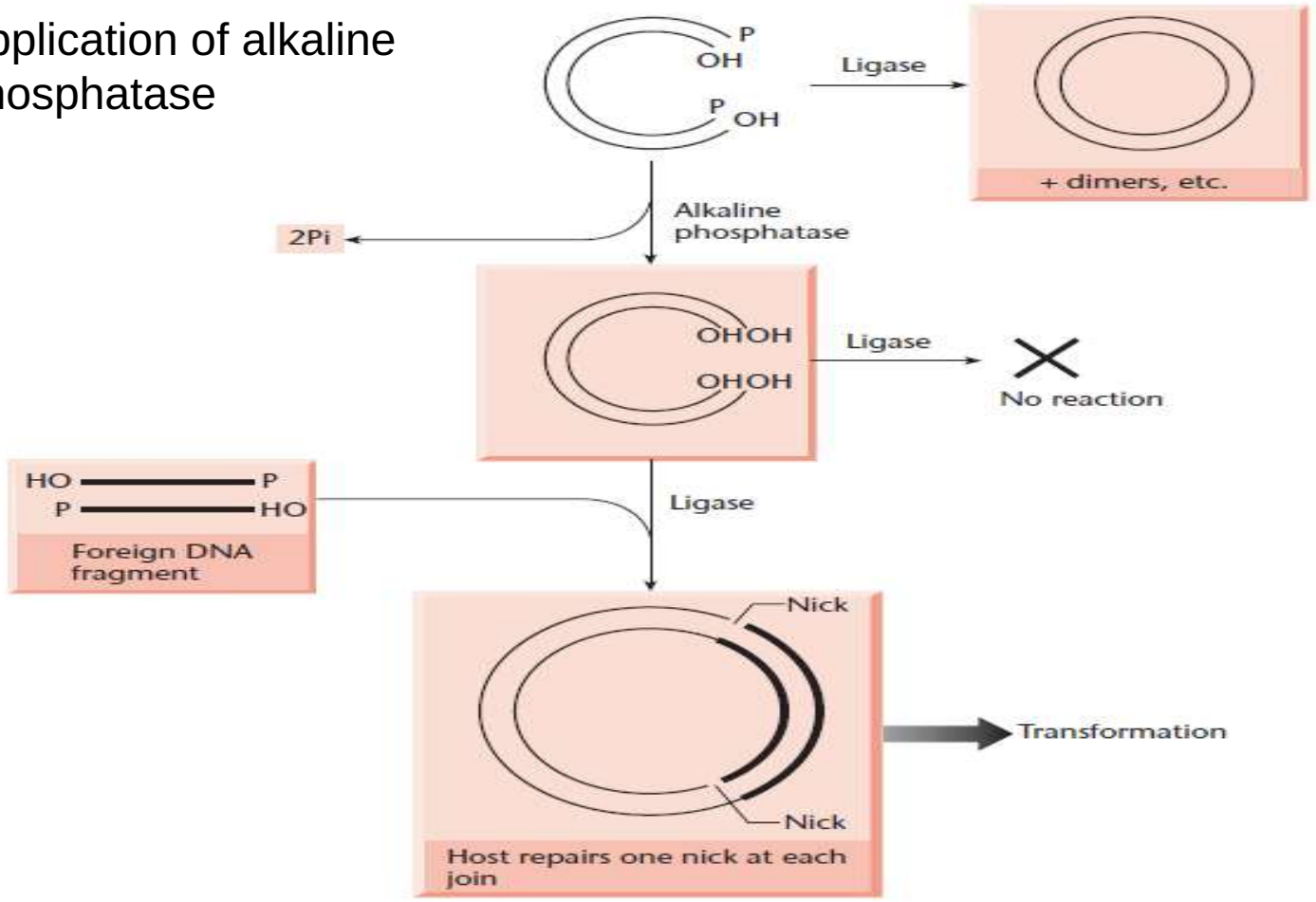
Module# 43

Alkaline Phosphatase

§ An enzyme responsible for removing phosphate groups from many types of molecules including DNA

§ Treatment of linearized plasmid vector DNA with alkaline phosphatase to remove 5'-terminal phosphate groups will prevent recircularization and plasmid dimer formation

Application of alkaline phosphatase



Joining DNA molecules

Module# 44

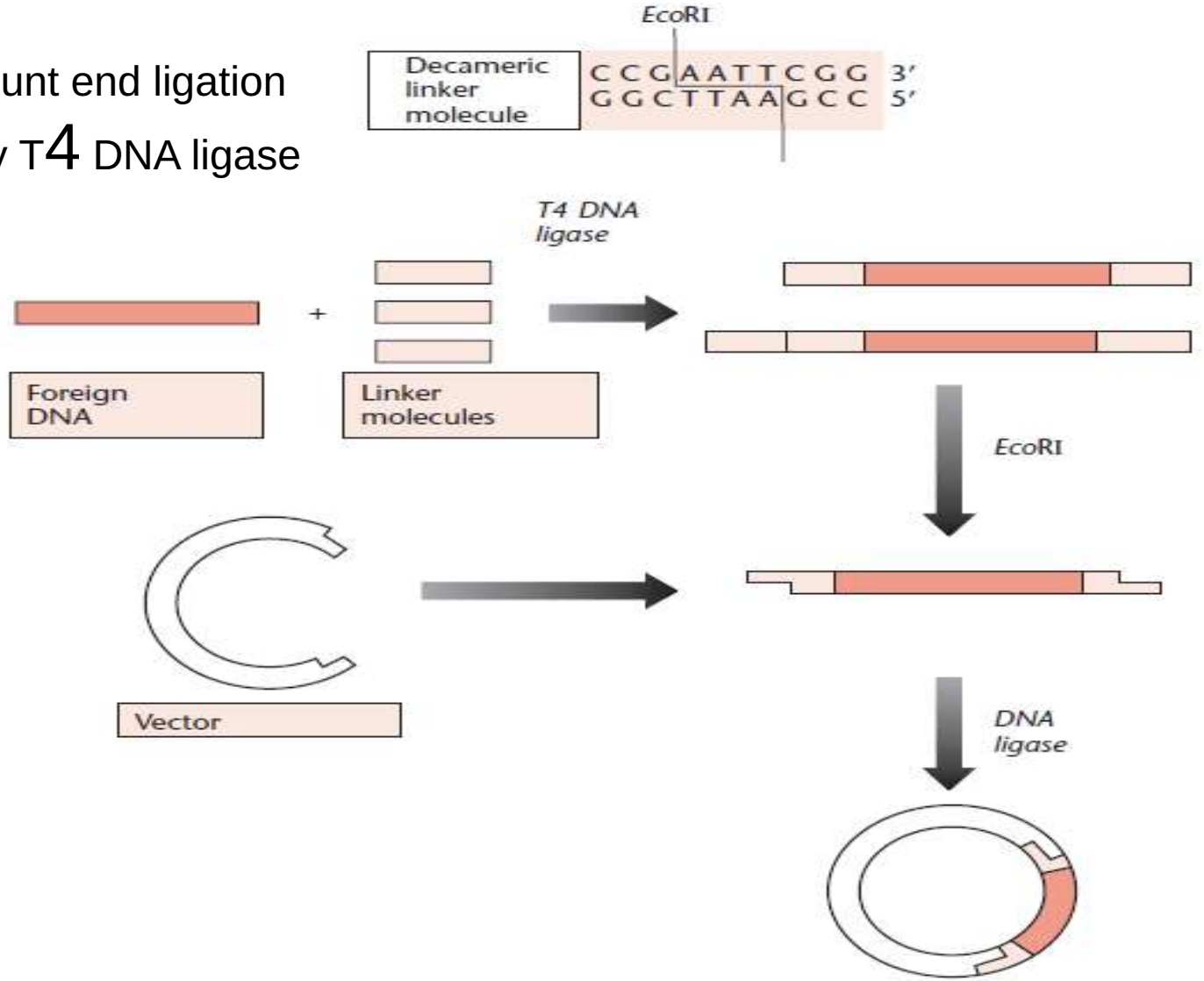
Blunt end ligation via linker molecules

§ Joining DNA fragments with cohesive ends by DNA ligase is a relatively efficient process

§ T4 DNA ligase has been used to joint blunt-ended DNA molecules

Blunt end ligation

by T4 DNA ligase

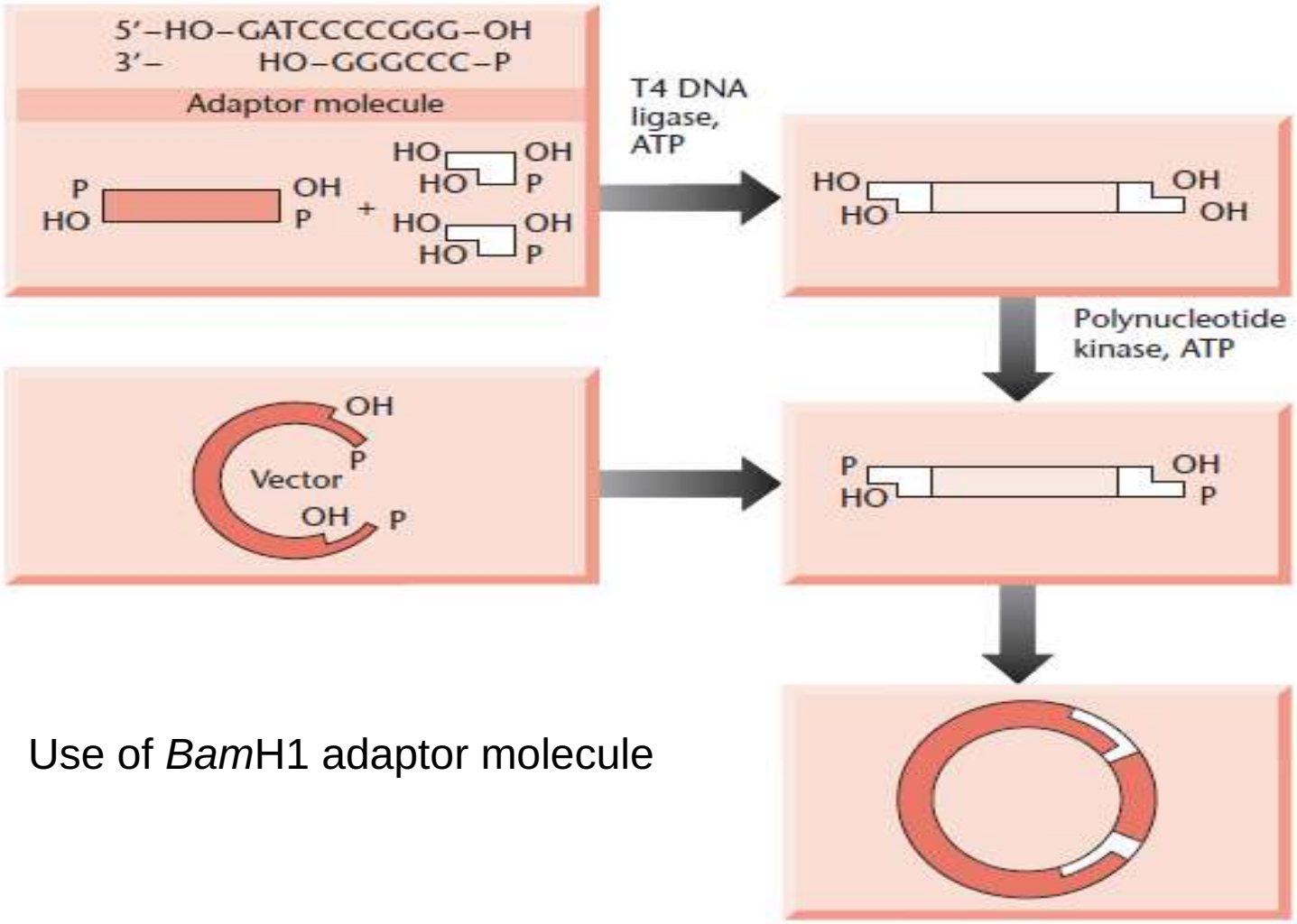


Joining DNA molecules

Module# 45

Adaptors

§ Chemically synthesized adaptor molecules which have a preformed cohesive end can be used to insert blunt ended foreign DNA into vector



Use of *Bam*H1 adaptor molecule

Joining DNA molecules

Adaptors

§ The main difference between linker and adaptor is that former having blunt ends while later have one cohesive end.

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§*BS ZOOLOGY 03336361244*

Essentials of Genetics

MODULE# 46

SOUTHERN BLOTTING

Southern Blotting

SOUTHERN BLOTTING

The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

Southern Blotting

SOUTHERN BLOTTING

Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975).

Southern Blotting

OTHER BLOTTING METHODS

Other blotting methods with similar principles, but using protein or RNA, have been named in reference to Edwin Southern's name.

Southern Blotting

BACKGROUND



Southern Blotting

SOUTHERN BLOTTING

Identify DNA sequence (gene) of interest.

Identified DNA may be a small piece of DNA or a mutation.

MODULE# 47

SOUTHERN BLOTTING PROCEDURE

Southern Blotting - Procedure

PROCEDURE

DNA is extracted from cells, leukocytes.

DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc)

The resulting fragments are separated on the basis of size by electrophoresis.

The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.

Southern Blotting - Procedure

PROCEDURE

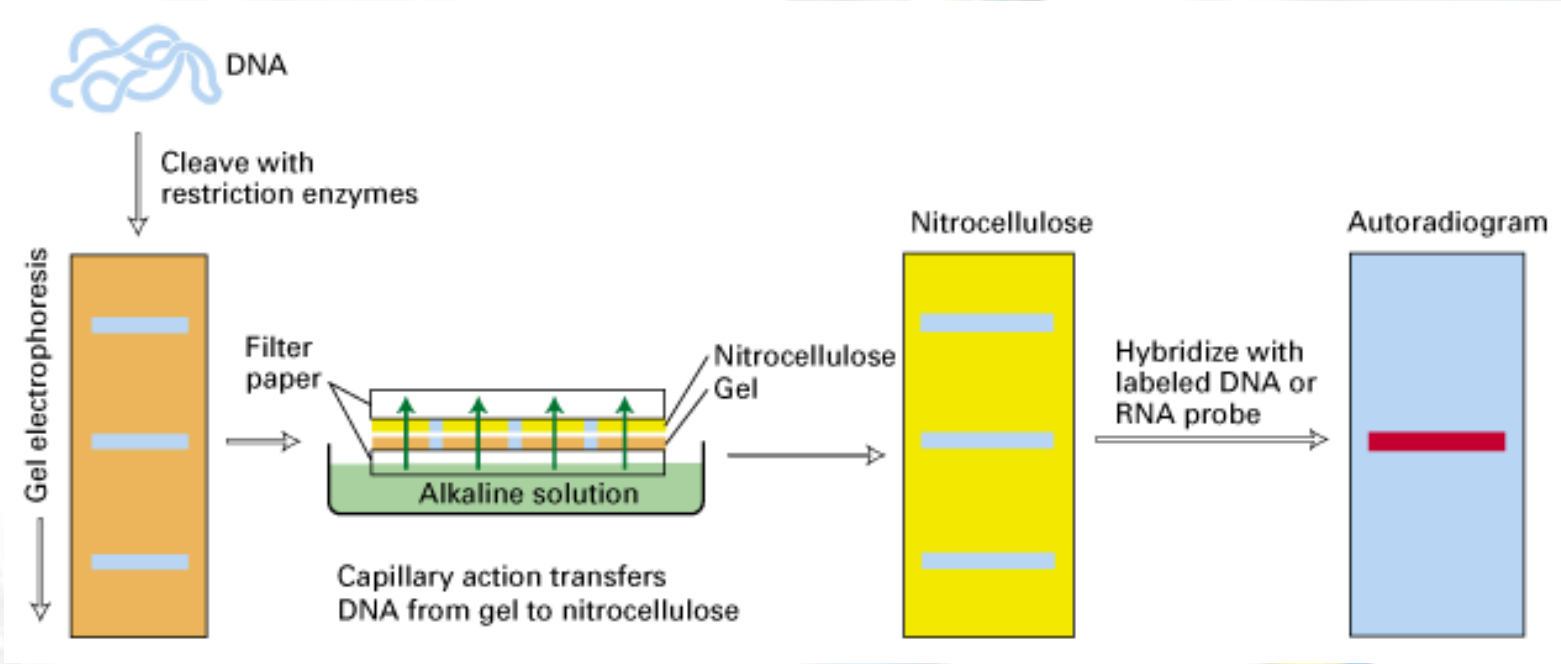
The labeled probe is added to the blocked membrane in buffer and incubated for several hours to allow the probe molecules to find their targets.

Blot is incubated with wash buffers containing NaCl and detergent to wash away excess probe.

Radioactive probes enable autoradiographic detection.

Southern Blotting - Procedure

PROCEDURE



Southern Blotting - Procedure

CONCLUSION

Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

Essentials of Genetics

MODULE# 48

SOUTHERN BLOTTING PROBES

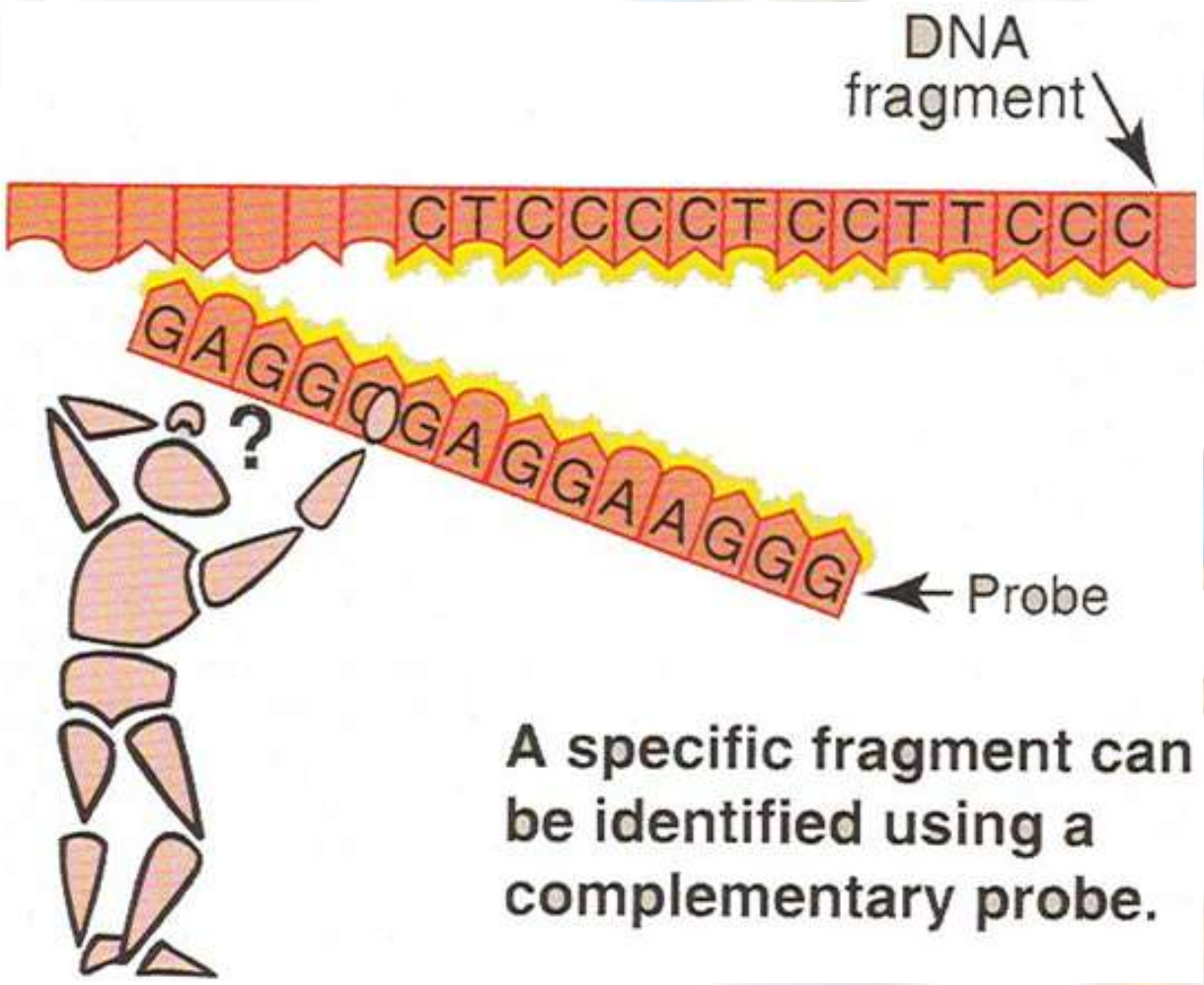
Southern Blotting - Probes

PROBES

Labeled material to detect a target.

For DNA: 20-30 nucleotides, complementary to a region in the gene or DNA.

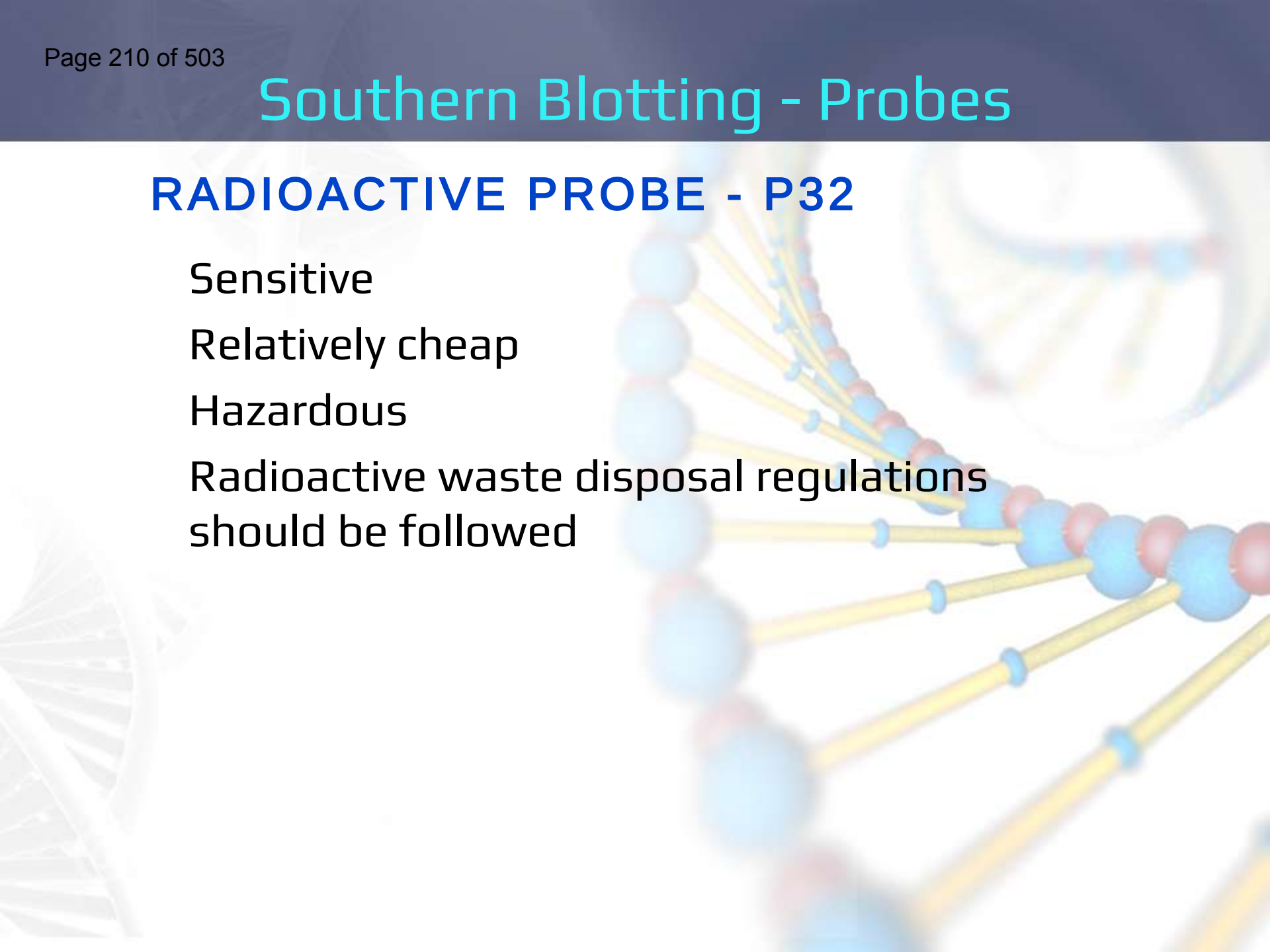
Southern Blotting - Probes



Southern Blotting - Probes

RADIOACTIVE PROBE - P32

- Sensitive
- Relatively cheap
- Hazardous
- Radioactive waste disposal regulations should be followed

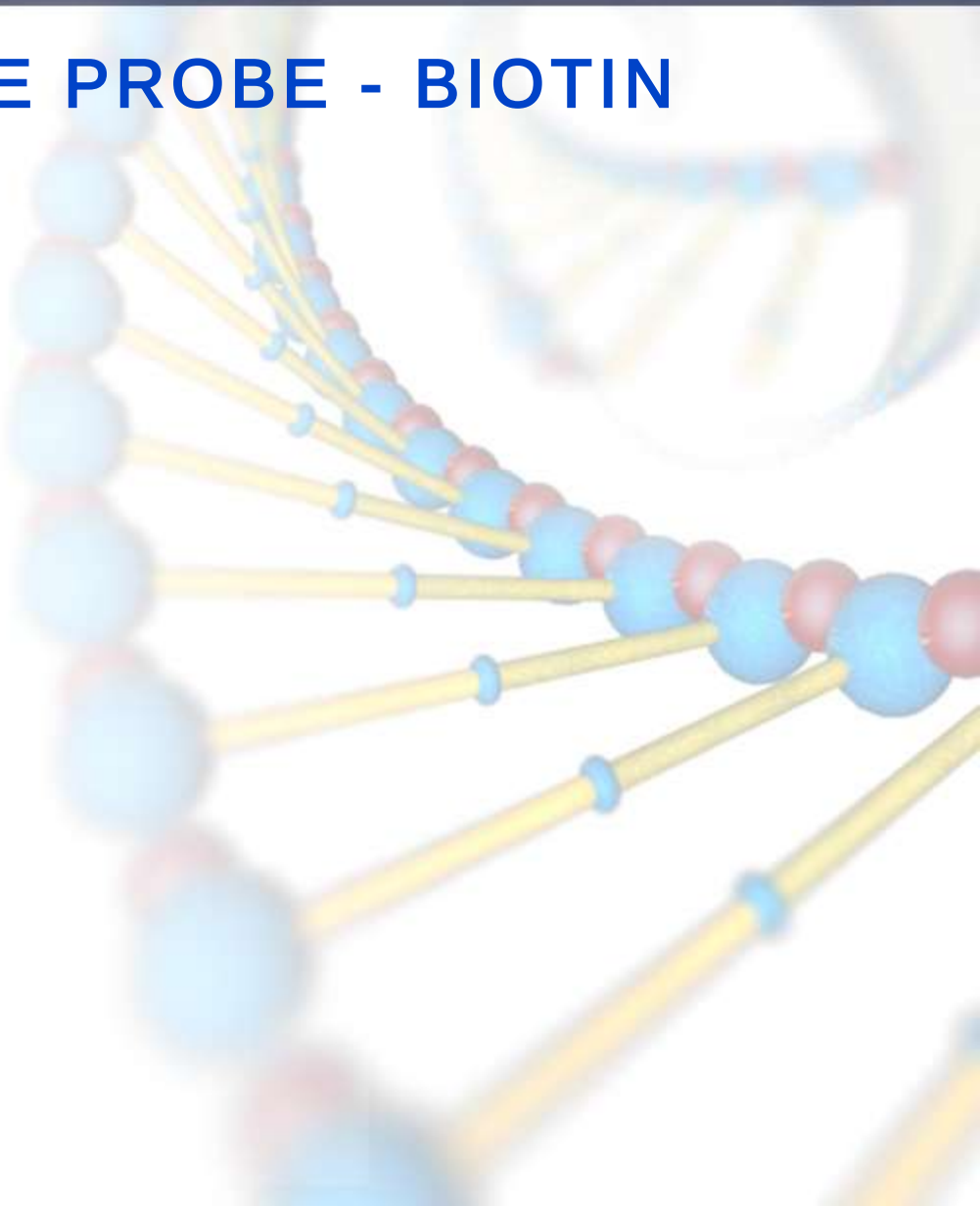


Southern Blotting - Probes

NON-RADIOACTIVE PROBE - BIOTIN

Sensitive

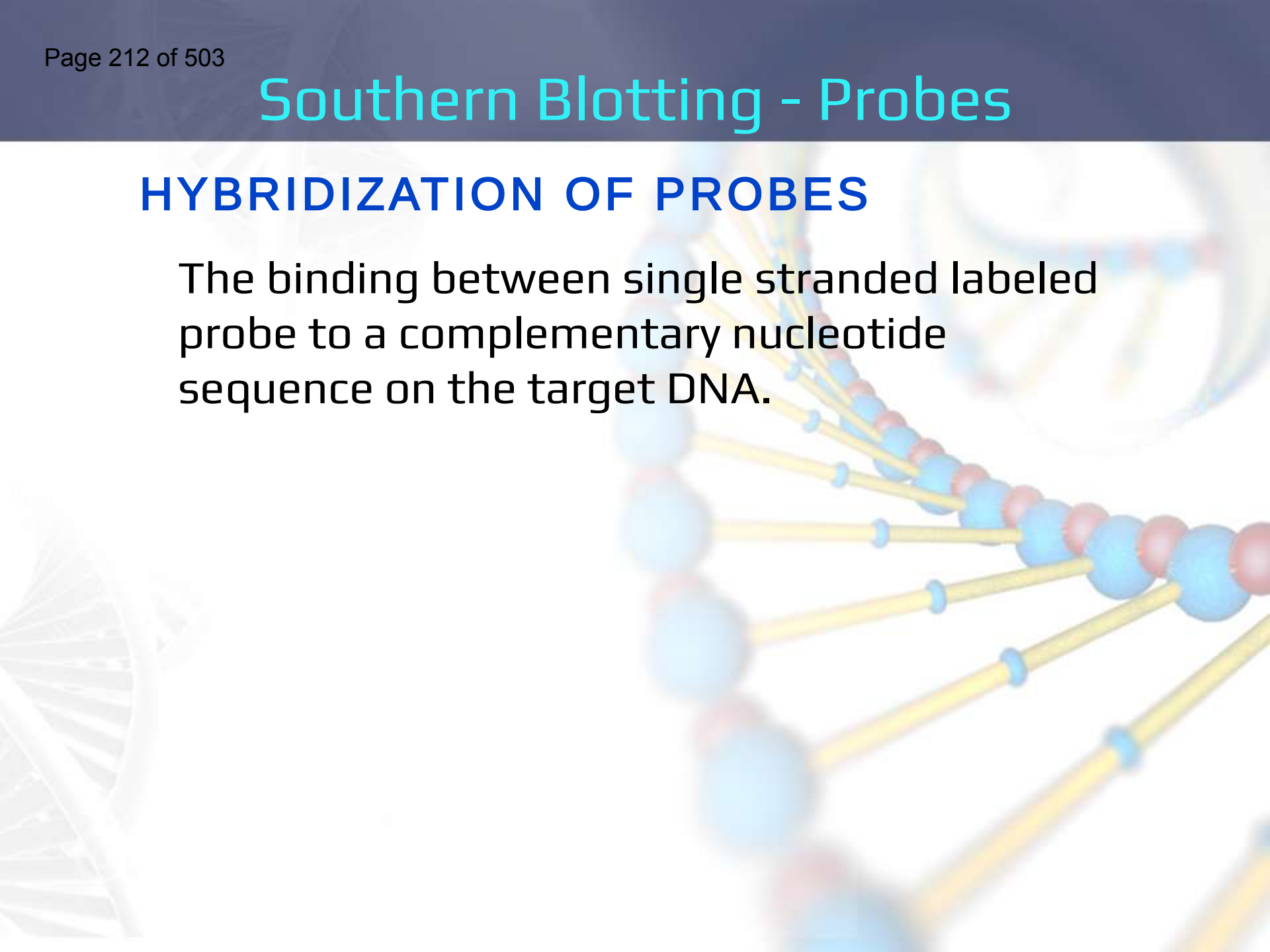
Relatively expensive



Southern Blotting - Probes

HYBRIDIZATION OF PROBES

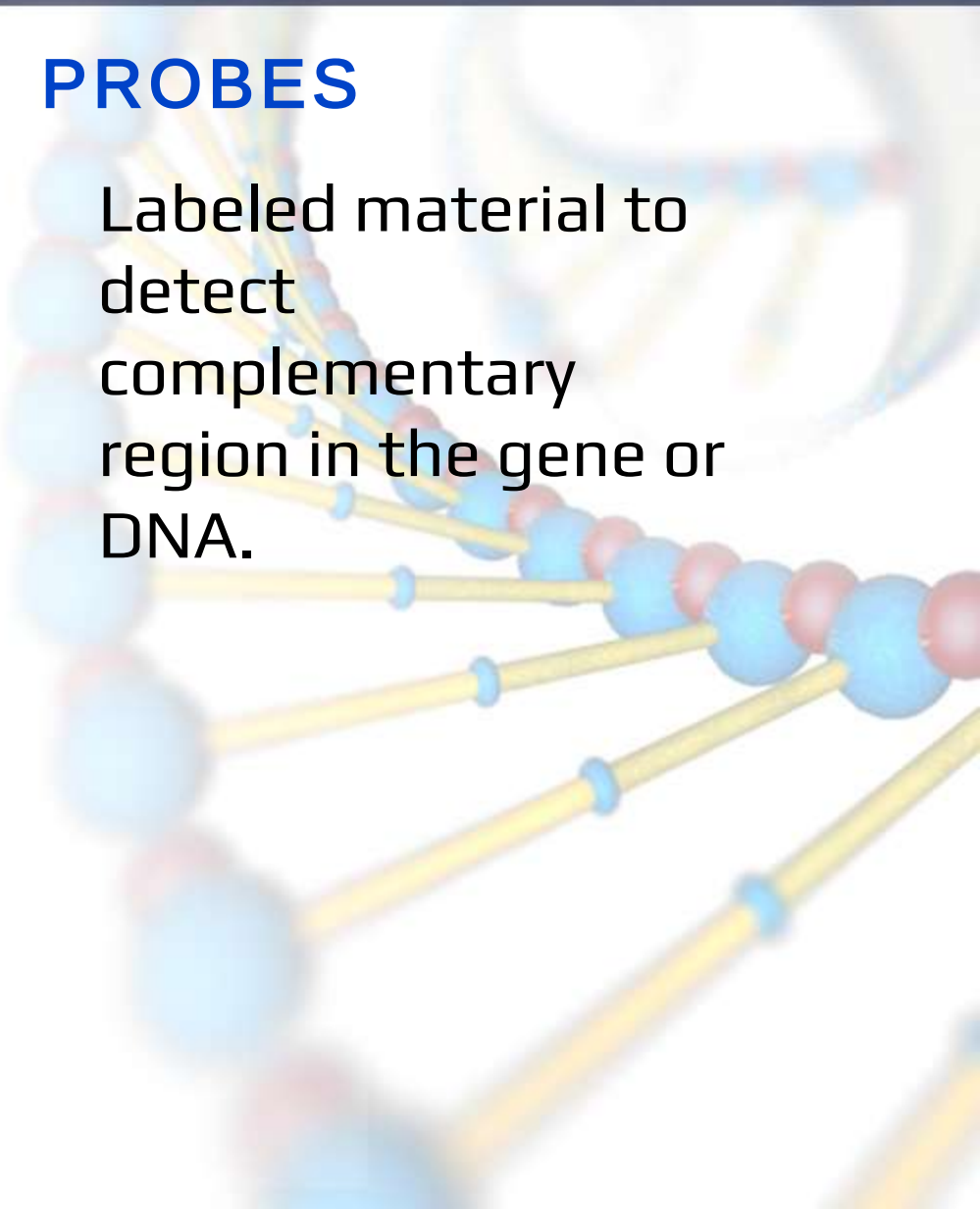
The binding between single stranded labeled probe to a complementary nucleotide sequence on the target DNA.



Southern Blotting - Probes

PROBES

Labeled material to detect complementary region in the gene or DNA.



MODULE# 49

TRANSFER METHODS IN SOUTHERN BLOTTING

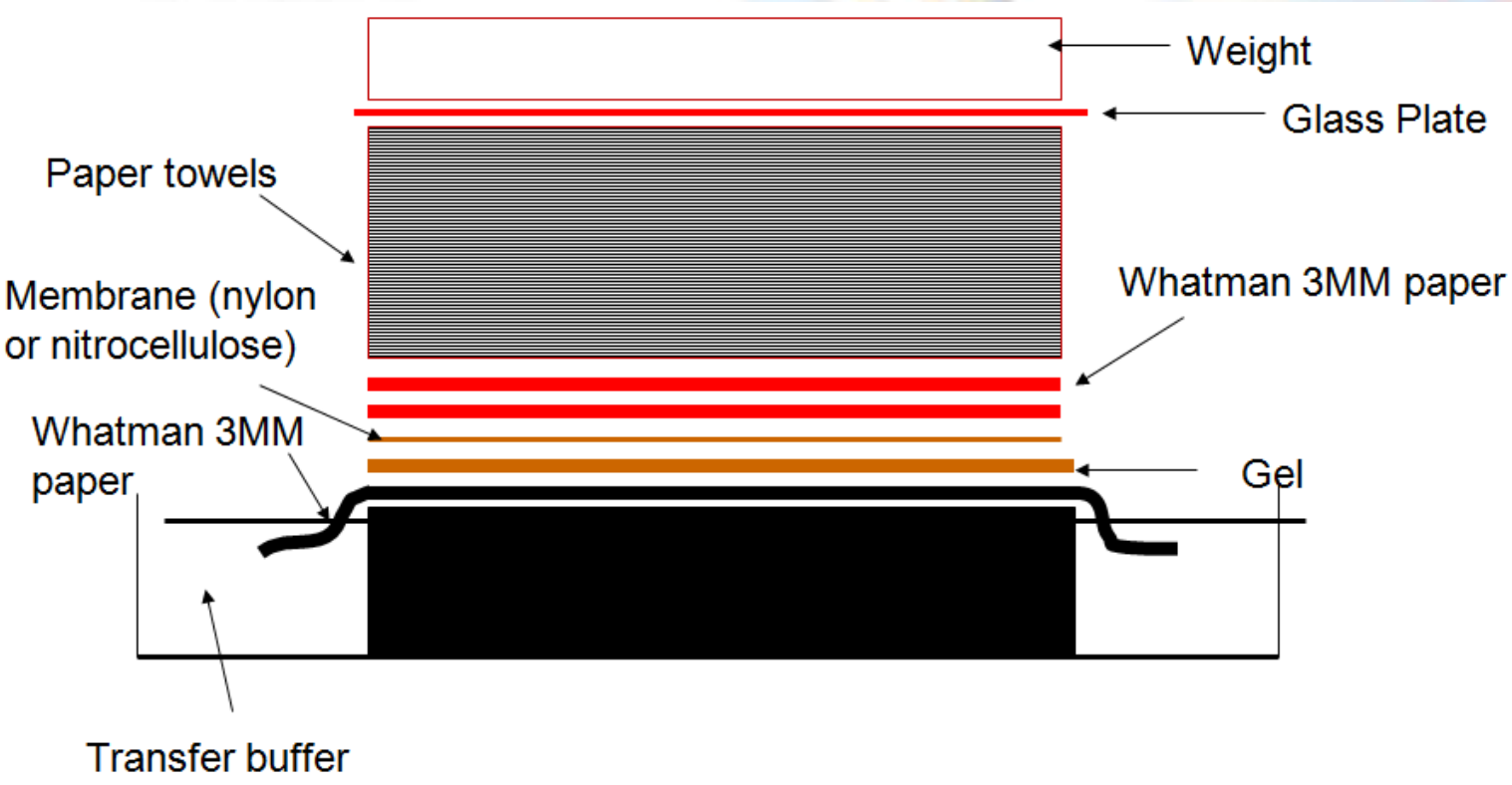
Transfer Methods - Southern

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

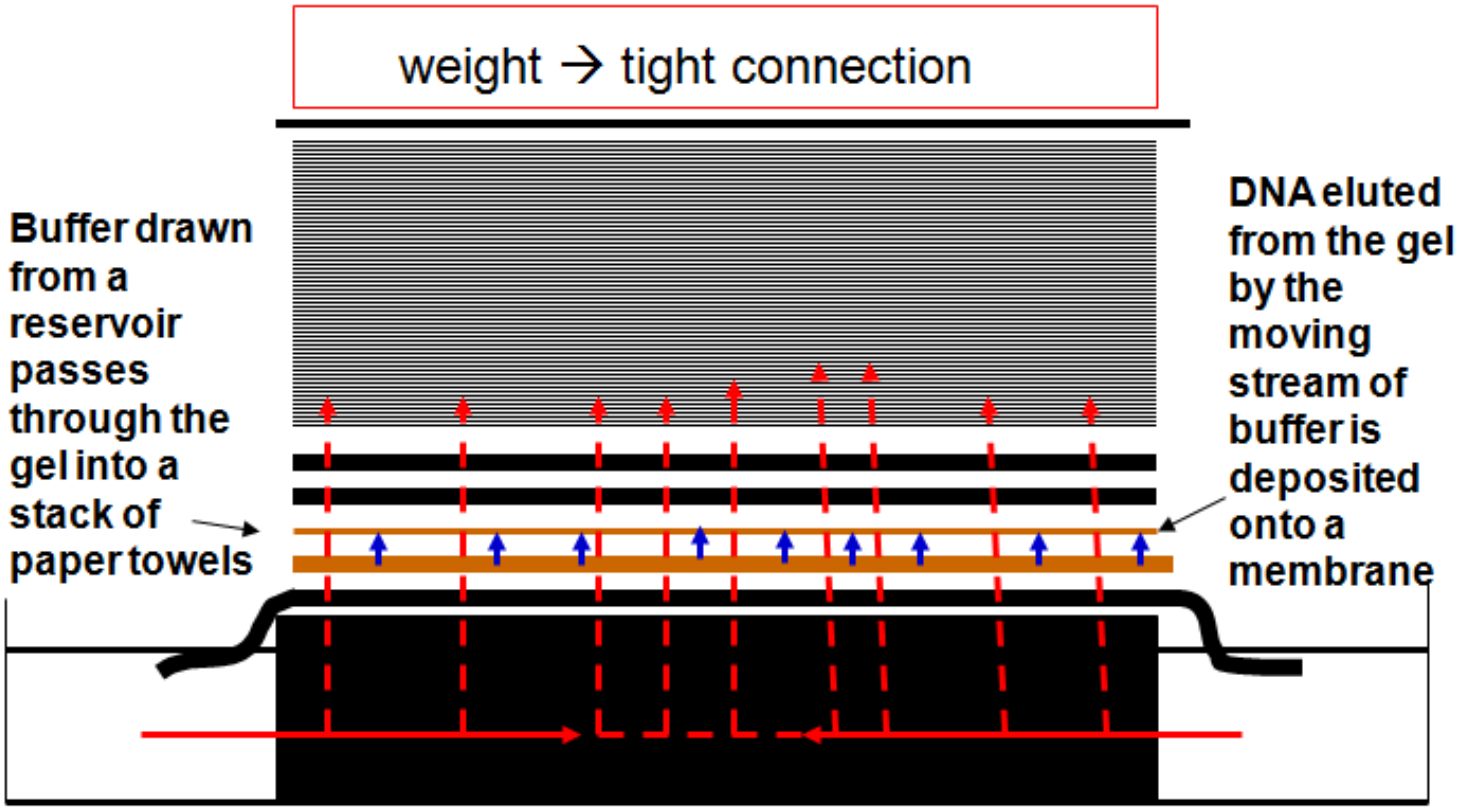
Transfer Methods - Southern

UPWARD TRANSFER



Transfer Methods - Southern

UPWARD TRANSFER OF DNA TO MEMBRANE



Transfer Methods - Southern

TRANSFER METHODS

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

MODULE# 50

MUTATIONS IDENTIFICATION BY SOUTHERN BLOTTING

Mutations Identification

MUTATION IDENTIFICATION

The presence of a mutation affecting a restriction site causes the pattern of bands to differ from those seen in a normal gene.

A change in one nucleotide may alter the nucleotide sequence so that the restriction endonuclease fails to recognize and cleave at that site.

Mutations Identification

MUTATION IDENTIFICATION

Digestion of genomic DNA to DNA fragments.

Size-separation of the fragments.

In situ denaturation of the DNA fragments.

Transfer of denatured DNA fragments into a solid support (nylon or nitrocellulose).

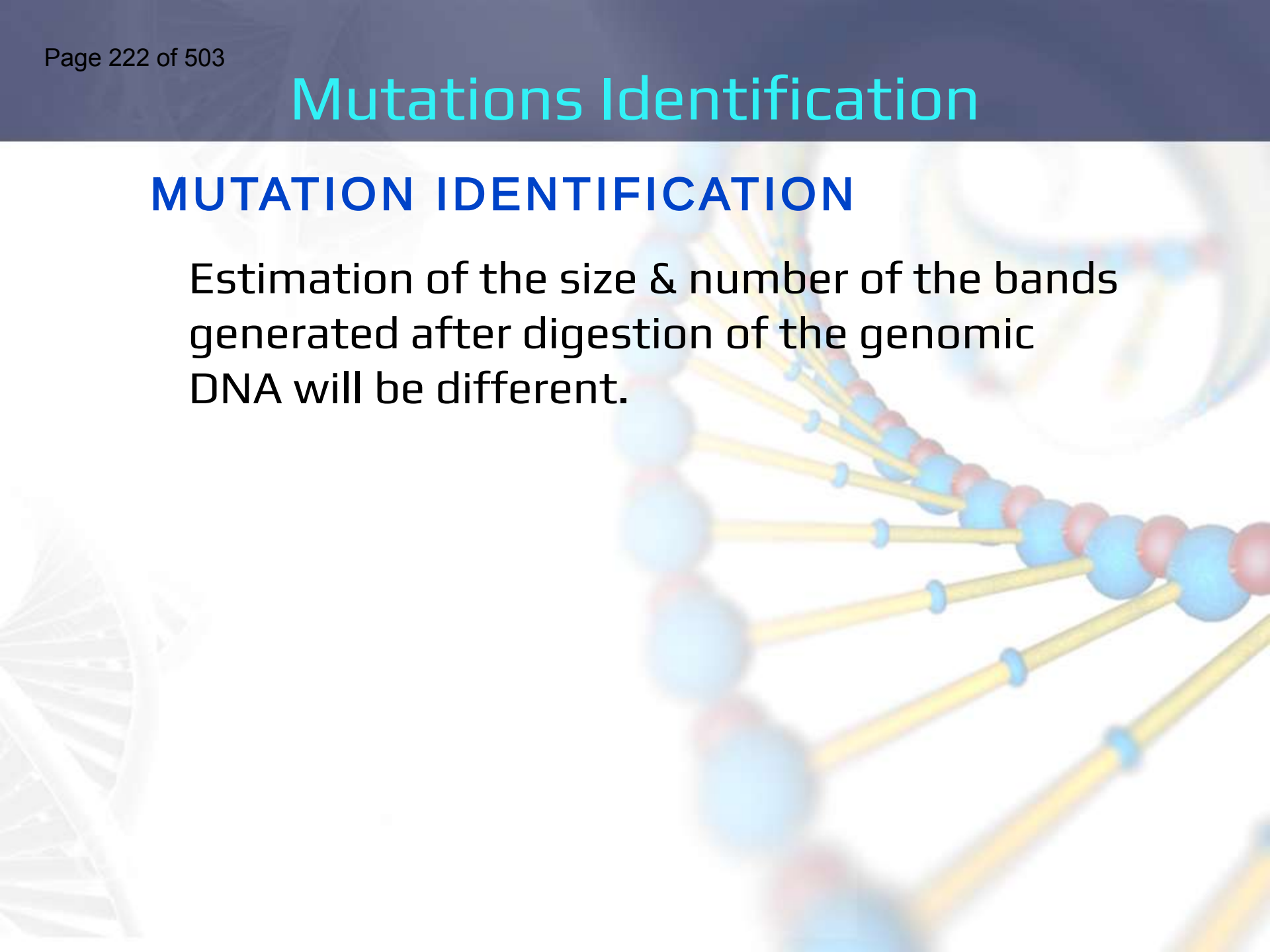
Hybridization of the immobilized DNA to a labeled probe (DNA, RNA).

Detection of the bands complementary to the probe (e.g. by autoradiography).

Mutations Identification

MUTATION IDENTIFICATION

Estimation of the size & number of the bands generated after digestion of the genomic DNA will be different.



Mutations Identification

MUTATION IDENTIFICATION

Mutations can be identified by Southern blotting.

MERGED BY BAQIR HUSSAIN BS ZOOLOGY 03336361244

MODULE# 51

NORTHERN BLOTTING

Northern Blotting

NORTHERN BLOTTING

Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

Northern Blotting

PROCEDURE

Isolation of intact mRNA.

Separation of RNA according to size (through a denaturing agarose gel.

Transfer of the RNA to a solid support.

Fixation of the RNA.

Hybridization of the immobilized RNA to probes complementary to the sequences of interest.

Northern Blotting

PROCEDURE

Removal of probe molecules that are nonspecifically bound to the solid matrix.

Detection, capture and analysis of an image of the specifically bound probe molecules.

Northern Blotting

NORTHERN BLOTTING

Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

MODULE# 52

NORTHERN BLOT APPLICATIONS

Northern Blot - Applications

APPLICATIONS

Study of gene expression in eukaryotic cells.

To measure the amount & size of RNAs transcribed from eukaryotic genes.

To estimate the abundance of RNAs.

Northern Blot - Applications

APPLICATIONS

To equalize the amounts of RNA loaded into lanes of gels.

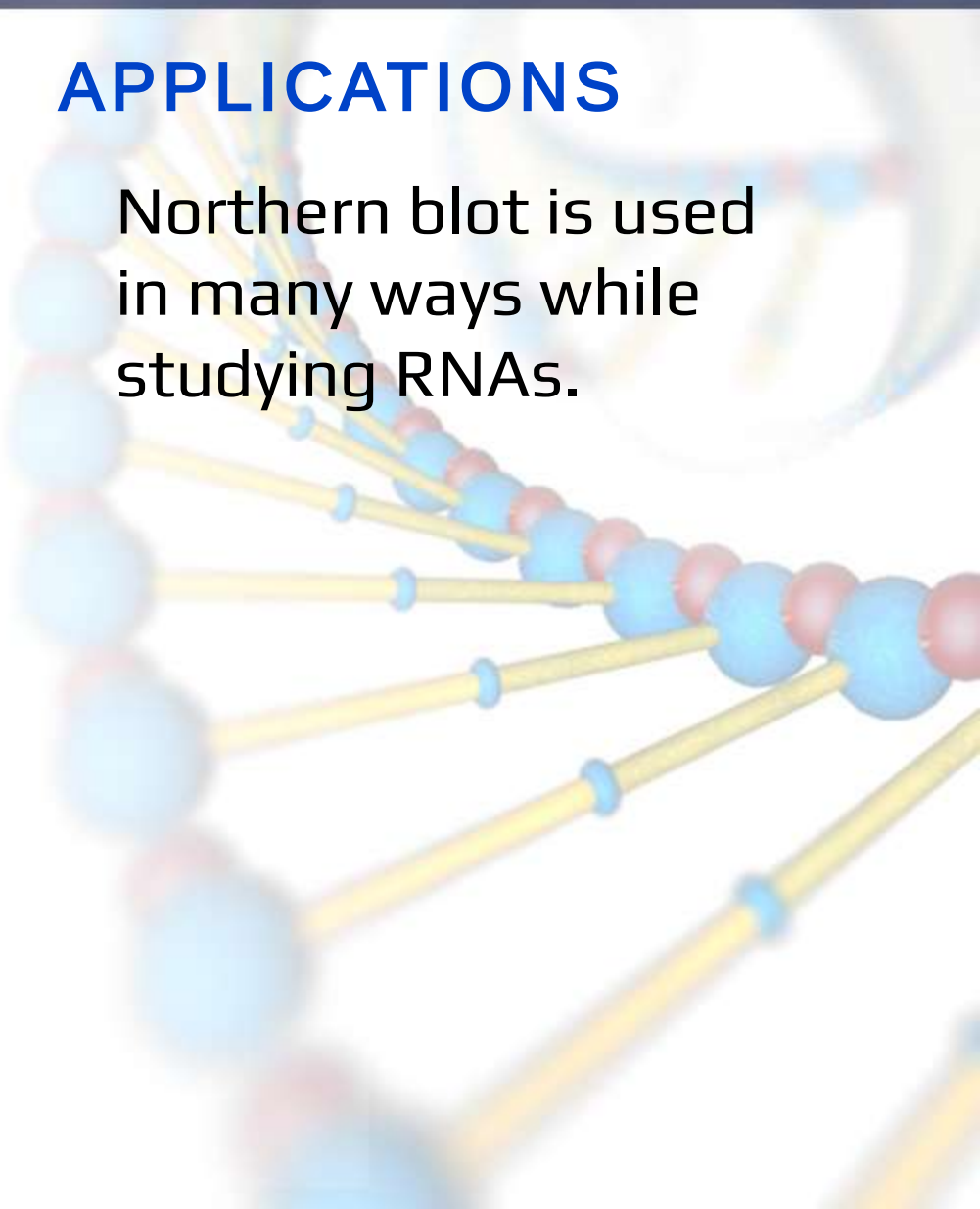
Use of housekeeping gene (endogenous constitutively-expressed gene).

Normalizing samples according to their content of mRNAs of the housekeeping gene.

Northern Blot - Applications

APPLICATIONS

Northern blot is used in many ways while studying RNAs.



Essentials of Genetics

MODULE# 53

WESTERN BLOTTING

Western Blotting

WESTERN BLOTTING

A technique used to detect the presence of a specific protein in a complex protein mixture.

Western Blotting

WESTERN BLOTTING

To determine the molecular weight of a protein.

To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled.

Western Blotting

WESTERN BLOTTING

Western blots have become one of the most common analytical tools for the detection of viral proteins.

Characterization of monoclonal and polyclonal antibody preparations and in determining the specificity of the immune response to viral antigens.

Western Blotting

WESTERN BLOTTING

A technique used to detect the presence of a specific protein in a complex protein mixture.

MODULE# 54

WESTERN BLOTTING PROCEDURE

Western Blotting - Procedure

PROCEDURE

1- Sample preparation

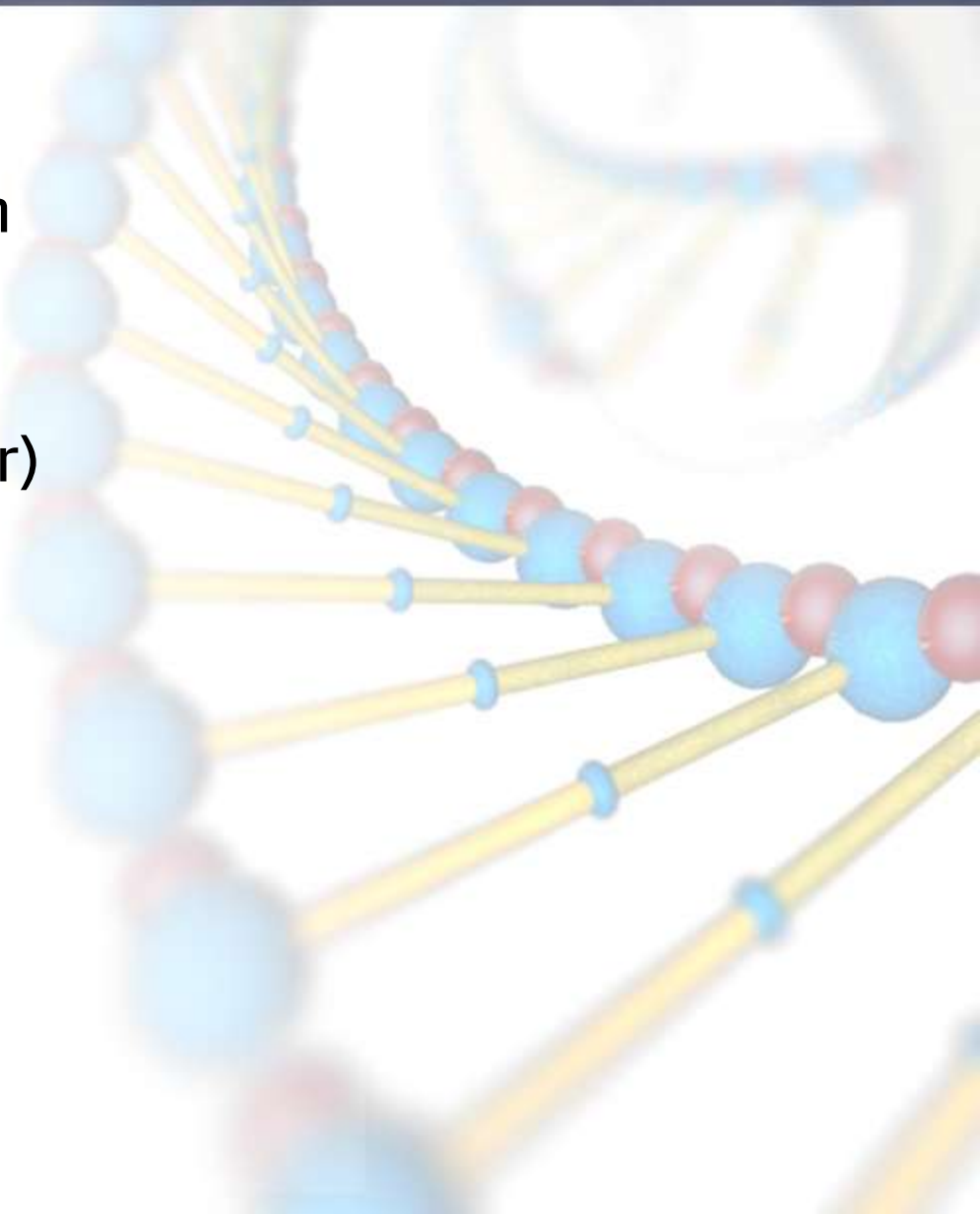
2- Gel Electrophoresis

3- Blotting (or transfer)

4- Blocking

5- Antibody probing

6- Detection



Western Blotting - Procedure

PROCEDURE

1- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.

2- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.

Western Blotting - Procedure

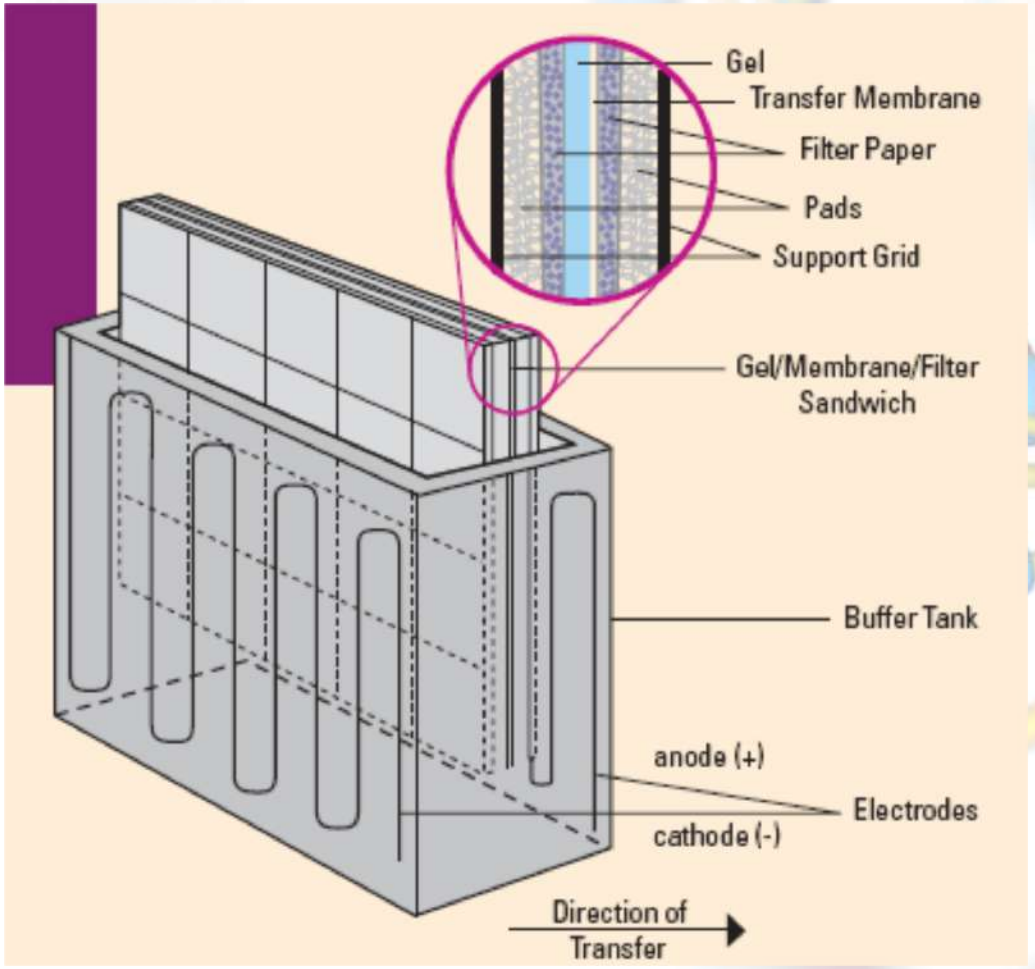
PROCEDURE

3- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD.

4- Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

Western Blotting - Procedure

ELECTROPHORETIC TRANSFER



Western Blotting - Procedure

PROCEDURE

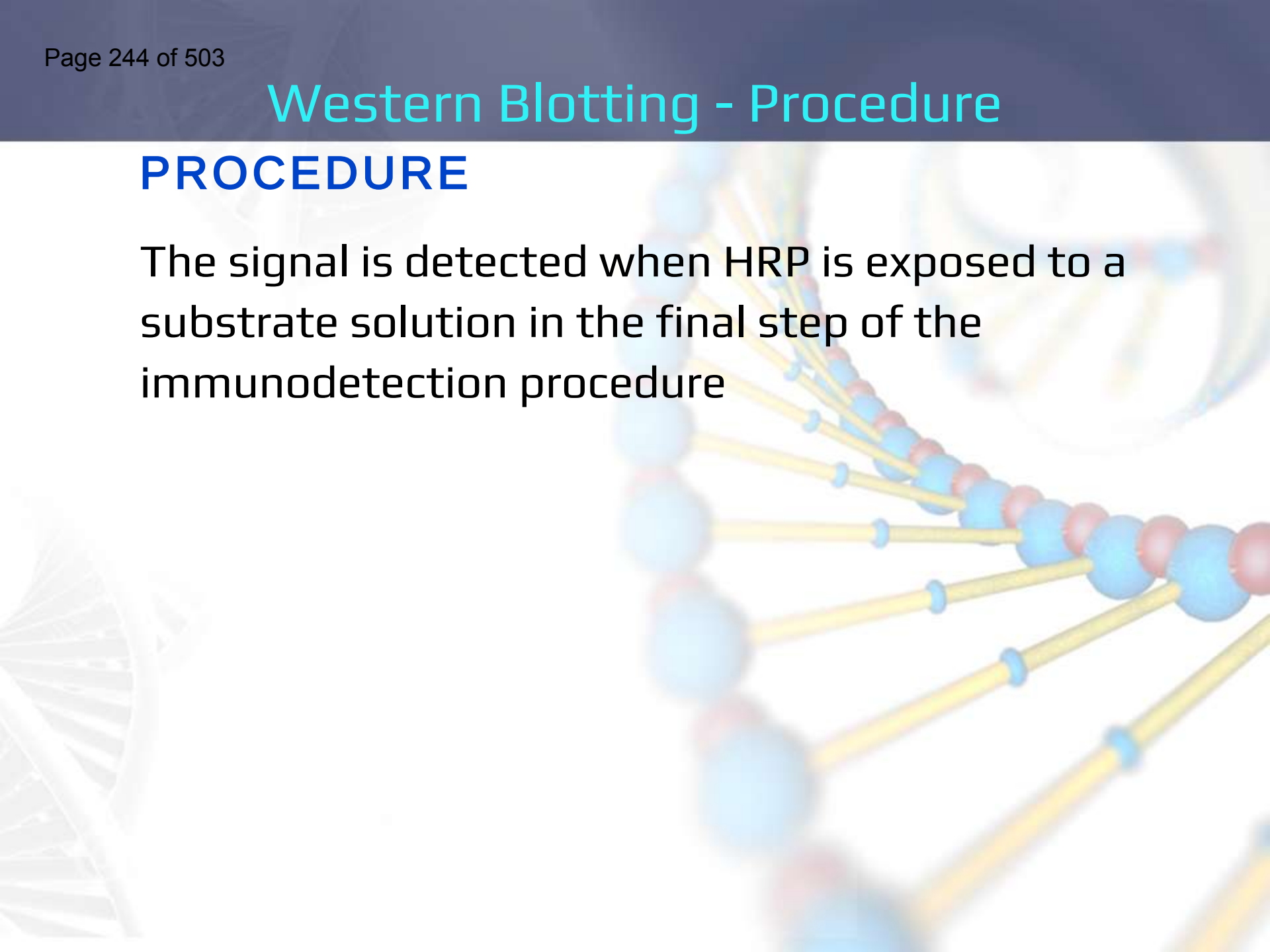
Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein.

A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.

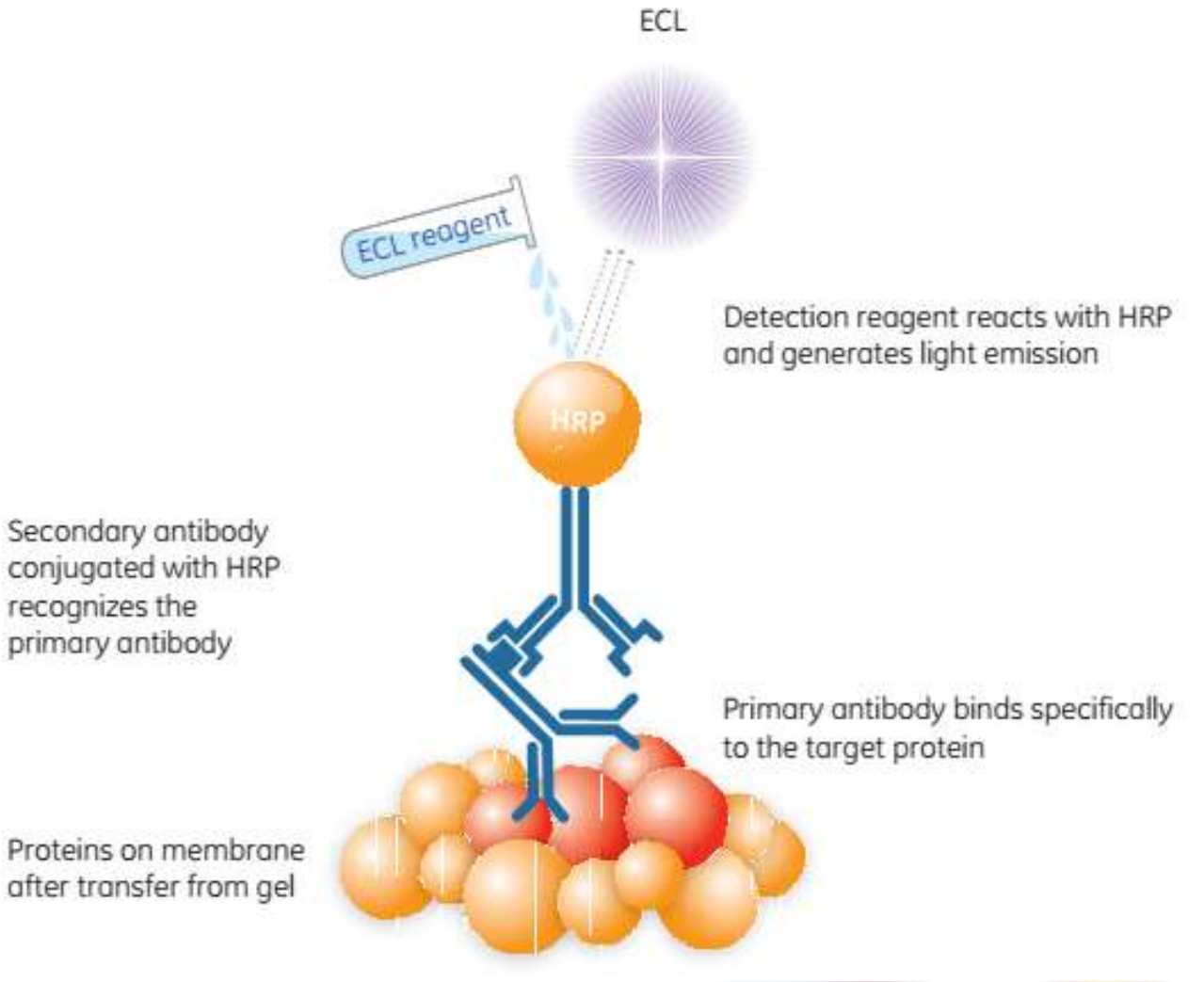
Western Blotting - Procedure

PROCEDURE

The signal is detected when HRP is exposed to a substrate solution in the final step of the immunodetection procedure



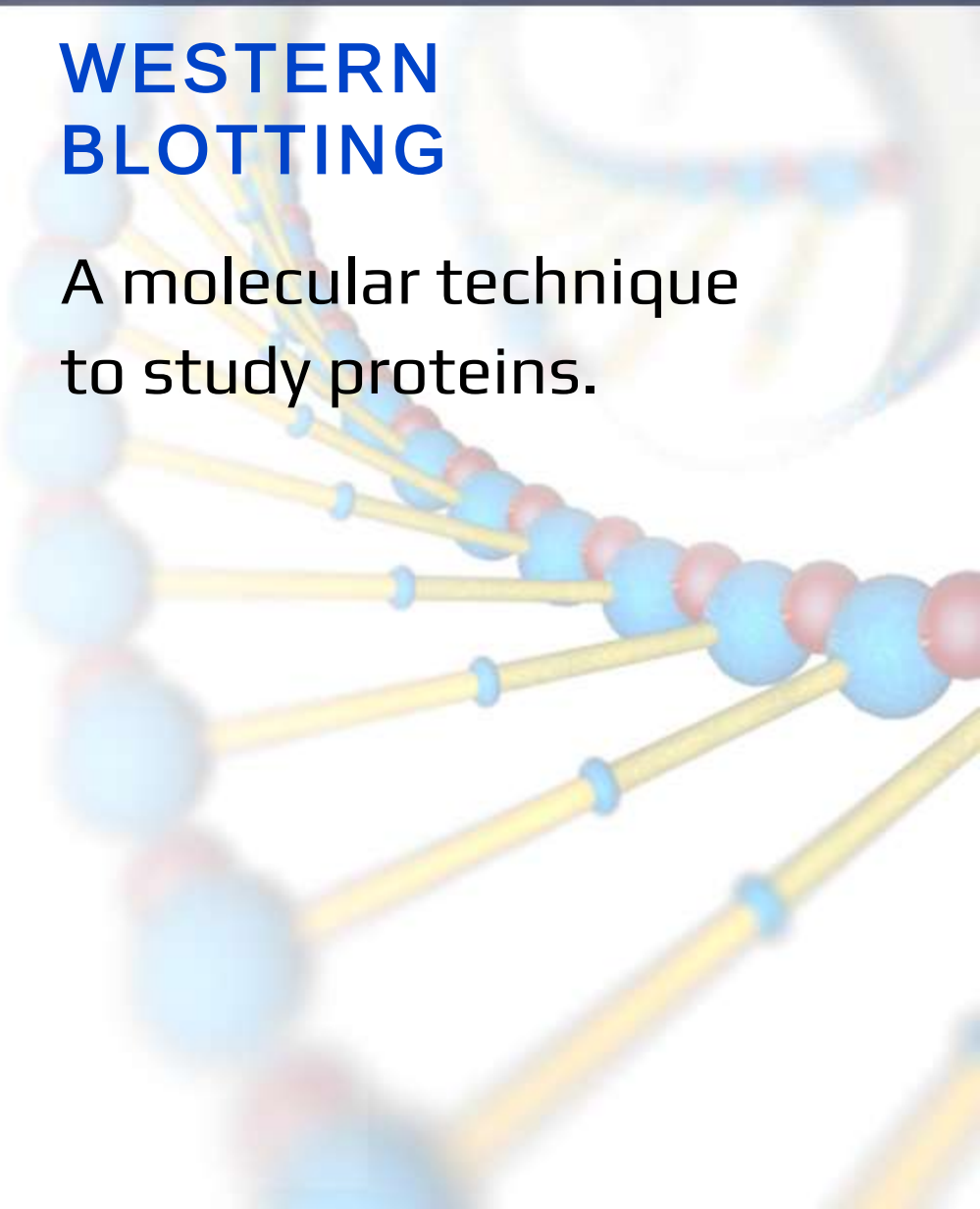
Western Blotting - Procedure



Western Blotting - Procedure

WESTERN BLOTTING

A molecular technique to study proteins.



Module# 55

Applications of western blotting

Analysis Of IgG Fractions purified from human plasma.

Diagnosis of HIV by ELISA, involves the western blotting technique.

Western blotting technique is also used to Detect Some Forms Of Lyme Disease.

Western blotting technique is used in Definitive Test For BSE, which is commonly know as Mad cow disease.

Confirmatory Test For Hepatitis-B involves western blotting technique.

Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.

This technique is also employed in The Gene Expression Studies

Limitations in western blotting

very delicate and time consuming process. A minute imbalance at any level of the procedure can skew the results of the entire process.

Incorrect labeling of the protein can happen due to the reaction of secondary antibody.

Cause erroneous in bands or no bands due to insufficient transfer.

Well trained technicians are required for this technique.

Primary antibody availability is crucial.

It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

Conclusion

Western blotting technique is simply a way to identify unknown proteins on a polyacrylamide gel.

It is sometimes called as protein blotting or immunoblotting.

It is a widely used analytical technique in the fields of molecular biology, immunogenetics, and other biochemistry disciplines.

Western blotting technique is also used in the field of medical diagnostics. i.e., in the analysis of various kinds of diseases.

Apart from the limitations of western blotting, it is more helpful now a days. Hence, The discovery of Western blotting technique has become boon in the field of science & technology

Module# 56

Difference between different blotting technique

Northern vs Southern vs Western Blotting

Type of Molecule Detected	
Northern Blotting	Northern blotting detects a specific RNA sequence from an RNA sample.
Southern Blotting	Southern blotting detects a specific DNA sequence from a DNA sample.
Western Blotting	Western blotting detects a specific protein from a protein sample.
Type of Gel	
Northern Blotting	This uses Agarose/formaldehyde gel.
Southern Blotting	This uses an Agarose gel.
Western Blotting	This uses Polyacrylamide gel.
Blotting Method	
Northern Blotting	This is a capillary transfer.
Southern Blotting	This is a capillary transfer.
Western Blotting	This is an electric transfer.

Northern vs Southern vs Western Blotting

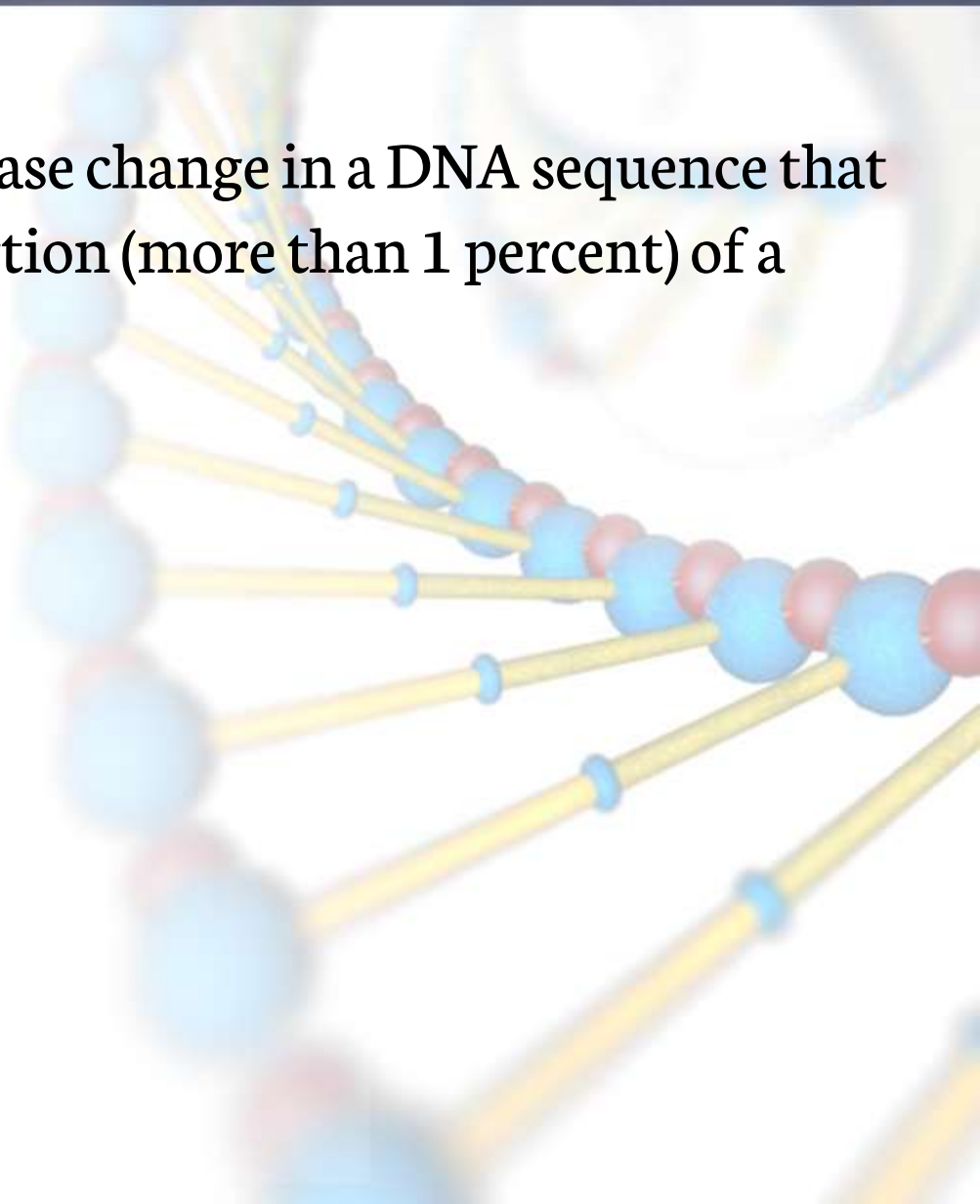
Probes Used	
Northern Blotting	cDNA or RNA probes labeled radioactively or nonradioactively.
Southern Blotting	DNA probes are labeled radioactively or non-radioactively.
Western Blotting	Primary antibodies are used as probes.
Detection System	
Northern Blotting	This is done using an autoradiograph, or detection of light or color change.
Southern Blotting	This is done using an autoradiograph, detection of light or color change.
Western Blotting	This is done using the detection of light or color change.

Module# 57

Single nucleotide polymorphism

What is SNP?

A **SNP** is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 percent) of a large population.



Some Facts

In human beings, 99.9 percent bases are same.

Remaining 0.1 percent makes a person unique.

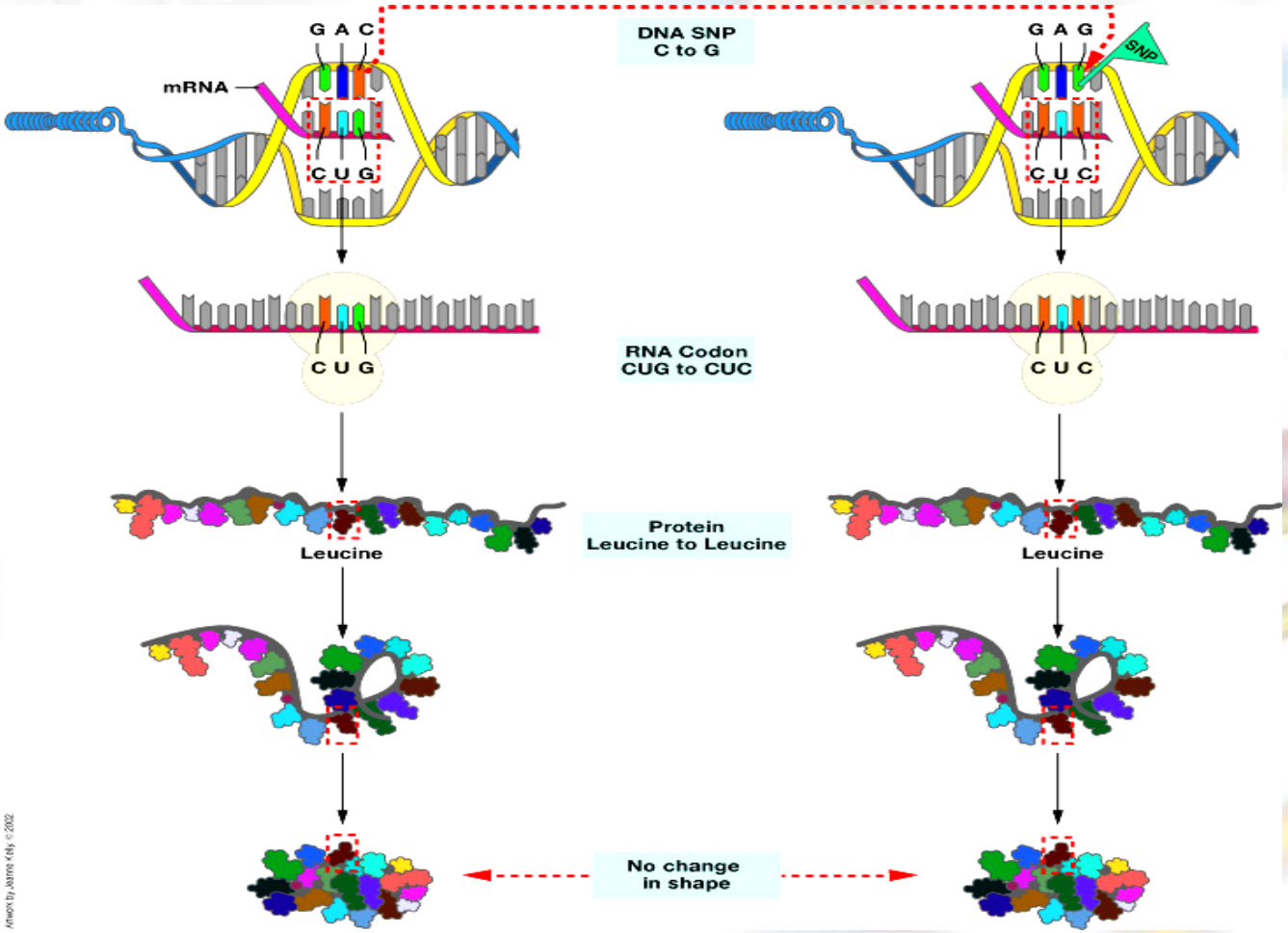
- Different attributes / characteristics / traits
  - how a person looks,
  - diseases he or she develops.

These variations can be:

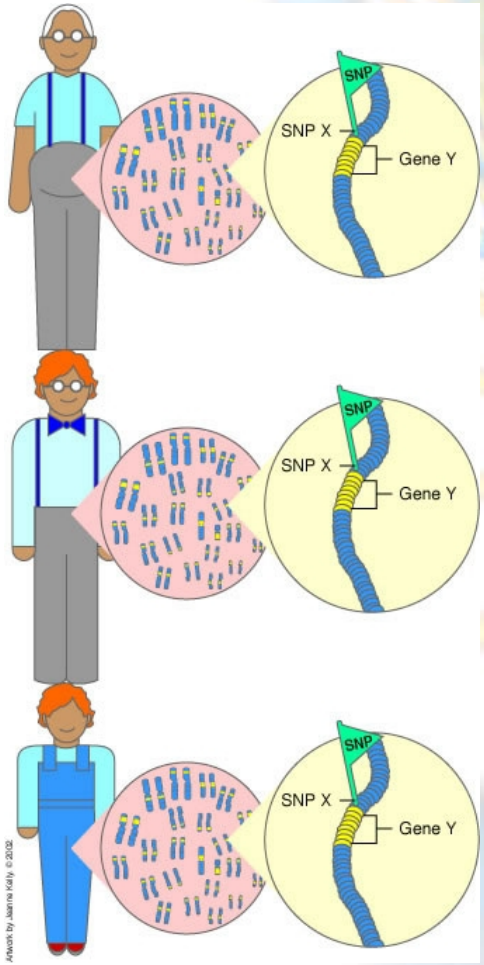
- Harmless (change in phenotype)
- Harmful (diabetes, cancer, heart disease, Huntington's disease, and hemophilia)
- Latent (variations found in coding and regulatory regions, are not harmful on their own, and the change in each gene only becomes apparent under certain conditions e.g. susceptibility to lung cancer)

- SNPs are found in
  - coding and (mostly) noncoding regions.
- Occur with a very high frequency
  - about 1 in 1000 bases to 1 in 100 to 300 bases.
- The abundance of SNPs and the ease with which they can be measured make these genetic variations significant.
- SNPs close to particular gene acts as a marker for that gene.
- SNPs in coding regions may alter the protein structure made by that coding region.

SNPs may / may not alter protein structure



SNPs act as gene markers



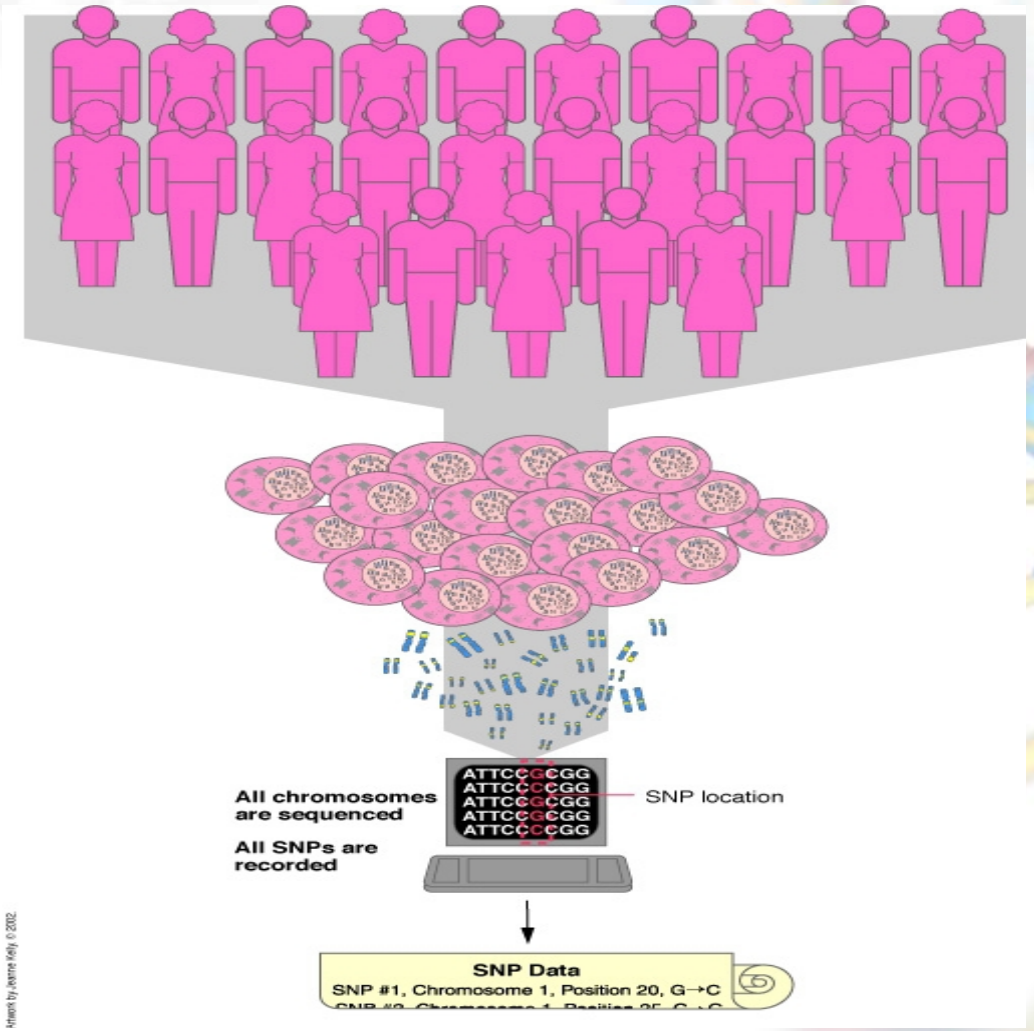
Module# 58

Single nucleotide polymorphism-2

SNP maps

- Sequence genomes of a large number of people
- Compare the base sequences to discover SNPs.
- Generate a single map of the human genome containing all possible SNPs => SNP maps

SNP Maps


All chromosomes are sequenced
All SNPs are recorded
ATTCCGCGG
ATTCCCCGG
ATTCCGCGG
ATTCCGCGG
ATTCCCCGG
SNP location
SNP Data
SNP #1, Chromosome 1, Position 20, G→C

SNP Profiles

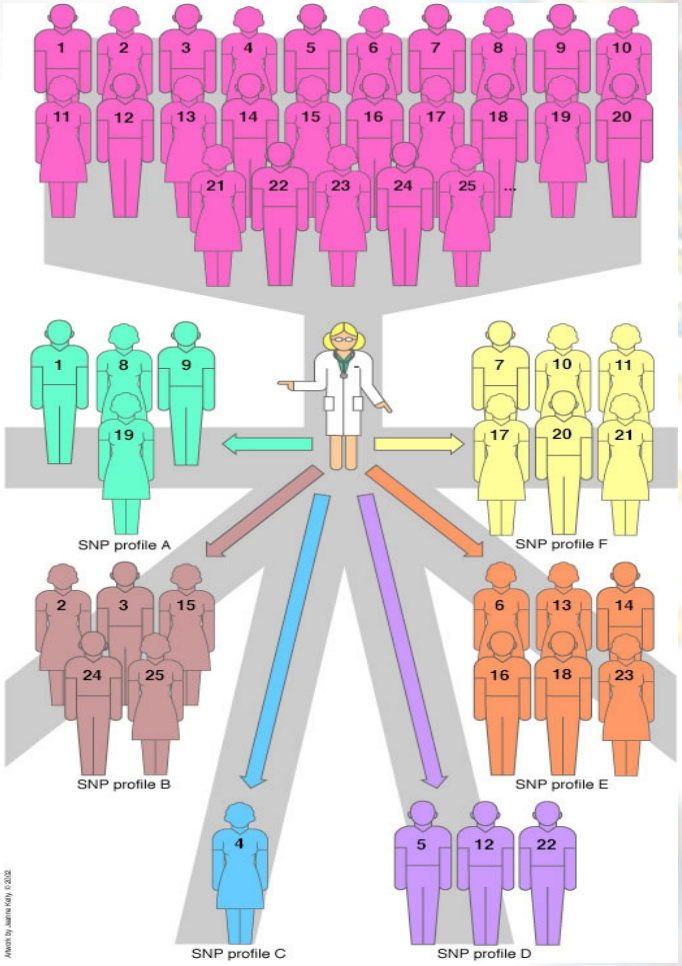
Genome of each individual contains distinct SNP pattern.

People can be grouped based on the SNP profile.

SNPs Profiles important for identifying response to Drug Therapy.

Correlations might emerge between certain SNP profiles and specific responses to treatment.

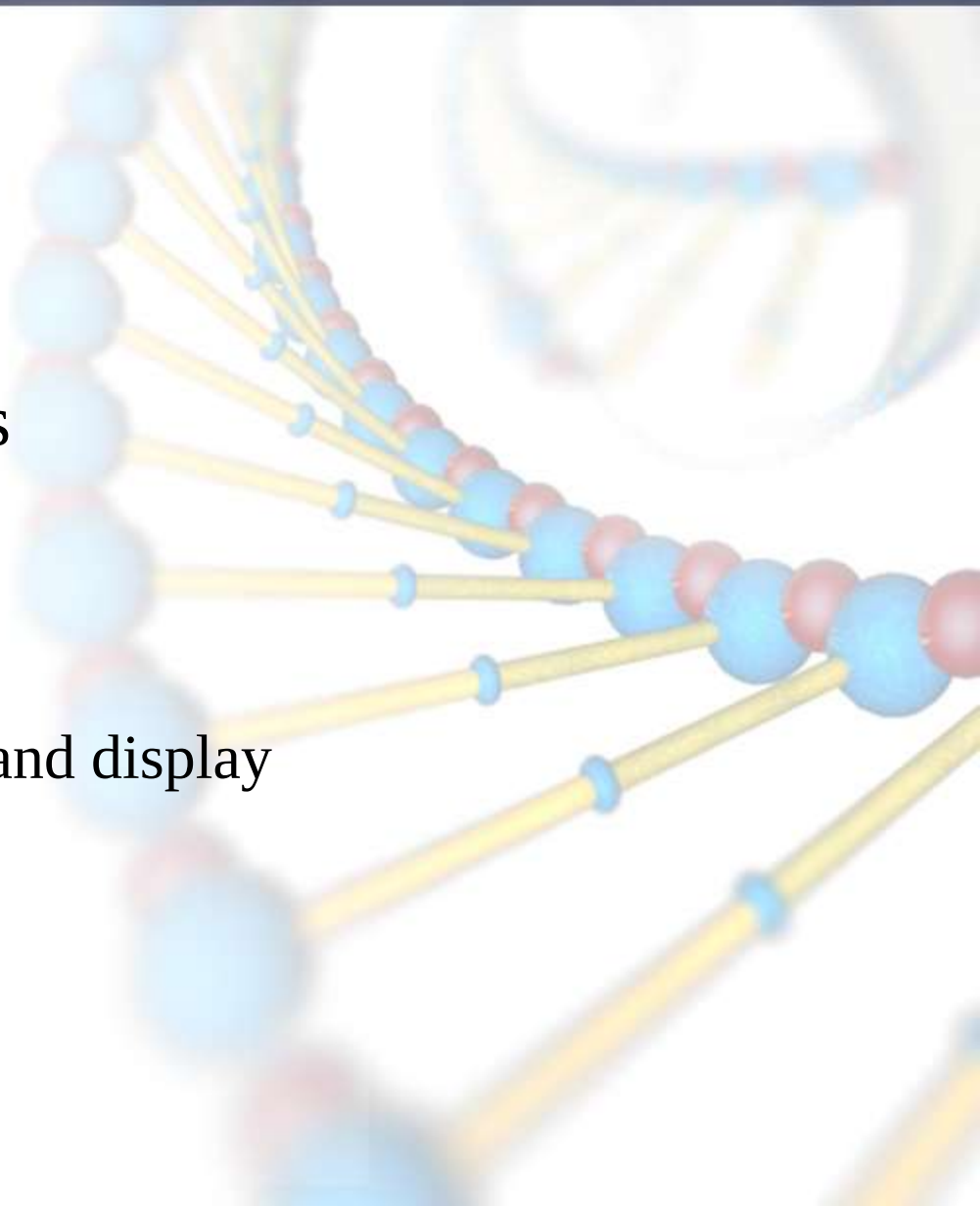
SNP Profiles



Module# 59

How to detect SNPs

Techniques to detect known Polymorphisms



- Hybridization Techniques
  - Micro arrays
  - Real time PCR
- Enzyme based Techniques
  - Nucleotide extension
  - Cleavage
  - Ligation
  - Reaction product detection and display

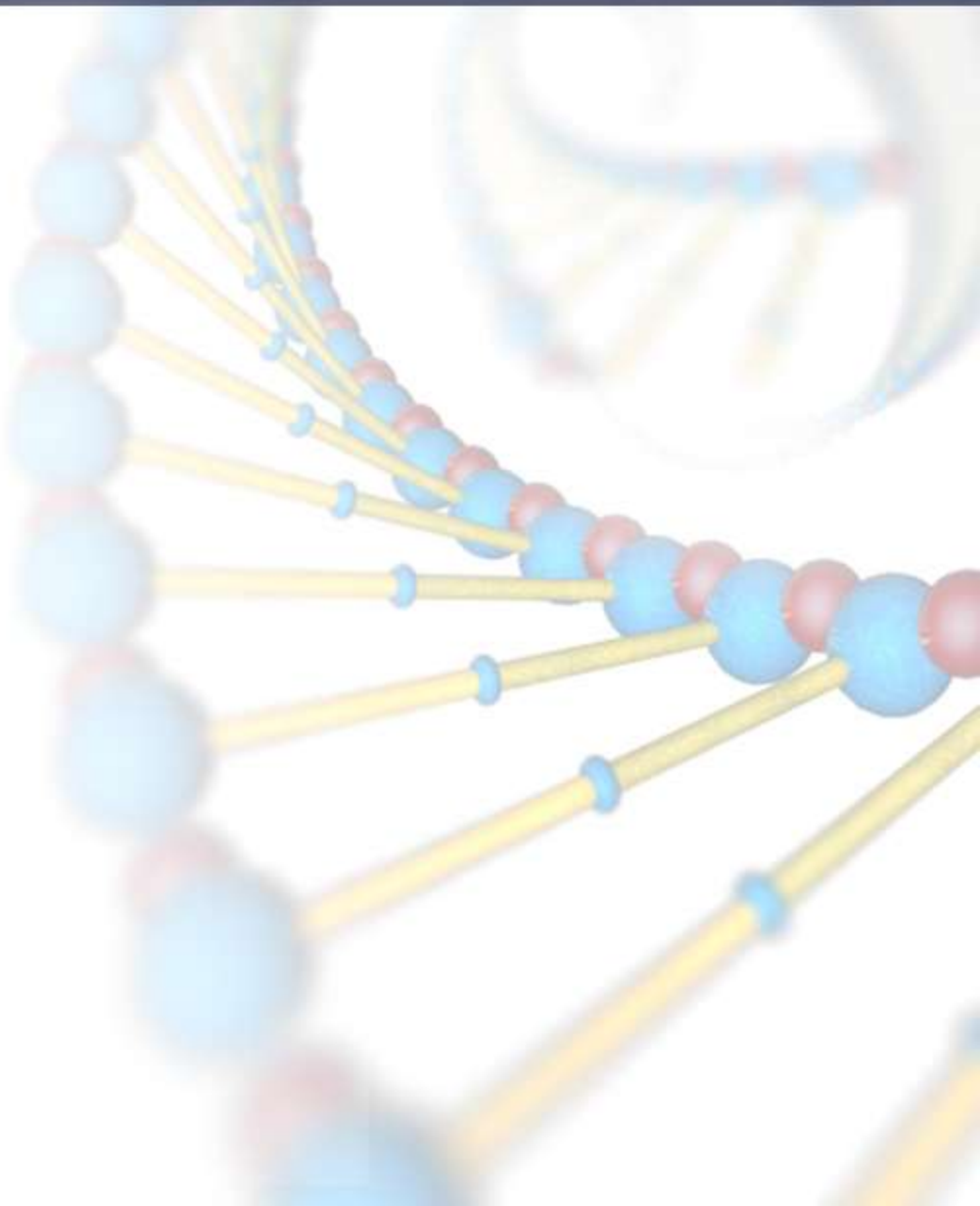
Techniques to detect unknown Polymorphisms

Direct Sequencing

Microarray

Cleavage / Ligation

Electrophoretic mobility assays



Direct Sequencing

- Sanger dideoxysequencing can detect any type of unknown polymorphism and its position, when the majority of DNA contains that polymorphism.
- Misses polymorphisms and mutations when the DNA is heterozygous
- limited utility for analysis of solid tumors or pooled samples of DNA due to low sensitivity
- Once a sample is known to contain a polymorphism in a specific region, direct sequencing is particularly useful for identifying a polymorphism and its specific position.
- Even if the identity of the polymorphism cannot be discerned in the first pass, multiple sequencing attempts have proven quite successful in elucidating sequence and position information.

Module# 60

SNP Screening

SNP Screening

- Two different screening strategies
  - Many SNPs in a few individuals
  - A few SNPs in many individuals
- Different strategies will require different tools
- Important in determining markers for complex genetic states

SNP genotyping methods for detecting genes contributing to susceptibility or resistance to multifactorial diseases, adverse drug reactions:

=> case-control association analysis

case GCCGTTGAC….

….GCCATTGAC….



Control …GCCATTGAC….

….GCCATTGAC….



SIGNIFICANCE OF SNPs

- IN DISEASE DIAGNOSIS
- IN FINDING PREDISPOSITION TO DISEASES
- IN DRUG DISCOVERY & DEVELOPMENT
- IN DRUG RESPONSES
- INVESTIGATION OF MIGRATION PATTERNS

ALL THESE ASPECT WILL HELP TO LOOK FOR MEDICATION & DIAGNOSIS AT INDIVIDUAL LEVEL

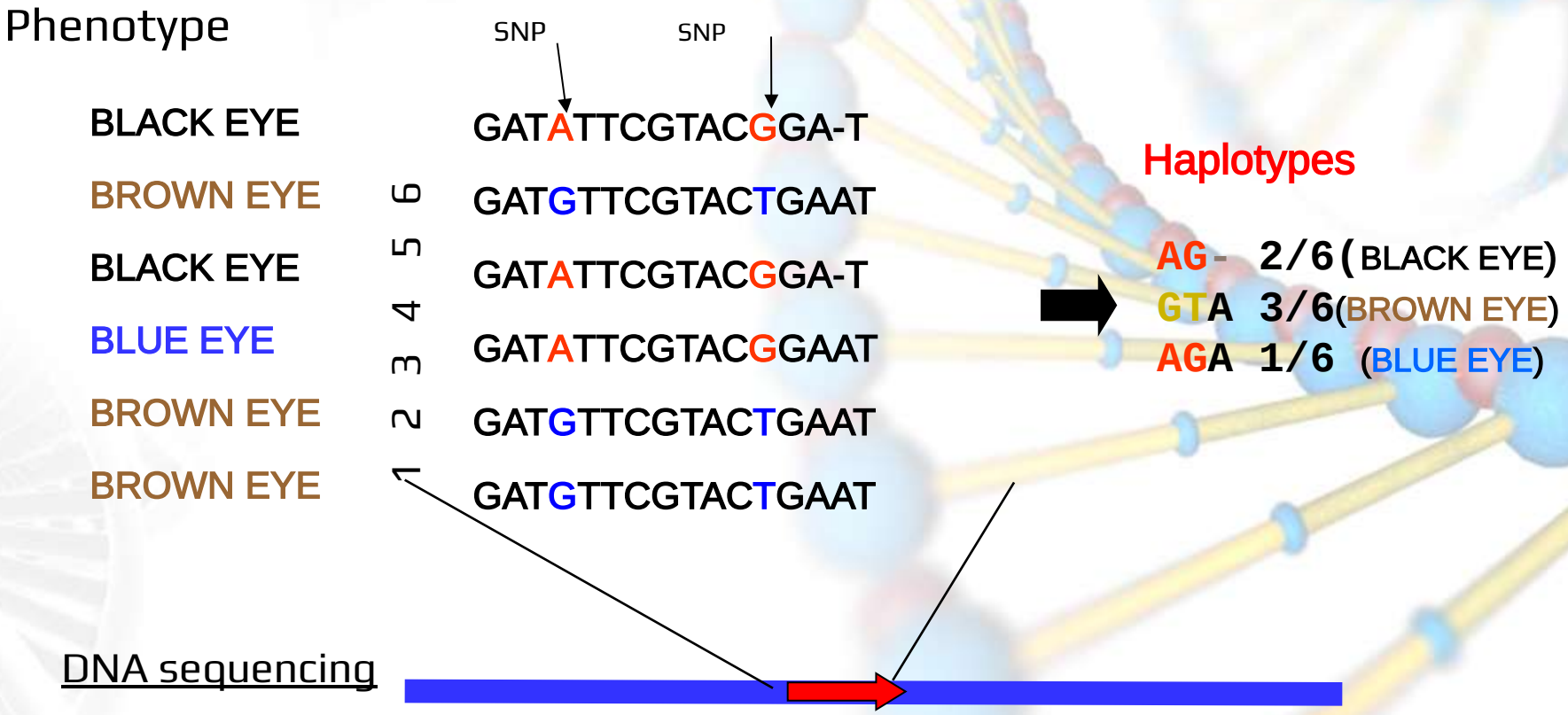
Module# 61

SNP-Haplotype

HAPLOTYPE

A set of closely linked genetic markers present on one chromosome which tend to be inherited together (not easily separable by recombination)

SNP-Haplotype



HAPLOTYPE CORRELATION WITH PHENOTYPE

- The “Haplotype centric” approach combines the information of adjacent SNPs into composite multilocus haplotypes
- Haplotypes are not only more informative but also capture the regional LD information, which is assumed to be robust and powerful
- Association of haplotype frequencies with the presence of desired phenotypic frequencies in the population will help in utilizing the maximum potential of SNP as a marker.

Some important SNP database Resources

1. dbSNP (http://www.ncbi.nlm.nih.gov/SNP/)

   LocusLink (http://www.ncbi.nlm.nih.gov/LocusLink/list.cgi)

2. TSC (http://snp.cshl.org/)

3. SNPper (http://snpper.chip.org/bio/)

4. JSNP (http://snp.ims.u-tokyo.ac.jp/search.html)

5. GeneSNPs (http://www.genome.utah.edu/genesnps/)

6. HGVbase (http://hgvbase.cgb.ki.se/)

7. PolyPhen (http://dove.embl-heidelberg.de/PolyPhen/)

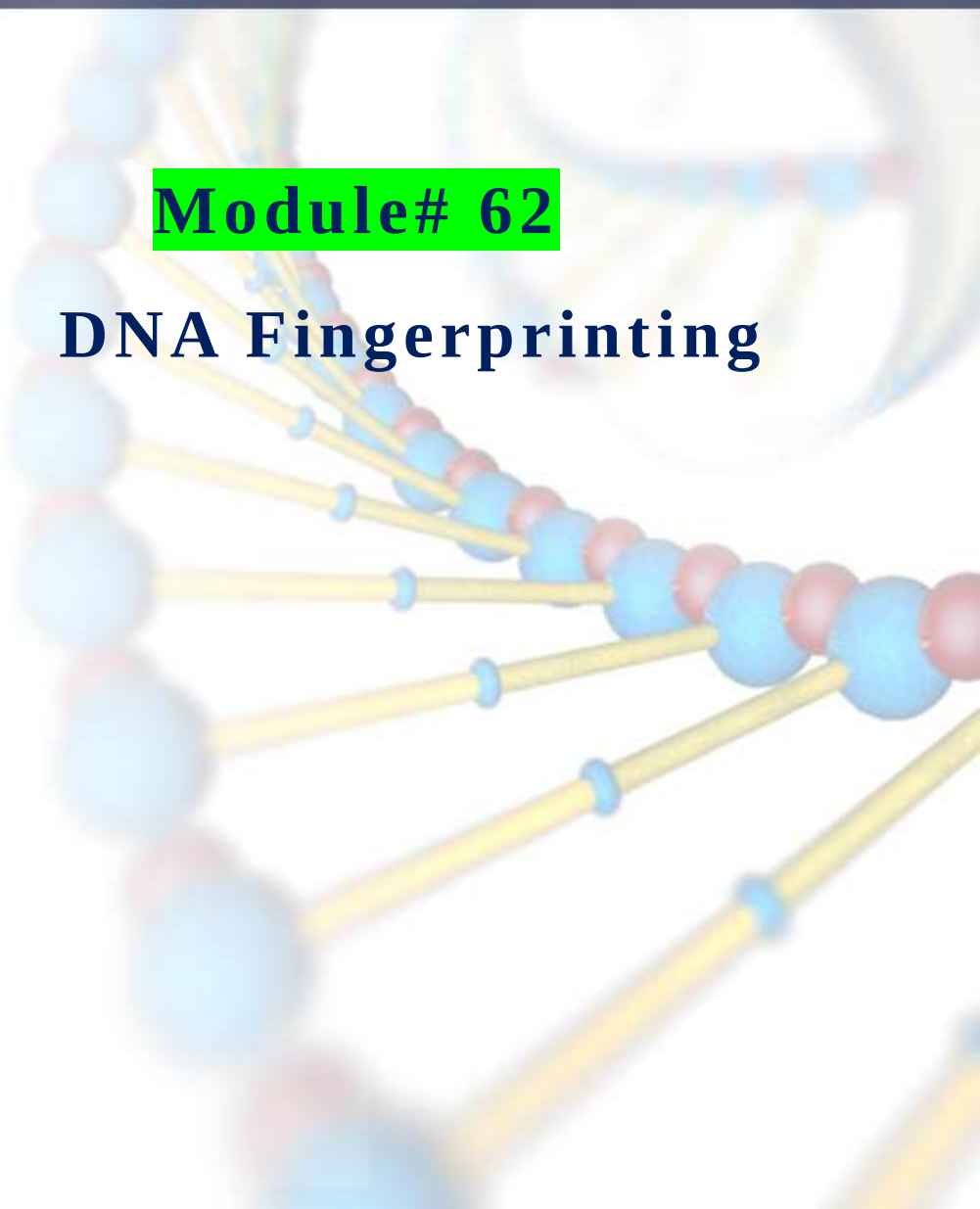
   OMIM (http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?db=OMIM)

8. Human SNP database

   (http://www-genome.wi.mit.edu/snp/human/)

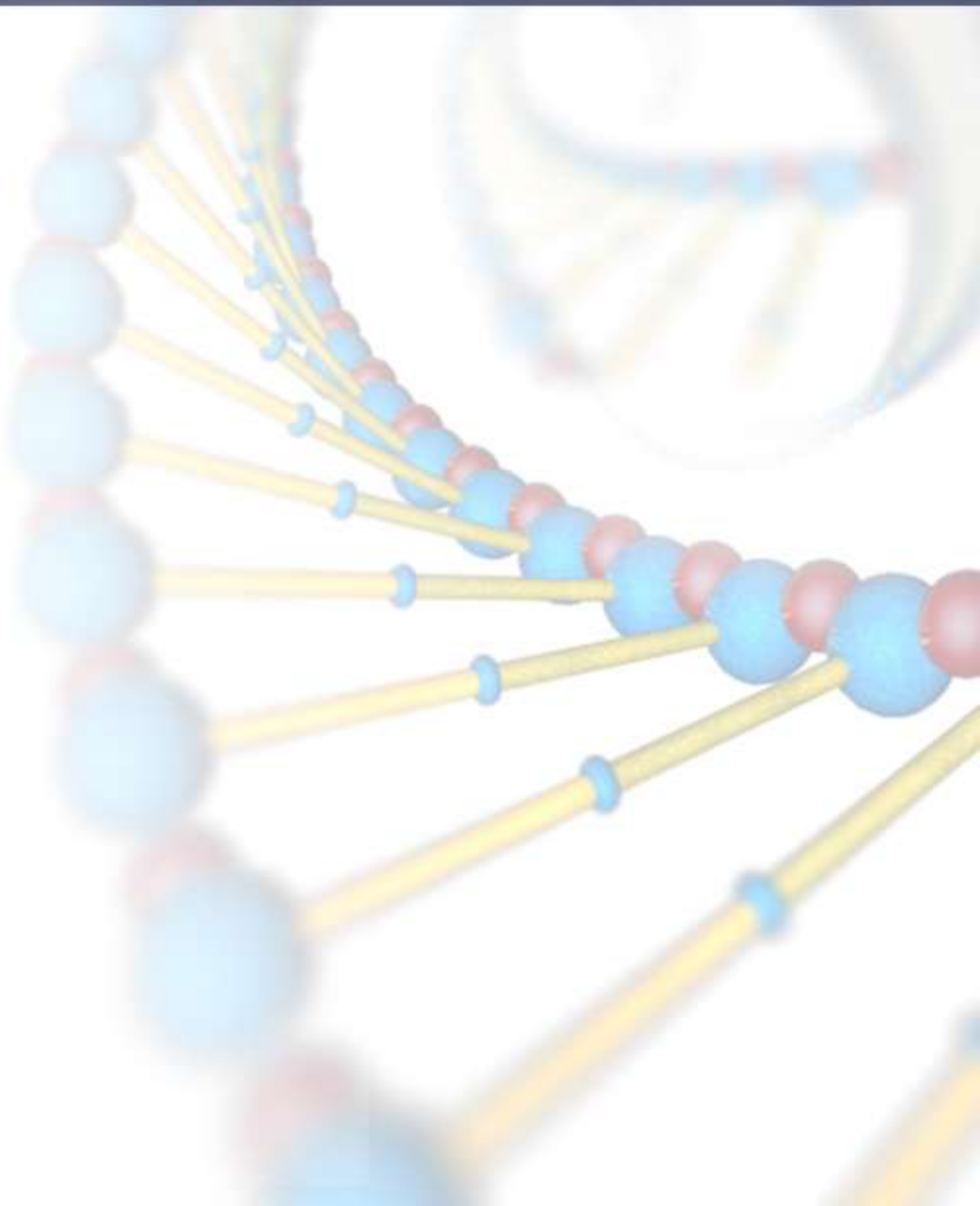
Module# 62

DNA Fingerprinting



Synonyms...

- DNA profiling.
- DNA testing.
- DNA typing.
- Genetic fingerprinting



Introduction

The process of DNA fingerprinting was developed by Professor Alec Jeffreys at Leicester University in 1984 as a form of genetic analysis. • It was first used in the law courts of England in 1987 to convict a man in a rape case. • It has now been used successfully in many crime and paternity cases in worldwide.



Although 99.9% of human DNA sequences are the same in every person, enough of the DNA is different to distinguish one individual from another, unless they are monozygotic twins.

- DNA profiling uses repetitive sequences that are highly variable, called variable number tandem repeats (VNTRs), particularly short tandem repeats (STRs). VNTR loci are very similar between closely related humans.
- The analysis of variable number of tandem repeats (VNTRs), to detect the degree of relatedness to another sequence of oligonucleotides , making them ideal for DNA fingerprinting.

Variable Number Tandem Repeats

A Variable Number Tandem Repeat (or VNTR) is a location in a genome where a short nucleotide sequence is organized as a tandem repeat.

These can be found on many chromosomes, and often show variations in length between individuals.

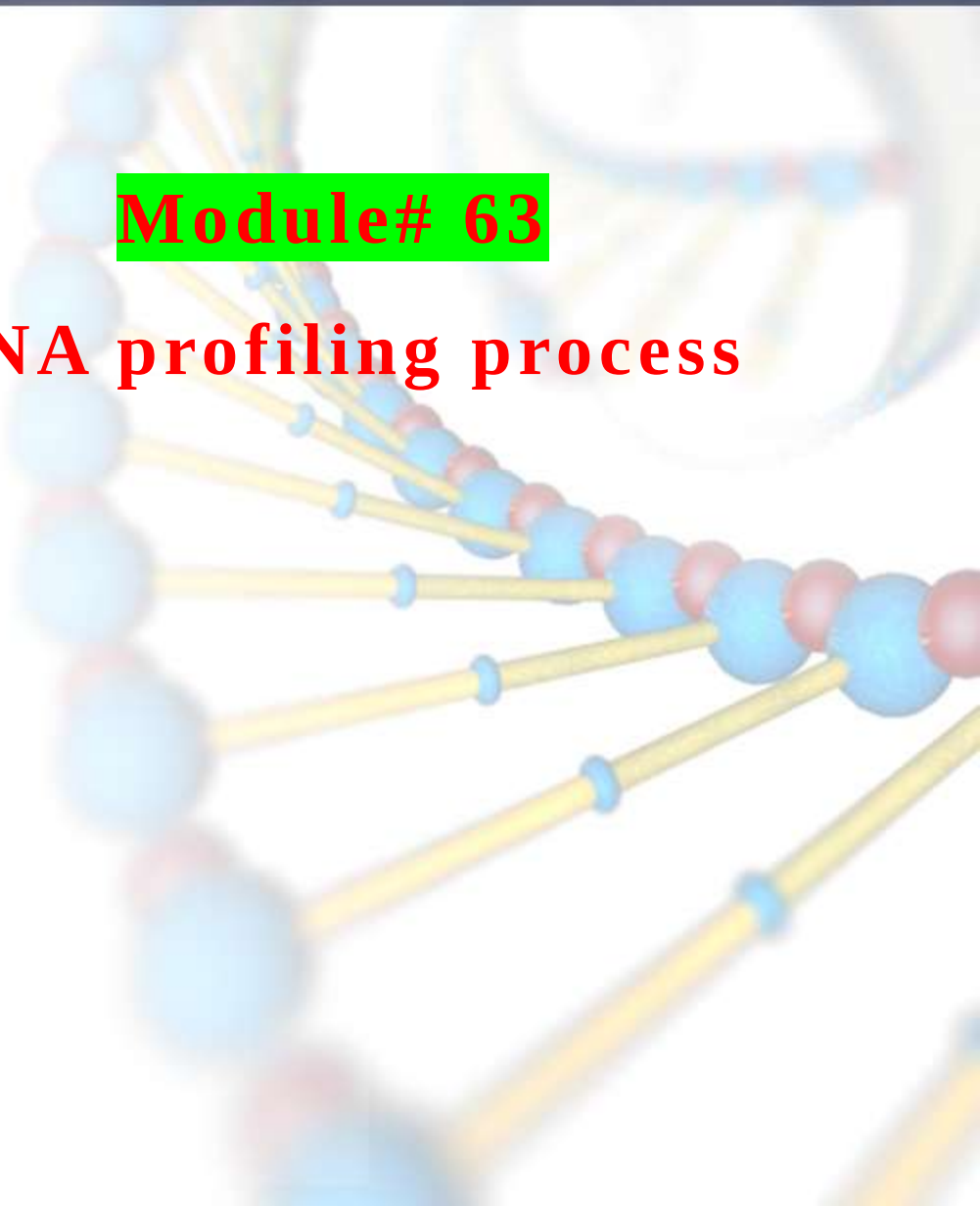
Each variant acts as an inherited allele, allowing them to be used for personal or parental identification.

There are two principal families of VNTRs:

- Microsatellites.
- Minisatellites. Microsatellites, also known as Simple Sequence Repeats (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA. • A minisatellites (also referred as VNTR) is a section of DNA that consists of a short series of bases 10–60 base pairs. • Their analysis is useful in genetics and biology research, forensics, and DNA fingerprinting.

Module# 63

DNA profiling process



The process begins with a sample of an individual's DNA (typically called a "reference sample"). • The most desirable method of collecting a reference sample is the use of a buccal swab, as this reduces the possibility of contamination. • When this is not available may need to be used to collect a sample of blood, saliva, semen, or other appropriate fluid or tissue from personal items (e.g. toothbrush, razor, etc.) or from stored samples (eg: banked sperm or biopsy tissue).

Procedure

1) Isolation of DNA :

- DNA must be recovered from the cells or tissues of the body.
- Only a small amount of tissue, like blood, hair, or skin, is needed.
- For example, the amount of DNA found at the root of one hair is usually sufficient.

2) Cutting, sizing, and sorting :

- Special enzymes called restriction enzymes are used to cut the DNA at specific sites.
- For example, an enzyme called EcoR1, found in bacteria, will cut DNA only when the sequence 5'..GAATTC..3' occurs.
- The DNA pieces are sorted according to size by a sieving technique called electrophoresis.
- The DNA pieces are passed through a gel agarose. This technique is the DNA equivalent of screening sand through progressively finer mesh screens to determine particle sizes.

3) Transfer of DNA to nylon :

The distribution of DNA pieces is transferred to a nylon sheet by placing the sheet on the gel and soaking them overnight.

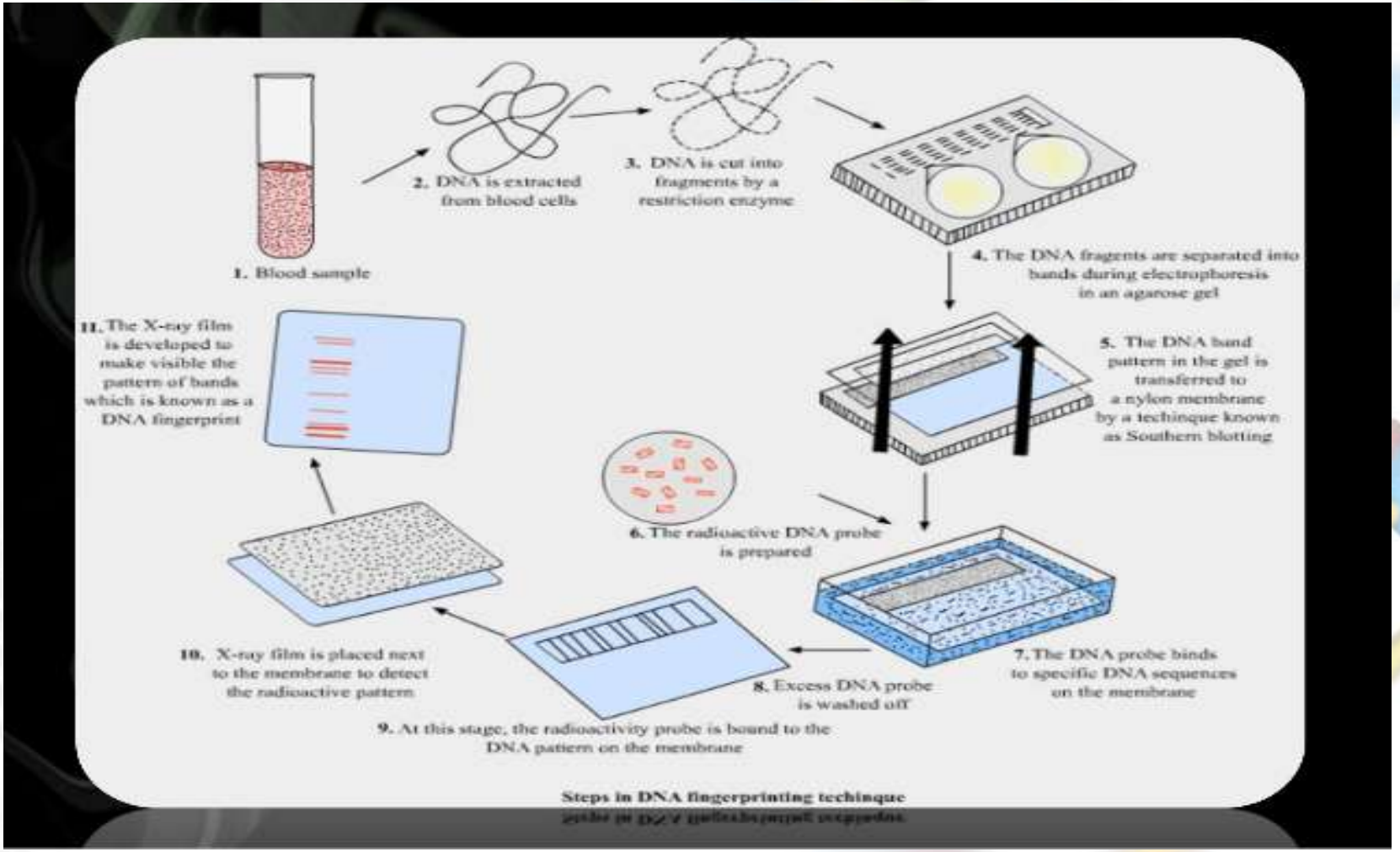
4) Probing :

Adding radioactive or colored probes to the nylon sheet produces a pattern called the DNA fingerprint.

- Each probe typically sticks in only one or two specific places on the nylon sheet.

5) DNA fingerprint :

The final DNA fingerprint is built by using several probes (5-10 or more) simultaneously. It resembles the bar codes used by grocery store scanners.



Steps in DNA fingerprinting technique

Module# 64

Types of DN fingerprinting methods

Types of DNA fingerprinting methods are…

- Electrophoresis.
- Polymerase chain reaction (PCR).
- Restriction fragment length polymorphism (RFLP).
- Random Amplified Polymorphic DNA(RAPD).
- Amplified fragment length polymorphism (AFLP).

Electrophoresis

Electrophoresis is a separations technique that is based on the mobility of ions in an electric field. Positively charged ions migrate towards a negative electrode and negatively-charged ions migrate toward a positive electrode. Ions have different migration rates depending on their total charge, size, and shape, and can therefore be separated.

Polymerase chain reaction…

chain reaction (PCR) was developed

- The polymerase by Karry Mullis of the Cetus Corporation in 1983.
- In this process, the DNA sample is denatured into the separate individual strands.
- Specific DNA primers are used to hybridize to two corresponding nearby sites on opposite DNA strands in such a fashion that the normal enzymatic extension of the active terminal of each primer (that is, the 3’ end) leads toward the other primer.
- In this fashion, two new copies of the sequence of interest are generated.

Repeated denaturation, hybridization, and extension in this fashion produce an exponentially growing number of copies of the DNA of interest.

- The PCR analysis amplified isolated regions on the strands of the DNA under examination.

Restriction fragment length polymorphism...

- RFLP analyzes the length of the strands of the DNA molecules with repeating base pair patterns.
- The basic technique for detecting RFLPs involves fragmenting a sample of DNA by a restriction enzyme, which can recognize and cut DNA wherever a specific short sequence occurs, in a process known as a restriction digest.
- The resulting DNA fragments are then separated by length through a process known as agarose gel electrophoresis, and transferred to a membrane via the Southern blot procedure.

Hybridization of the membrane to a labeled DNA probe then determines the length of the fragments which are complementary to the probe.

• An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

Module# 65

Types of DNA fingerprinting methods-2

Random Amplified Polymorphic DNA

Random Amplified Polymorphic DNA... of PCR reaction, but the segments of DNA

- It is a type that are amplified at random.
- RAPD creates several arbitrary, short primers (8–12 nucleotides), then proceeds with the PCR using a large template of genomic DNA, hoping that fragments will amplify.
- By resolving the resulting patterns, a semi-unique profile can be gleaned from a RAPD reaction.
- RAPD does not require any specific knowledge of the DNA sequence of the target organism: the identical 10-mer primers will or will not amplify a segment of DNA, depending on positions that are complementary to the primers' sequence.

ADVANTAGES OF RAPD

- It requires no DNA probes and sequence information for the design of specific primers.
- It involves no blotting or hybridization steps, hence, it is quick, simple and efficient.
- It requires only small amounts of DNA (about 10 ng per reaction) and the procedure can be automated.
- High number of fragments.
- Arbitrary primers are easily purchased.
- Unit costs per assay are low compared to other marker technologies.

DISADVANTAGES OF RAPD...

- Nearly all RAPD markers are dominant, i.e. it is not possible to distinguish whether a DNA segment is amplified from a locus that is heterozygous (1 copy) or homozygous (2 copies).
- Co -dominant RAPD markers, observed as different-sized DNA segments amplified from the same locus, are detected only rarely.
- PCR is an enzymatic reaction, therefore, the quality and concentration of template DNA, concentrations of PCR compo-nents, and the PCR cycling conditions may greatly influence the outcome. Thus, the RAPD technique is notoriously labo-ratory-dependent and needs carefully developed laboratory protocols to be repro-ducible.

Mismatches between the primer and the template may result in the total absence of PCR product as well as in a merely decreased amount of the product. Thus, the RAPD results can be difficult to interpret.

- Lack of a prior knowledge on the identity of the amplification products.
- Problems with reproducibility (sensitive to changes in the quality of DNA, PCR components and PCR conditions).
- Problems of co-migration. Gel electrophoresis can separate DNA quantitatively, cannot separate equal-sized fragments qualitatively (i.e. according to base sequence).

Amplified fragment length polymorphism... RFLP

- This technique was also faster than analysis and used PCR to amplify DNA samples. It relied on variable number tandem repeat (VNTR) polymorphisms to distinguish various alleles, which were separated on a polyacrylamide gel.
- By using the PCR analysis to amplify the minisatellite loci of the human cell, this method proved quicker in recovery than the RFLP.
- However, due to the use of gel in its analysis phase, there are issues of bunching of the VTRN's, causing misidentifications in the process.

Module# 66

Applications of Fingerprinting

Applications...

1) Diagnosis and Developing cures for inherited disorders :

- DNA fingerprinting is used to diagnose inherited disorders in both prenatal and newborn babies in hospitals around the world.
- These disorders may include cystic fibrosis, hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and many others.
- Early detection of such disorders enables the medical staff to prepare themselves and the parents for proper treatment of the child.
- In some programs, genetic counselors use DNA fingerprint information to help prospective parents

In other programs, prospective parents use DNA fingerprint information in their decisions concerning affected pregnancies.

2)Biological Evidence to Identify Criminals:

- Where fingerprints are not available but biological specimens are available like blood or semen stains, hair, or items of clothing at the scene of the crime then these items may prove to be valuable sources of DNA of the criminal.
- Since the year 1987, innumerable cases have been solved with the help of DNA fingerprint evidence.

3) Paternity disputes :

• Another important use of DNA fingerprints in the court system is to establish paternity in custody and child support litigation. In these applications, DNA fingerprints bring an unprecedented, nearly perfect accuracy to the determination.

4) Personal Identification :

• DNA maybe the best way to identify a person as all body tissues and organs contain the same DNA type. The specimen required also is very small. In fact the US army has been doing DNA fingerprinting of all its soldiers and has a huge databank.

Module # 067

Making Antibodies: Polyclonal

Antibodies

- An **antibody (Ab)**, also known as an **immunoglobulin (Ig)**, is a large, Y-shaped protein used by the immune system to identify and neutralize foreign objects such as pathogenic bacteria and viruses.
- The antibody recognizes a unique molecule of the pathogen, called an antigen.
- Each tip of the "Y" of an antibody contains a paratope (analogous to a lock) that is specific for one particular epitope (analogous to a key) on an antigen, allowing these two structures to bind together with precision.

Polyclonal and Monoclonal Antibodies

- **Polyclonal antibodies** (pAbs) are mixture of heterogeneous antibodies which are usually produced by different B cell clones in the body. They can recognize and bind to many different epitopes of a single antigen.
- **Monoclonal antibodies (mAbs)** are generated by identical B cells which are clones from a single parent cell. This means that the monoclonal antibodies have monovalent affinity and only recognize the same epitope of an antigen.

Making Antibodies: Polyclonal

MAKING ANTIBODIES

Antibodies can be made in various ways. The choice of which method to use is dependent on final assay format

Making Antibodies: Polyclonal

Qualities of Antibody

(to determine its usefulness in the method to be used)

- defined specificity
- affinity and avidity
  - Antibody affinity is defined as **strength of the binding interaction between antigen and antibody.**
  - Antibody avidity **describes the strength with which an antibody binds to a complex antigen**.

Making Antibodies: Polyclonal

POLYCLONAL ANTIBODIES

These are raised in appropriate donor animals

- Generally rabbits for smaller amounts
- Sheep or goats for larger quantities

Making Antibodies: Polyclonal

Occasionally rats or mice can be used for small research quantities of antibody

Making Antibodies: Polyclonal

ADJUVANTS

Substances which increase the immunogenicity of the antigen

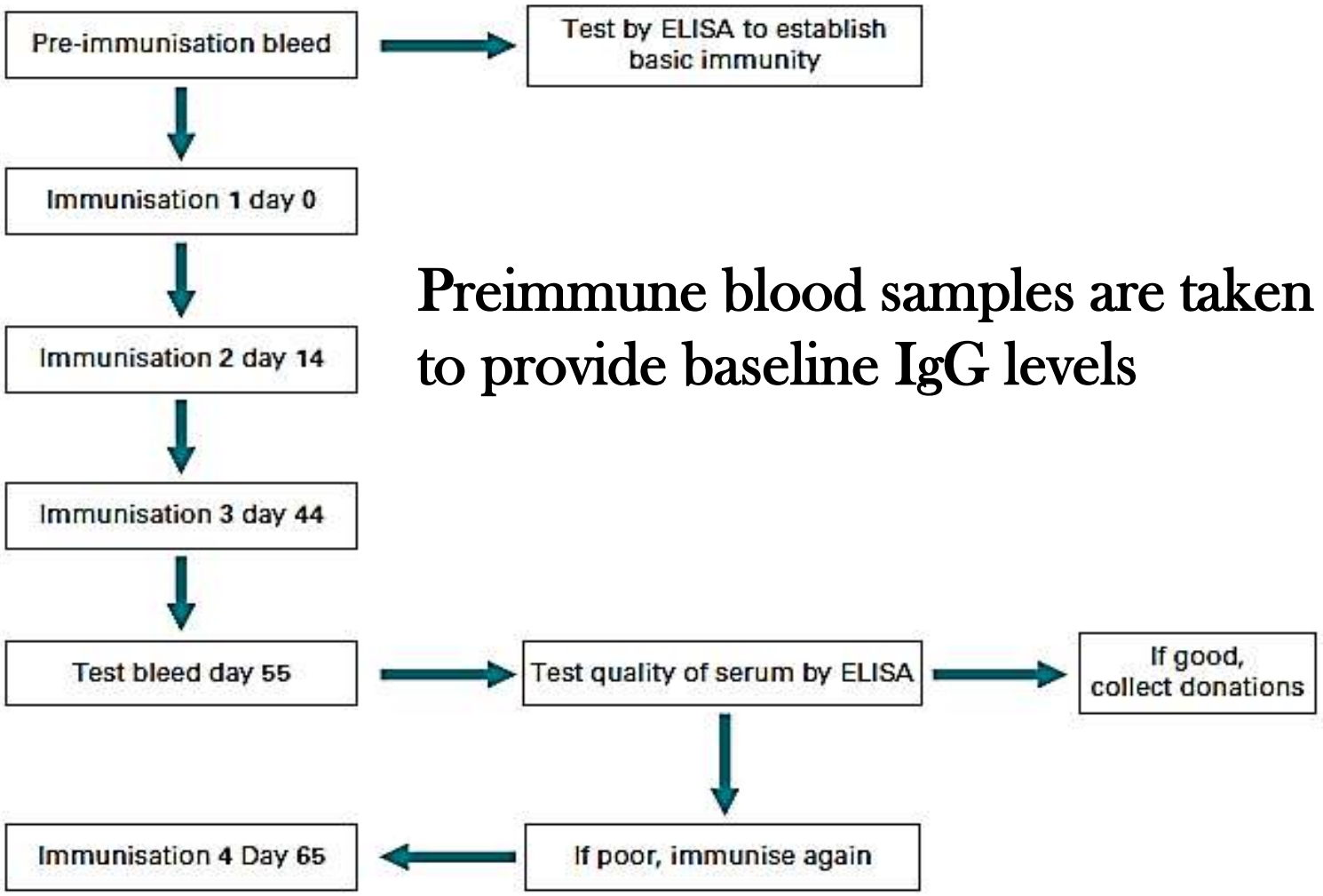
These are used to reduce the amount of antigen required

Making Antibodies: Polyclonal

EXAMPLE OF ADJUVANTS

Adjuvants may be chemicals (detergents and oils) Complex proprietary products (containing bacterial cell walls)

Making Antibodies: Polyclonal



Preimmune blood samples are taken to provide baseline IgG levels

Making Antibodies: Polyclonal

Usually a range of doubling serum dilutions are made (1/100–1/12 800) and tested against the antigen

Making Antibodies: Polyclonal

Serum from a satisfactory course of immunisations will detect antigen at 1/6400 dilution

This indicates the high levels of circulating antibody

Making Antibodies: Polyclonal

Once high level of circulating antibody is detected in test bleeds then donations can be taken

Making Antibodies: Polyclonal

- Blood donations are allowed to clot and the serum collected
- Serum can be stored at 4° C or lower for longer periods

Making Antibodies: Polyclonal

It is also possible to produce antibodies in chicken eggs

Making Antibodies: Polyclonal

Immunoglobulin Y (IgY)

Avian immunoglobulin is known as immunoglobulin Y (IgY) and chickens secrete it into eggs to provide protection for the developing embryo

Making Antibodies: Polyclonal

This can be utilised for effective polyclonal antibody production

Making Antibodies: Polyclonal

- Chickens are immunized three or four times with the antigen
- Eggs are collected and can yield up to 50 mg antibody per yolk

Making Antibodies: Polyclonal

The antibody has to be purified from the egg yolks

Making Antibodies: Polyclonal

Small quantities of very pure polyclonal antibodies can be produced in rats and mice in ascitic fluid

Making Antibodies: Polyclonal

ASCITES

It is a mammalian response to a tumour within the peritoneum

Making Antibodies: Polyclonal

Animals are immunised with the antigen of interest

Once a high serum level is detected then ascitic fluid production is induced

Making Antibodies: Polyclonal

NON-SECRETORY MYELOMA CELLS

Such as NS-0 are introduced into the peritoneal cavity of the animal by injection and allowed to grow there

Making Antibodies: Polyclonal

END

TUMOR CELLS

The presence of the tumour cells causes the animal to produce ascitic fluid which contains high levels of immune- globulins to the original antigen

Module # 068

Monoclonal Antibodies

Monoclonal

MONOCLONAL ANTIBODY

Mice are usually the animals of choice for monoclonal antibody production

Monoclonal

BALB/C

- Ausual mouse strain
- Used for monoclonal antibody production

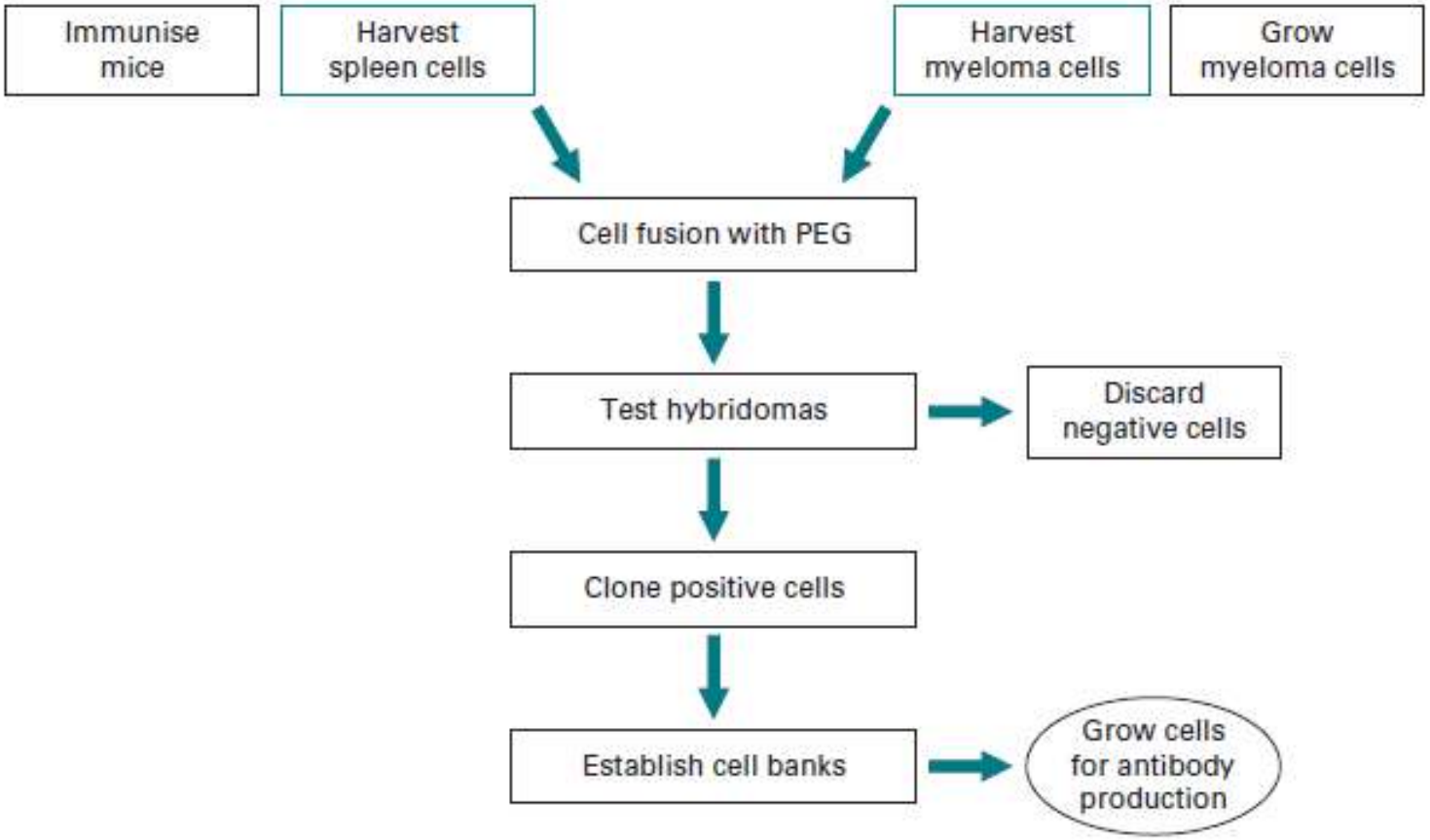
Monoclonal

Mice are immunised, usually three or four times over the course of 3–4 months, by the intraperitoneal route using antigen mixed with adjuvant.

Antigen: a toxin or other foreign substance which induces an immune response in the body, especially the production of antibodies.

Adjuvant: a substance which enhances the body's immune response to an antigen.

Monoclonal



Monoclonal antibody production

Monoclonal

Once the mice are sufficiently immune they are left for 2–3 months to ‘rest’

Monoclonal

Mice are sacrificed and the spleens removed

Single spleen will provide sufficient cells for two or three cell fusions

Monoclonal

EXTRACTION OF MOUSE SPLEEN CELLS

Extract mouse spleen cells (containing B-lymphocytes) and culture them in the lab

Monoclonal

Extract Mouse Tumor Cells Which grow continuously, and culture them later on.

Monoclonal

Mix spleen cells and tumor cells on the same plate and culture

Monoclonal

Add polyethylene glycol

This causes some

B-lymphocytes to fuse with tumour cells to produce a hybrid cell called a hybridoma

Monoclonal

Grow the cells
Under conditions
that allow only
hybridoma cells to
survive

Monoclonal

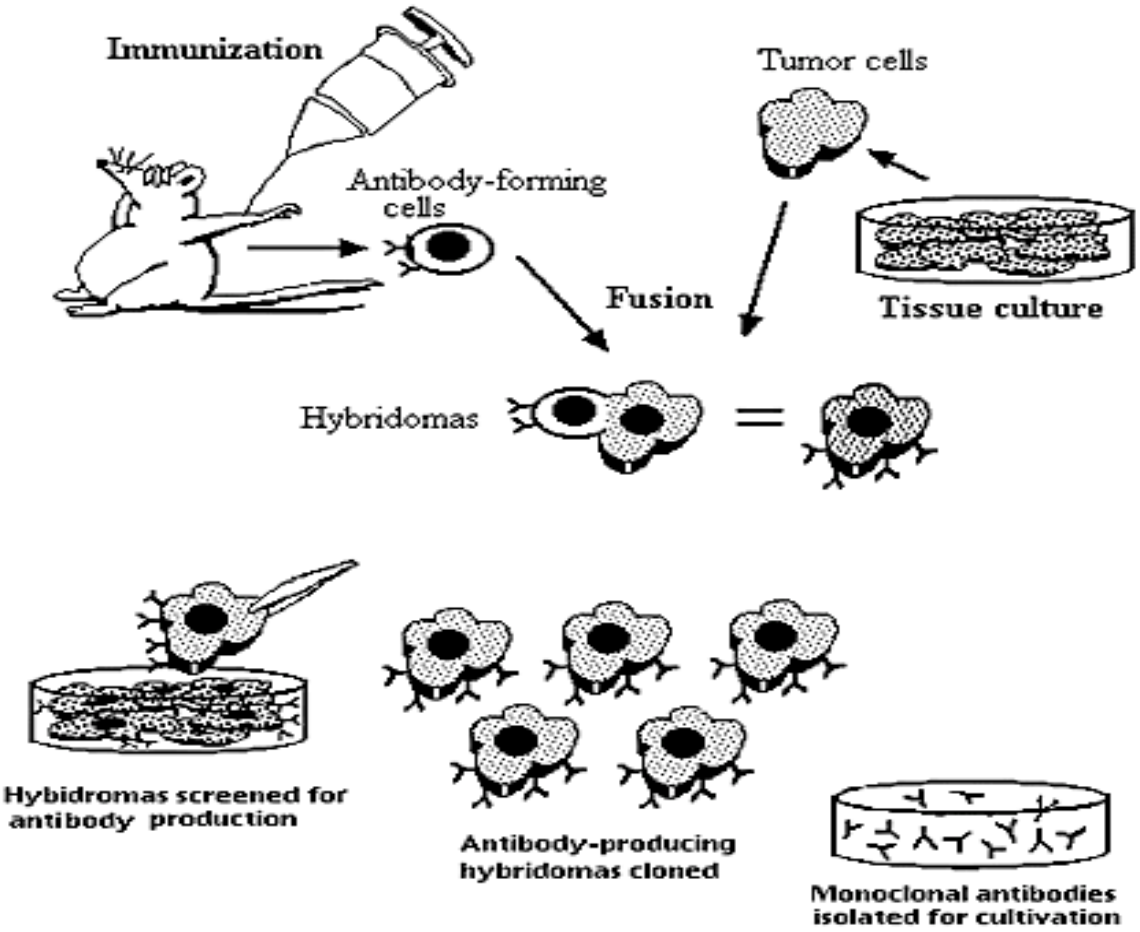
Extract the cells,
culture them
separately

Monoclonal

Test the medium

Around each cell for the specific antibody of interest

Monoclonal



Monoclonal Antibody Production

Monoclonal

Culture the cells

Culture the cell, making the desired antibody and use as needed

END

Module # 069

Freezing Cells and Cell Banking

Freezing Cells & Cell Banking

- Cell lines are cultures of animal cells that can be propagated repeatedly and sometimes indefinitely.
- They arise from primary cell cultures. Primary cultures are initiated directly from the cells, tissues, or organs of animals and are typically used in experiments within a few days.

Freezing Cells & Cell Banking

Cell lines are frozen to provide a source of inoculum for future cultures

Freezing Cells & Cell Banking

Cells stored at the low temperatures achieved using liquid nitrogen vapour are stable for many years

Freezing Cells & Cell Banking

Cells Transferred into Specialist Medium prior to freezing to protect them both when the temperature is lowered and also as the temperature is raised when thawed

Freezing Cells & Cell Banking

Serum containing 10% DMSO

Works well as a freezing medium

Freezing Cells & Cell Banking

A typical freezing should contain around 1×10^6 cells

Freezing Cells & Cell Banking

The cells are harvested from the flask by tapping to dislodge them

Pelleted by centrifugation to remove the culture medium

Freezing Cells & Cell Banking

Cells are Resuspended

In 1.0 ml freezing medium chilled to 4° C, placed into a cryogenic vial and transferred to a cell freezing container

Freezing Cells & Cell Banking

Butan-1-ol

Freezing container contains butan-1-ol which when placed into a -70° C freezer controls the rate of freezing to 1° C per minute

Freezing Cells & Cell Banking

The cells are left for a minimum of 24 h and a maximum of 72 h prior to transfer into cryogenic storage

Freezing Cells & Cell Banking

Transfer to Liquid Nitrogen Storage

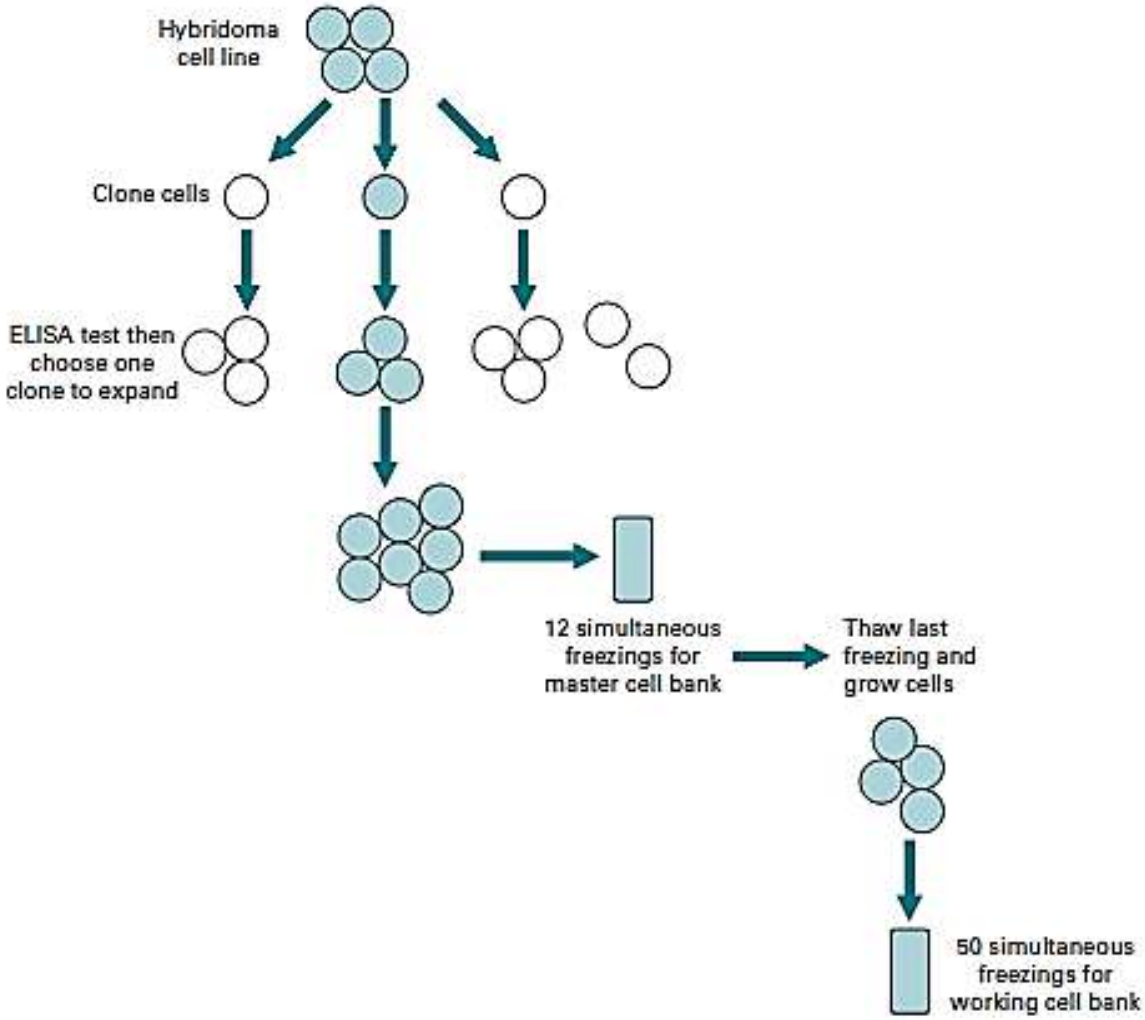
It should be rapid, to prevent thawing of the cells

Freezing Cells & Cell Banking

Cell Banking

Cell banks are established from known positive clones and are produced in a way that maximizes reproducibility between frozen cell stocks and minimizes the risk of cellular change

Freezing Cells & Cell Banking



Master and working cell banks

Freezing Cells & Cell Banking

A positive clone derived from a known positive clone is rapidly expanded in tissue culture

Freezing Cells & Cell Banking

This is the master cell bank and is stored at -196° C under liquid nitrogen vapour

Freezing Cells & Cell Banking

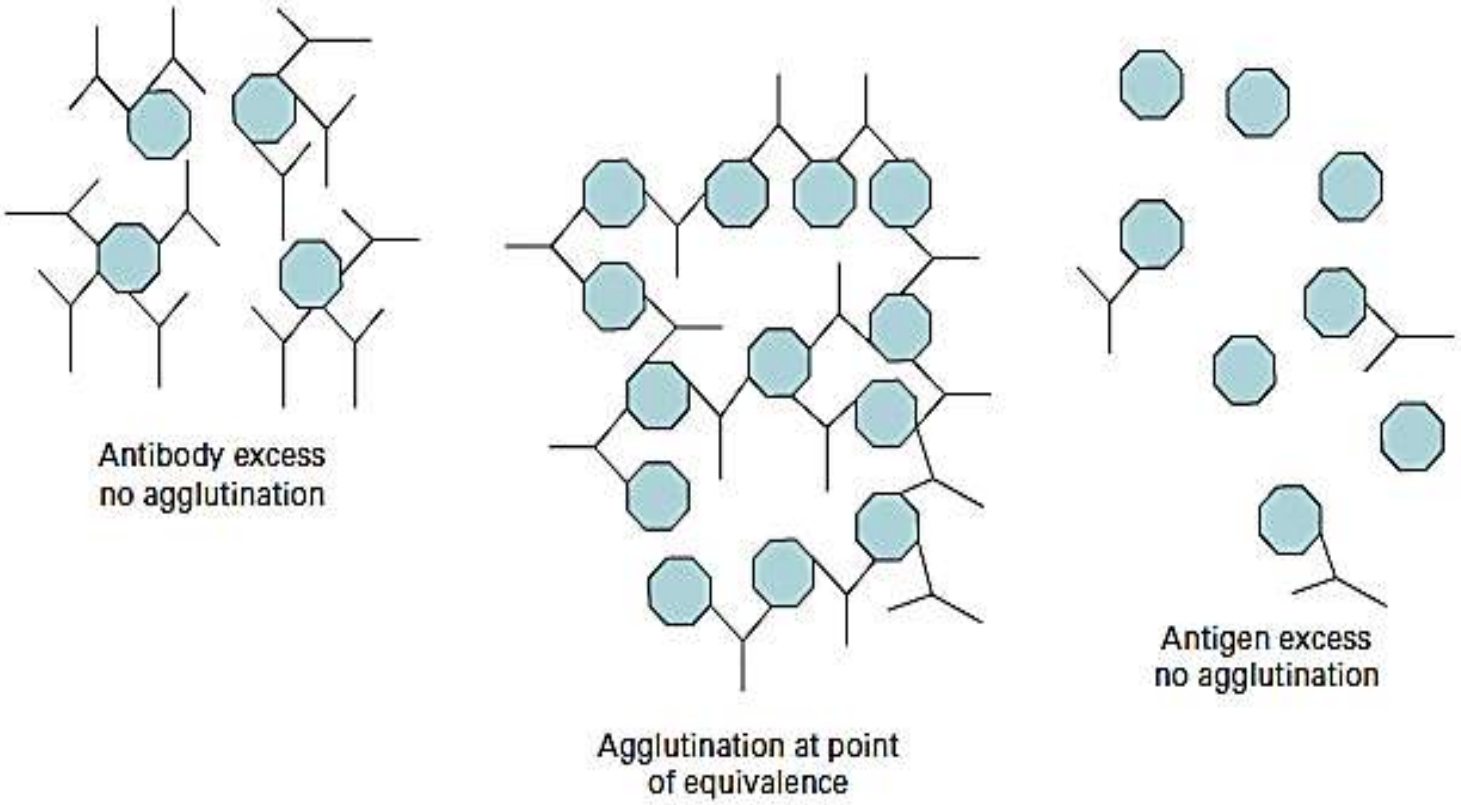
This strategy ensures that all of the vials of the working cell bank are identical

Module # 070
Enzyme Immunosorbent Assays

Enzyme Immunosorbent Assays

The initial immunoassay formats described were methods based on the agglutination reaction

Enzyme Immunosorbent Assays



Enzyme Immunosorbent Assays

The reaction between antibody and antigen can be observed when agglutination occurs

Enzyme Immunosorbent Assays

the vast majority of immunoassays carried out fall into the category of enzyme immunosorbent assays

Enzyme Immunosorbent Assays

Routinely used for the diagnosis of infectious agents such as viruses, and other substances in blood

Enzyme Immunosorbent Assays

The antigen is the substance or agent to be measured

Enzyme Immunosorbent Assays

IMMUNO-SORBENT ASSAY (ELISA) PLATE

The most commonly used solid phase is the enzyme linked immunosorbent assay (ELISA) plate

Enzyme Immunosorbent Assays

Immobilisation is achieved by the use of a coating antibody which actively traps antigen to the solid phase

Enzyme Immunosorbent Assays

Reporter Enzyme

A second antibody which is labelled with a reporter enzyme is allowed to bind to the immobilised antigen

Enzyme Immunosorbent Assays

Enzyme Substrate

Is then added to the antigen/ antibody/ enzyme complex and a reaction

Enzyme Immunosorbent Assays

There are many permutations of this method but all of them rely on the antibody–antigen complex being formed and the presence of it being confirmed by the reactions of the reporter enzyme

Enzyme Immunosorbent Assays

The antigen is central to the assay as it provides the bridge between the solid phase and the signal generating molecule

Enzyme Immunosorbent Assays

The coating antibody also concentrates the antigen from the sample as it binds the antigen irreversibly

END

Module # 071
Triple Antibody Sandwich ELISA

Triple Antibody Sandwich ELISA

- Also known as indirect ELISA

Triple Antibody Sandwich ELISA

It is often used to identify antibodies in patient blood

This may be signof infection

Triple Antibody Sandwich ELISA

With other immunoassays, layers of reagents are built up, each dependent on the binding of the previous one

Triple Antibody Sandwich ELISA

System is used to test patient blood for the presence of hepatitis B virus (HBV) antibodies

Triple Antibody Sandwich ELISA

HBV coating antibody

It is bound to the wells of a microtitre plate and HBV coat protein added to them

Triple Antibody Sandwich ELISA

The live virus is not used as antigen as this would be too dangerous to use in the laboratory

Triple Antibody Sandwich ELISA

HBV coat protein

It is made synthetically for use as antigen in this type of test

Triple Antibody Sandwich ELISA

- After incubation and washing, patient's serum is added
- If it contains

antibodies it reacts to the antigen

Triple Antibody Sandwich ELISA

Anti-human antibody

Conjugated to an enzyme marker and then added which will bind to the patient antibodies

Triple Antibody Sandwich ELISA

Substrate is then added to identify samples which were positive

Triple Antibody Sandwich ELISA

The test works well for the diagnosis of HBV infection

Triple Antibody Sandwich ELISA

Another use of Test

To ensure that blood donations given for transfusion are free from this virus

END

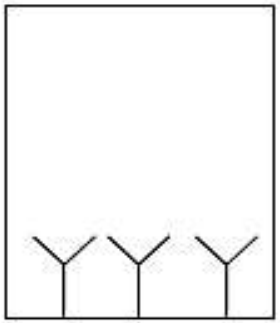
Module # 072

Double Antibody Sandwich ELISA

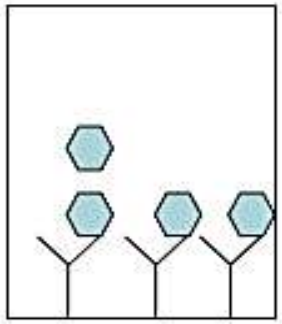
Double Antibody Sandwich ELISA

THE MOST WIDELY USED IMMUNOCHEMICAL TECHNIQUE IN DIAGNOSTICS

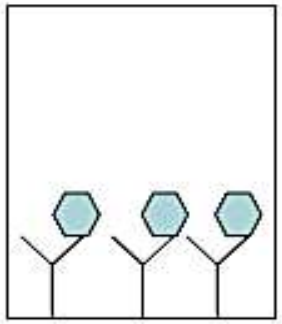
Double Antibody Sandwich ELISA



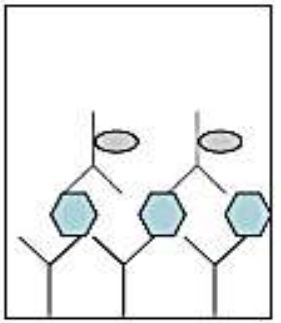
ELISA plate coated with antibody



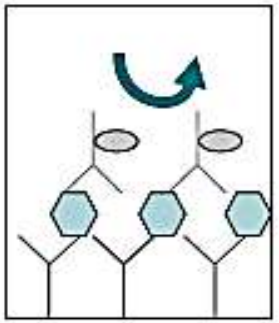
Sample incubated on plate



Antigen trapped by antibody



Antibody/enzyme conjugate incubated on plate



Substrate added to plate causing colour change in positive wells

DAS ELISA

Double Antibody Sandwich ELISA

This technique is:

- RAPID
- ROBUST
- RELIABLE

Double Antibody Sandwich ELISA

The principle is the same as other ELISA techniques

Double Antibody Sandwich ELISA

PRINCIPLE

- The antigen is immobilised to a solid phase by a primary antibody
- Detected with a second antibody which has been labelled with a marker enzyme

Double Antibody Sandwich ELISA

The antigen creates a bridge between the two antibodies

Double Antibody Sandwich ELISA

Change in Colour of Chromogenic Substrate

Presence of the enzyme causes a colour change in the chromogenic (colour-producing) substrate

Double Antibody Sandwich ELISA

Marker Enzyme used is:

- Horseradish peroxidase (HRP)
- Alkaline phosphatase (AP)

Double Antibody Sandwich ELISA

IRMA

In some systems the enzyme is replaced with a radioactive label and this format is known as IMMUNE-RADIOMETRIC ASSAY (IRMA)

Double Antibody Sandwich ELISA

USE OF DAS ELISA

In horticulture

and agriculture

- To ensure that plant material is free of virus

Double Antibody Sandwich ELISA

There are many potato viruses but potato leafroll virus (PLRV) in particular causes considerable problem

Double Antibody Sandwich ELISA

PLRV Antibiodies

Are coated onto the wells of ELISA plates and then the sap to be tested is added

Double Antibody Sandwich ELISA

After incubation, the plates are washed

PLRV antibody conjugated to alkaline phosphatase is added

Merge by Baqir Hussain BS ZOOLOGY

03336361244 Date - 04/09/2024

Double Antibody Sandwich ELISA

The plates are incubated and after washing, substrate is used to identify the positive wells

Double Antibody Sandwich ELISA

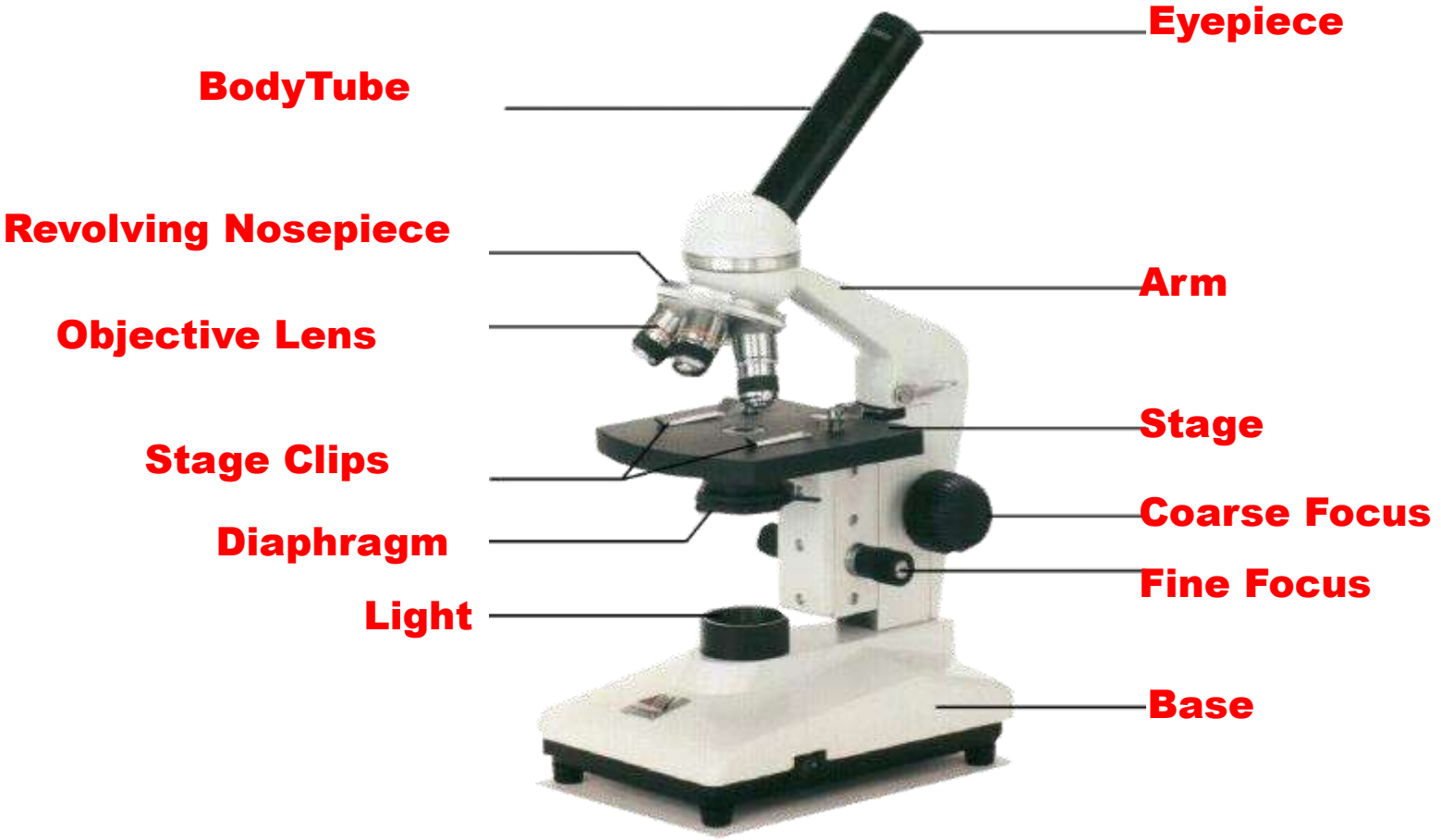
The system again requires the presence of antigen (PLRV) for the sandwich of antibodies to be built up

END

Module # 073 WHAT IS A MICROSCOPE?

- A microscope is an optical instrument used to view small objects by enlarging them with two convex lenses.
- Microscopes use lenses to magnify objects
- Most photographs of cells are taken using a microscope, and these pictures can also be called **micrographs.**

Parts of a Compound Microscope:



Parts of a Compound Microscope:

Eyepiece (or Ocular lens): Contains a 10X lens. What you look through.

The body tube goes from the eyepiece to the lenses.

Arm: Connects the base and the upper part of the microscope.

Nosepiece: Part that contains the lenses

The low power (10X) objective lens. Used to magnify the image.

High power objective lens. The magnification is 40X.

Course adjustment: Used to focus the image under low power.

Fine adjustment: used to focus the image under high power.

Stage: Where you place the slide or specimen. There is an opening for light to pass through to the eyepiece.

Stage clips: Hold the slide containing the specimen on the stage.

Diaphragm. Used to adjust the amount of light going through the specimen.

Light source. A lamp or mirror is used to direct light from the base through the hole in the stage and through the specimen.

Base: The bottom support of the microscope. Always hold this when carrying the microscope.

Working Principle

- The most commonly used microscope for general purposes is the standard compound microscope. It magnifies the size of the object by a complex system of lens arrangement.
- It has a series of two lenses; (i) the objective lens close to the object to be observed and (ii) the ocular lens or eyepiece, through which the image is viewed by eye. Light from a light source (mirror or electric lamp) passes through a thin transparent object.
- The objective lens produces a magnified 'real image' first image) of the object. This image is again magnified by the ocular lens (eyepiece) to obtain a magnified 'virtual image' (final image), which can be seen by eye through the eyepiece. As light passes directly from the source to the eye through the two lenses, the field of vision is brightly illuminated. That is why; it is a bright-field microscope.

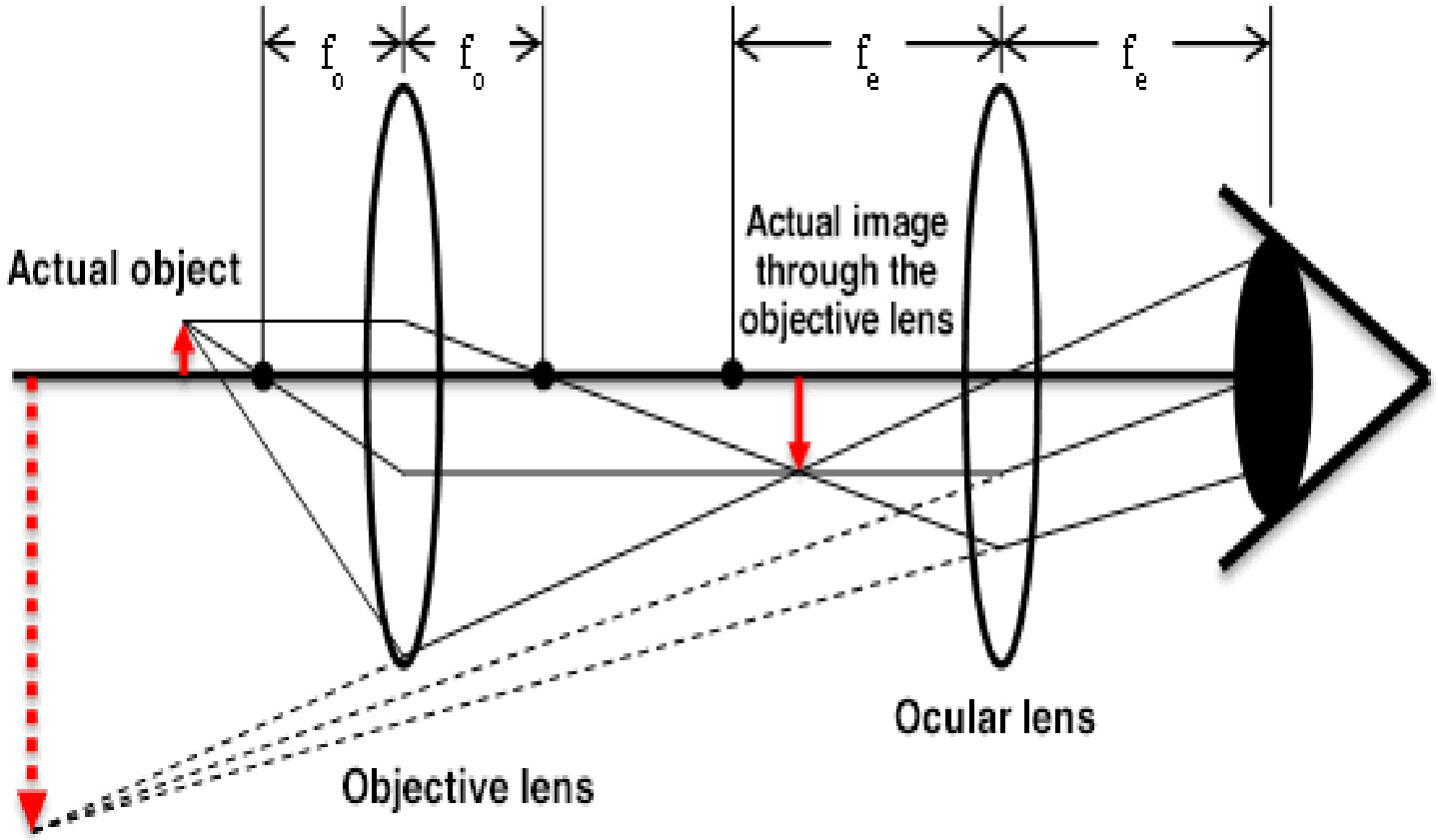


Image observed through the ocular lens
(virtual image magnified by the ocular lens)

Magnification and Resolution

- **Magnification** is the ability to make small objects seem larger, such as making a microscopic organism visible.
- **Resolution** is the ability to distinguish two objects from each other.
- **Resolution** is the shortest distance between two points that can still be visually distinguished as separate.
- The resolution of a typical unaided human eye is about 200 µm. Using a microscope decreases the resolution to distances as short as 0.2 µm.

MAGNIFICATION

- Three objectives with different magnifying powers are screwed to the revolving nosepiece.
  - **(a) Low power objective (X 10)**
  - **(b) High dry objective (X 40)**
  - **(c) Oil-immersion objective (X100)**
- Each microscope is provided with two types of eyepieces with different magnifying powers (X10 and X25).
- **Total magnification:**

The total magnification obtained in a compound microscope is the product of objective magnification and ocular magnification.

$M_t = M_{ob} \times M_{oc}$

Where,

M_t = Total magnification, M_{ob} = Objective magnification and M_{oc} = Ocular magnification

Difference between Simple and Compound Microscope

S.N.	Simple Microscope	Compound Microscope
1	Single lens	3 to 5 objective lenses
2	Has only one lens for magnifying objects.	Has two sets of lenses for magnifying objects: eyepiece lens and objective lenses
3	Low Magnification	Higher magnification.
4	Condenser lens is absent	Condenser lens is present which is used to adjust the intensity of light for magnification of object.
5	Light source is natural	Illuminator is a source of light
6	Has only one adjustment screw that is used to move the limb up and down for focusing an object.	Has coarse adjustment screw (for rapid focusing an object) and fine adjustment screw (for fine and sharp focusing).
7	Can only be used in simple ways such as enlarging small letters while reading.	Has a wide range of use such as in studying the structure of different objects

Types of Microscopy

- Simple Microscope
- Compound Microscope
- Dark Field Microscope
- Phase Contrast Microscope
- Flourescent Microscope
- Interference Microscope
- Electron Microscope
- Atomic Force Microscope
- Polarization Microscope

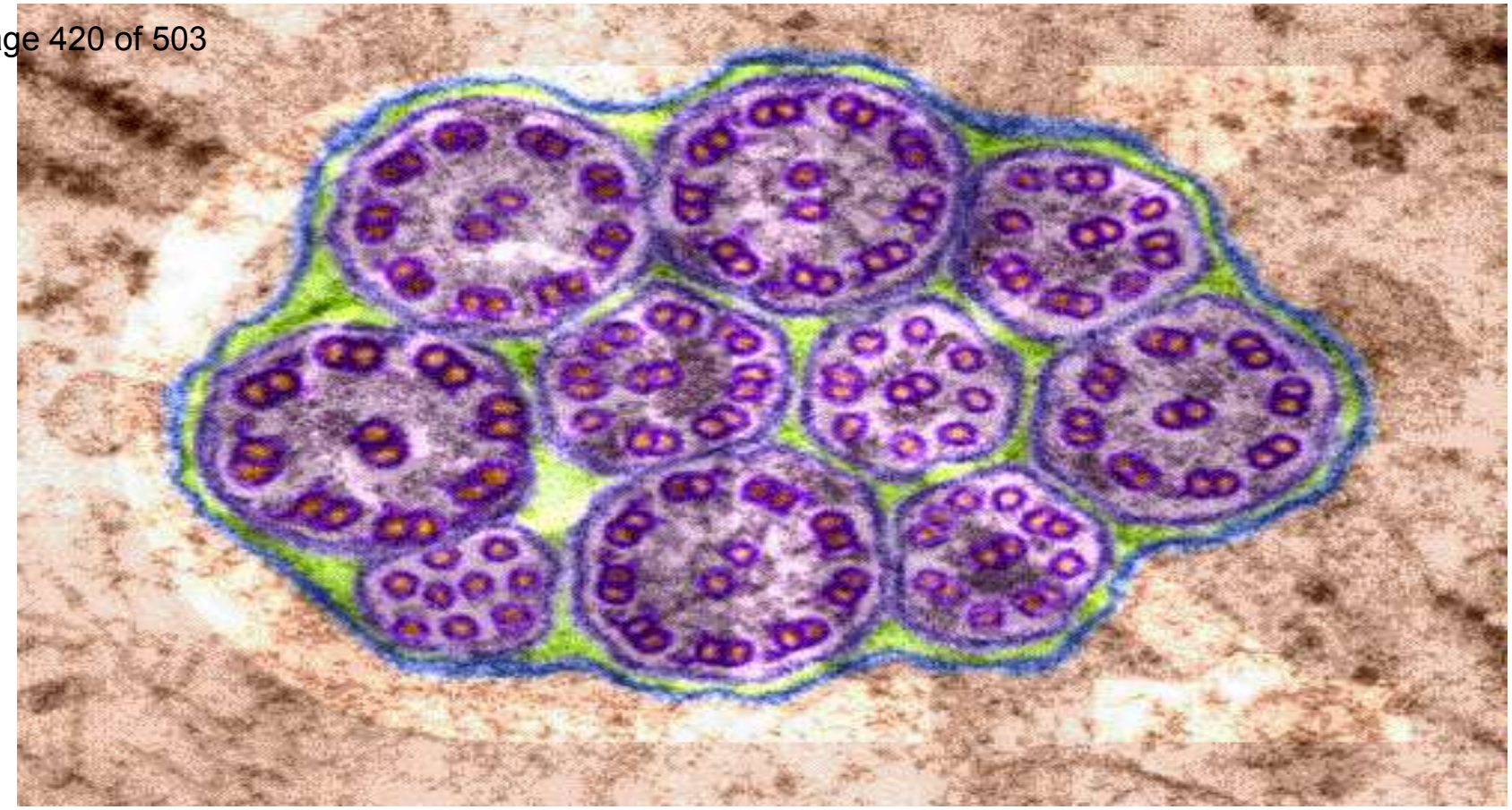
Types of Microscopy

- Compound Light Microscope
  - 1st type of microscope, most widely used
  - light passes through 2 lenses
  - Can magnify up to 2000x
- Electron Microscope
  - Used to observe VERY small objects: viruses, DNA, parts of cells
  - Uses beams of electrons rather than light
  - Much more powerful

Types of Microscopy

- Transmission Electron Microscope (TEM)
  - –Can magnify up to 250,000x





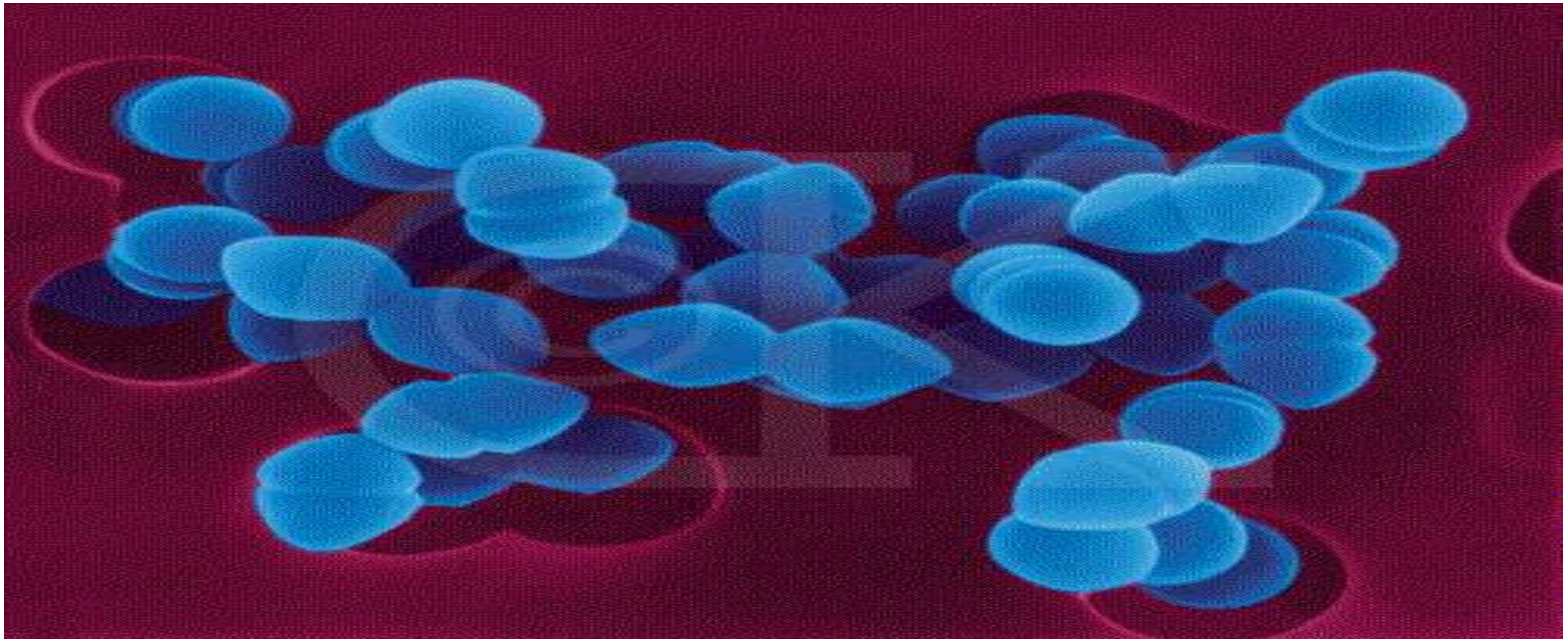
<u>Transmission electron microscopes</u>:
shows internal details of a specimen

Types of Microscopy

- Scanning Electron Microscope (SEM)
  - –Can magnify up to 100,000x



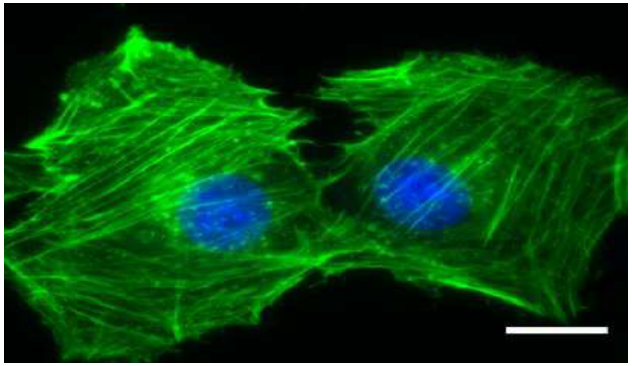
©Eric B. Workman



<u>Scanning electron microscopes</u>: show the surfaces of a specimen in 3 dimensions.

Types of Microscopy

- **Fluorescence microscopy** is used to image samples that fluoresce (absorb one wavelength of light and emit another). Light of one wavelength is used to excite the fluorescent molecules, and the light of a different wavelength that they emit is collected and used to form a picture. In most cases, the part of a cell or tissue that we want to look at isn't naturally fluorescent, and instead must be labeled with a fluorescent dye or tag before it goes on the microscope.
- Examples of Fluorescent **stains:** DAPI and Hoechst



Microscopy

USES:

- Diagnostics
- Identification
- Histology – Tissue analysis
- Examining Forensic Evidence
- Studying cells
- Studying Micro-organisms
- Scientific Research

Module # 074

Immuno- fluorescent (IF) Microscopy

Immunofluorescent (IF) Microscopy

Immuno-fluorescent (IF) Microscopy

Uses antibodies conjugated to fluorescent markers to locate specific structures on specimens

Immunofluorescent (IF) Microscopy

It allows specific structures to be visualised by illuminating them with ultraviolet light

Immunofluorescent (IF) Microscopy

Fluorescein and Rhodamine

These are the usual labels used but alternative markers are available

Immunofluorescent (IF) Microscopy

Fluoroscein produces a green fluorescence

Rhodamine is red

Immunofluorescent (IF) Microscopy

Microscopes equipped to carry out IF have dual sources of light

Immunofluorescent (IF) Microscopy

This white light allows the operator to view the specimen before illuminating with ultraviolet to look for specific fluorescence

Immunofluorescent (IF) Microscopy

This Technique is used For looking at surface markers on eukaryotic cells

Another Use is

As a whole-cell

staining technique

in bacteriology

Immunofluorescent (IF) Microscopy

Endocytosis

Uptake of membrane-bound particles by cells

Immunofluorescent (IF) Microscopy

Studies include:

- Studies on whole mammalian cells
- Migration
- Endocytosis
- Fate of labelled receptors

Immunofluorescent (IF) Microscopy

Capping

Bound receptors in cell membranes frequently migrate to one end of the cell prior to being endocytosed

END

Module # 075

IMMUNO-SORBENT ELECTRON MICROSCOPY

Immunosorbent Electron Microscopy

ISEM is a diagnostic technique used primarily in virology

Immunosorbent Electron Microscopy

Virus-specific antibodies conjugated to gold particles

Immunosorbent Electron Microscopy

These virus antibodies are used to visualise virus particles on electron microscopes

Immunosorbent Electron Microscopy

- The gold is electron dense
- It is seen as a dark shadow against the light background of the specimen field

Immunosorbent Electron Microscopy

TECHNIQUE IS USED IN:

both transmission or scanning systems

Immunosorbent Electron Microscopy

If gold-labelled primary antibodies are not available then anti-IgG–gold conjugated antibodies can be used

Immunosorbent Electron Microscopy

These anti-IgG–gold conjugated antibodies can be use with the primary antibody in a double antibody system

Immunosorbent Electron Microscopy

Both monoclonal and polyclonal antibodies can be used for ISEM

END

Module # 076

Electrophoresis of Proteins

Electrophoresis of Proteins

Sodium dodecyl sulphate (SDS)–polyacrylamide gel electrophoresis (PAGE)

The most widely used method for analysing protein mixtures qualitatively

Electrophoresis of Proteins

USE OF SDS

It is particularly useful for monitoring protein purification

Electrophoresis of Proteins

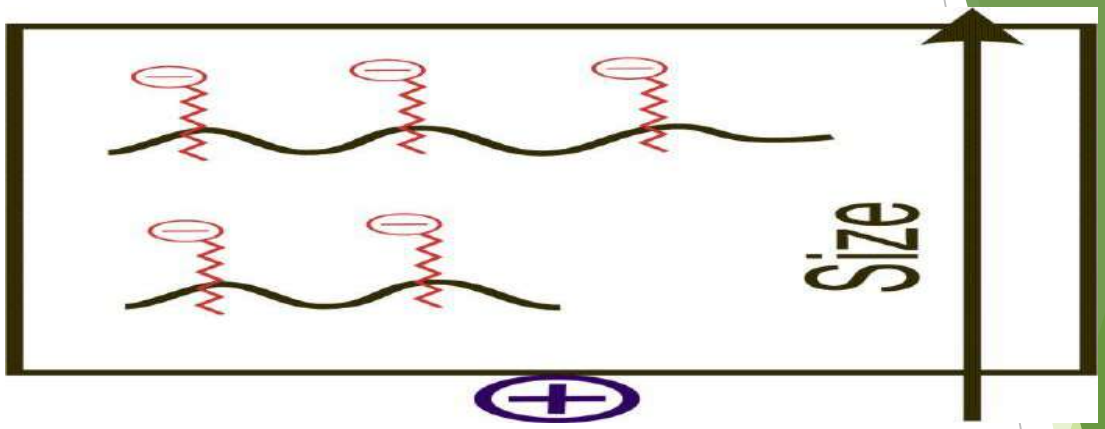
High density of binding of SDS to Proteins

The ratio size/charge is nearly the same for many SDS denatured proteins

Electrophoresis of Proteins

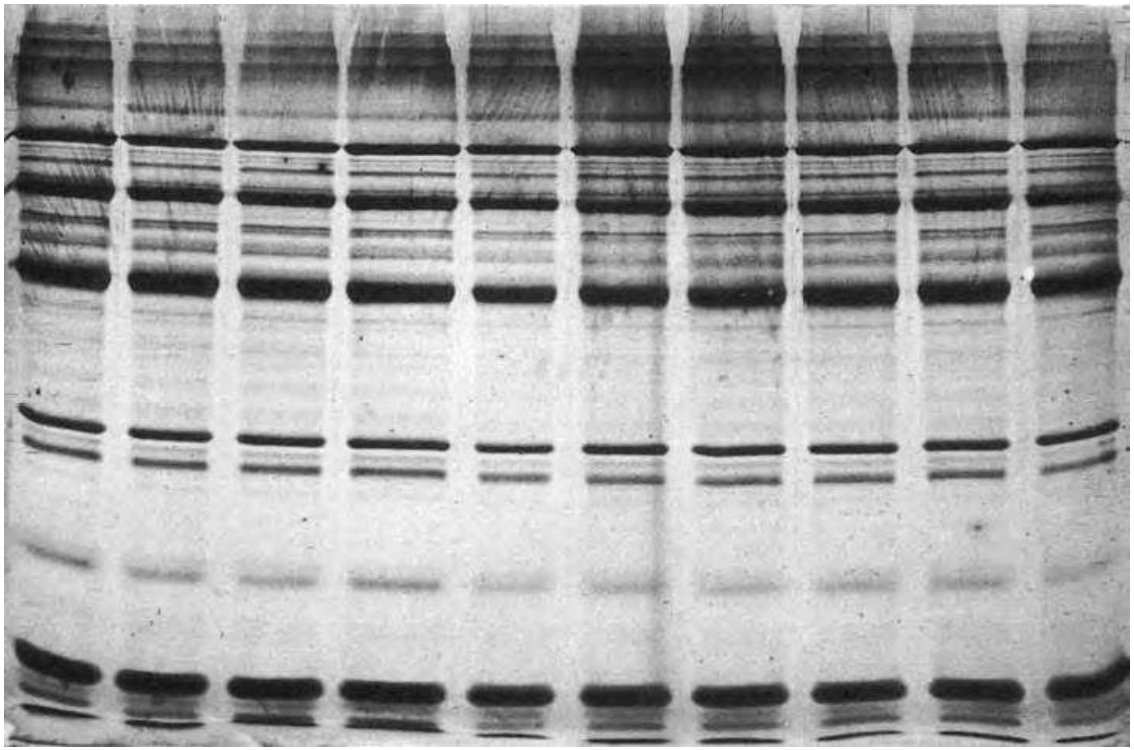
Hence proteins are separated only by length of their polypeptide chains (but not by differences in charge)

Electrophoresis of Proteins



Bands in SDS gradient gel are usually sharper than in homogeneous SDS PAGE

Electrophoresis of Proteins



Typical SDS-polyacrylamide gel

Electrophoresis of Proteins

Native gel electrophoresis

Polypeptides retain their higher-order structure and often retain enzymatic activity and interaction with other polypeptides

Electrophoresis of Proteins

Migration of proteins depends on many factors, including:

- Size
- Shape
- Native charge

Electrophoresis of Proteins

- Native gels omit the SDS and reducing agent (DTT)
- Do not put SDS or DTT in the sample buffer

Electrophoresis of Proteins

- Do not heat the samples
- prepare the gel and tank buffer solutions without SDS

Electrophoresis of Proteins

Gradient gels

Is a polyacrylamide gel system, but instead of running a slab gel of uniform pore size throughout, gradient gel is formed, where the acrylamide concentration varies uniformly

Electrophoresis of Proteins

Gradient gels are normally run as SDS gels with a stacking gel

Electrophoresis of Proteins

Isoelectric focusing gels

Electrophoretic method that separates proteins according to the iso-electric points

Electrophoresis of Proteins

Isoelectric focussing gels

- Is ideal for seperation of amphoteric substances

Electrophoresis of Proteins

Isoelectric focussing gels

- Seperation is achieved by applying a potential difference across a gel that contains a pH gradient

Module # 077
Electrophoresis of Nucleic Acids

Electrophoresis of Nucleic Acids

Agarose Gel Electrophoresis of DNA

For the majority of DNA samples, electrophoretic separation is carried out in agarose gels.

Electrophoresis of Nucleic Acids

DNA molecules and their fragments are larger than proteins

Most DNA fragments would be unable to enter a polyacrylamide gel

The larger pore size of an agarose gel is required

Electrophoresis of Nucleic Acids

Nucleotides

When talking about single-stranded DNA it is common to refer to size in terms of nucleotides (nt)

Electrophoresis of Nucleic Acids

Separation in Agarose Gels

It is achieved because of resistance to their movement caused by the gel matrix.

Electrophoresis of Nucleic Acids

The largest molecules will have the most difficulty passing through the gel pores. The smallest molecules will be relatively unhindered

Electrophoresis of Nucleic Acids

The mobility of DNA molecules during gel electrophoresis will depend on size

The smallest molecules moving fastest

Electrophoresis of Nucleic Acids

Gel Concentrations

Must be chosen to suit the size range of the molecules to be separated

Electrophoresis of Nucleic Acids

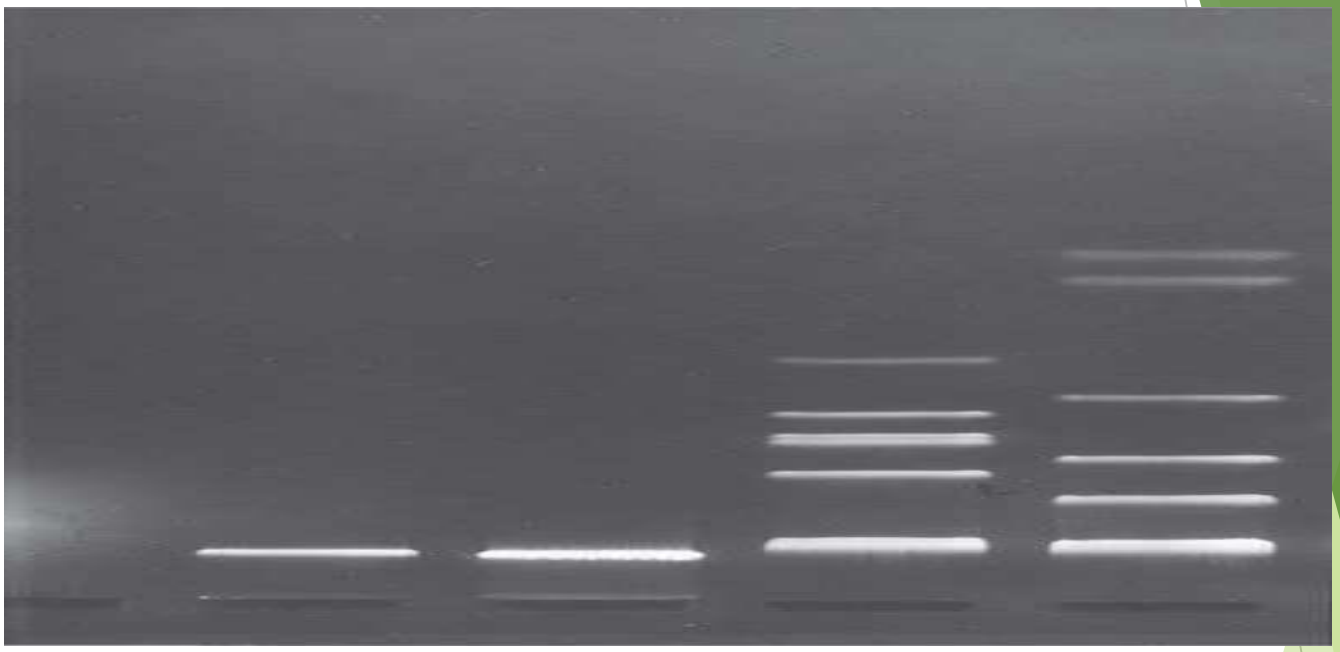
DNA gels are invariably run as:

Horizontal

Submarine

Submerged gels

Electrophoresis of Nucleic Acids



Photograph showing four tracks from a 0.8% agarose submarine gel

Electrophoresis of Nucleic Acids

DNA Sequencing Gels

Agarose gel electrophoresis of DNA is a 'workhorse' technique for the molecular biologist

A different form of electrophoresis has to be used when DNA sequences are to be determined

Electrophoresis of Nucleic Acids

DNA sequencing method

The final analysis usually involves separating single-stranded DNA molecules shorter than about 1000 nt and differing in size by only 1 nt

Electrophoresis of Nucleic Acids

Electrophoresis in agarose can be used as a preparative method for DNA

Electrophoresis of Nucleic Acids

Pulsed – Field Gel Electrophoresis

The method basically involves electrophoresis in agarose where two electric fields are applied alternately at different angles for defined time periods (e.g. 60 s)

Electrophoresis of Nucleic Acids

Pulsed – Field Gel Electrophoresis Useful for:

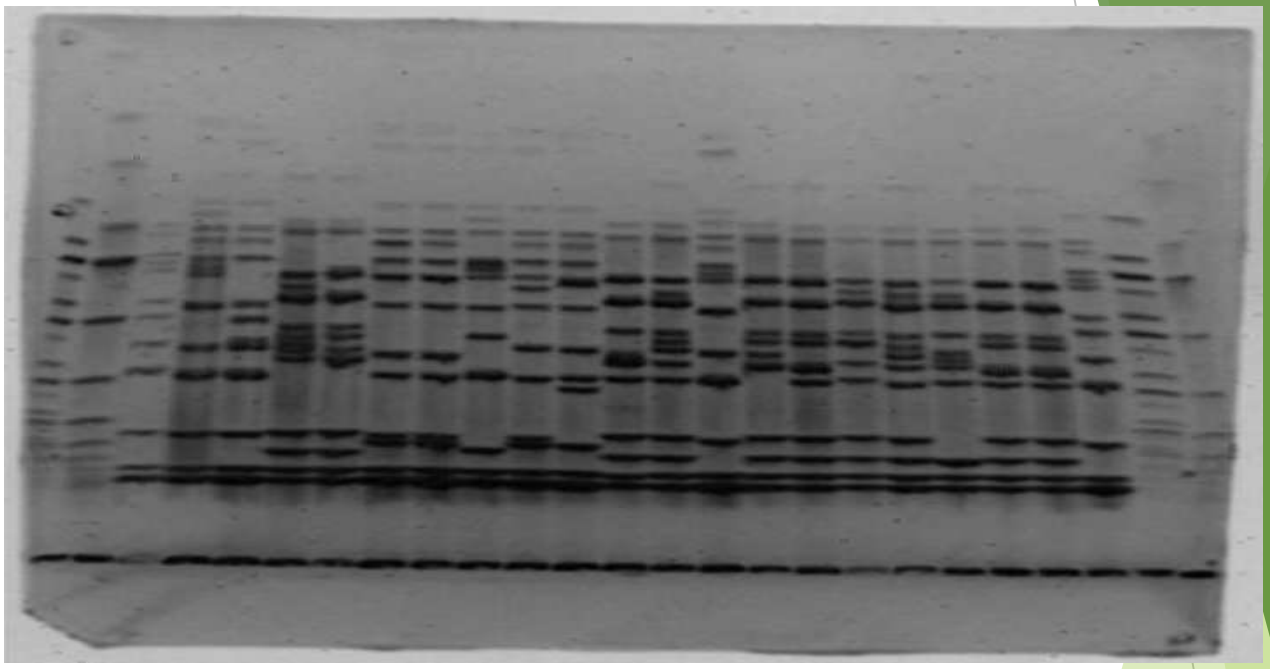
In identifying the course of outbreaks of bacterial foodborne illness (e.g. Salmonella infections).

Electrophoresis of Nucleic Acids

Example

DNA from Salmonella species, when digested with the restriction enzyme Xba1, gives around 15 fragments ranging from 25 kb to 680 kb

Electrophoresis of Nucleic Acids



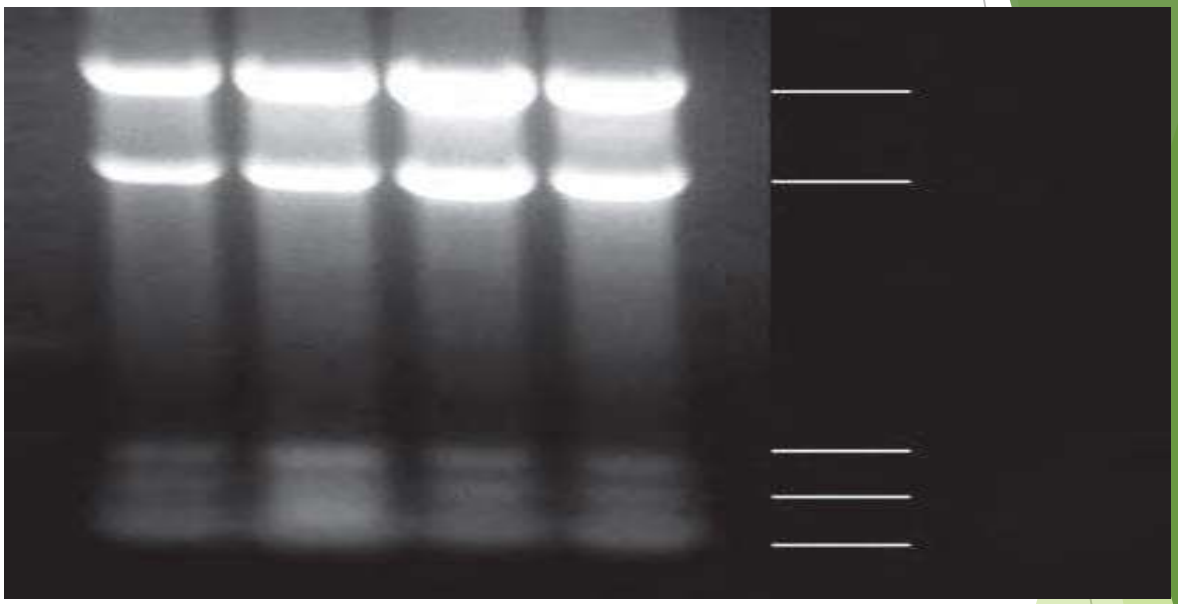
PFGE separation of the digestion pattern

Electrophoresis of Nucleic Acids

Electrophoresis of RNA

Like DNA, electrophoresis of RNA is usually carried out in agarose gels, and the principle of the separation, based on size, is the same

Electrophoresis of Nucleic Acids



Separation of yeast RNA on a 1.5% agarose gel

Electrophoresis of Nucleic Acids

Rapid Method for Checking Integrity of RNA

By electrophoresis in a 2% agarose gel in about 1 h. **Ribosomal RNAs** (18 S and 28 S) clearly resolved and any degradation (seen as a smear) or DNA contamination is seen easily

END

Module # 078 What is Chromatography?



Derived from the Greek word ***Chroma*** meaning colour, chromatography provides a way to identify unknown compounds and separate mixtures

Chromatography

- The separation of a mixture by distribution of its components between a mobile and stationary phase over time
  - mobile phase = solvent
  - stationary phase = column packing material
- Chromatography is a technique for
  - separating mixtures of compounds
  - identifying unknown compounds
  - establishing the purity or concentration of compounds
  - monitoring product formation in the pharmaceutical and biotechnology industries

Mobile Phase and Stationary Phase

- In all chromatography there is a mobile phase and a stationary phase.
- **The stationary phase is the phase that doesn't move and the mobile phase is the phase that does move**.
- The mobile phase moves through the stationary phase picking up the compounds to be tested.
- **Mobile phase** - phase that moves through chromatograph
- **Stationary phase** - column; phase that is stationary in chromatograph
- **Bonded phase** - reactive groups imparted to stationary phase in order to achieve selectivity

Purpose of Chromatography

- **Analytical** - determine chemical composition of a sample

- **Preparative** - purify and collect one or more components of a sample

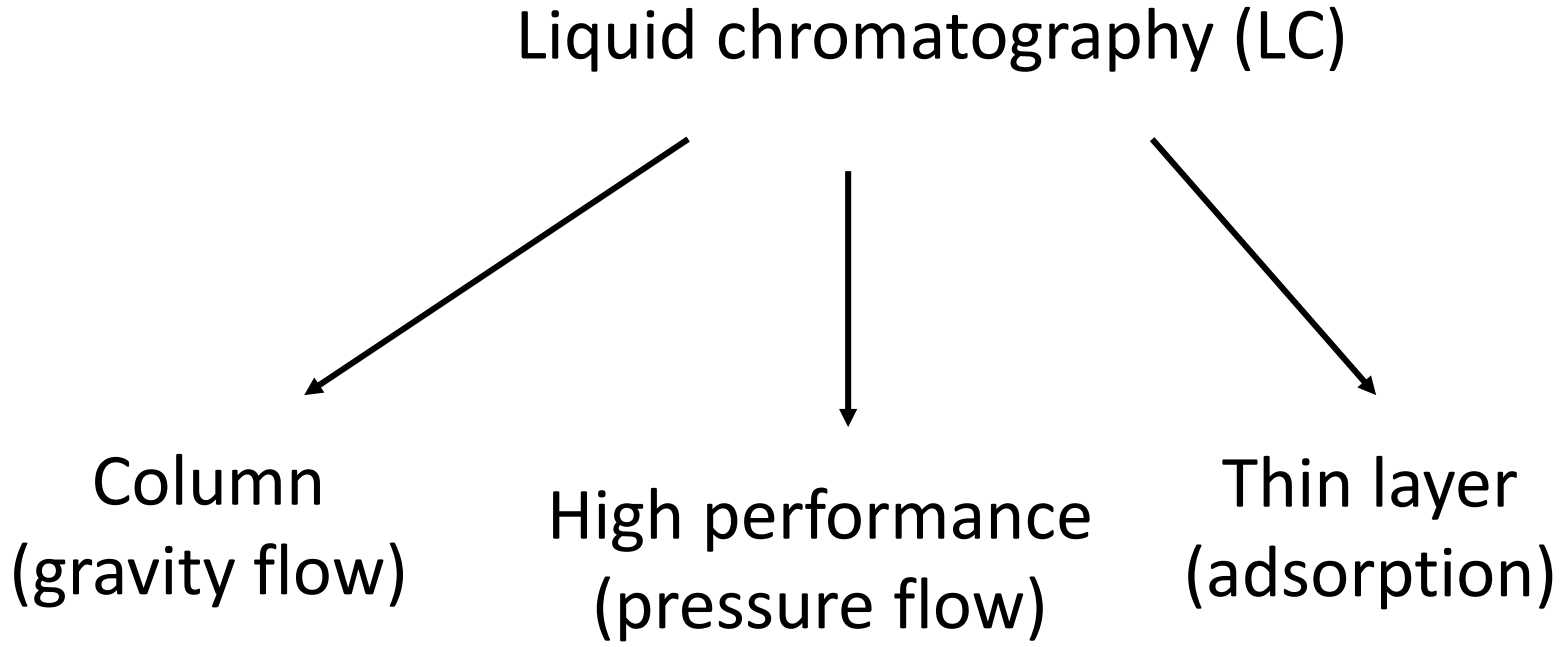
Classification of Methods

- There are two classification schemes:
  - mobile phase
  - attractive forces

Mobile Phase

- gas (GC)
- water (LC)
- organic solvent (LC)
- supercritical fluid (SCFC)

Classification based on Mobile Phase



Classification based on Attractive Forces

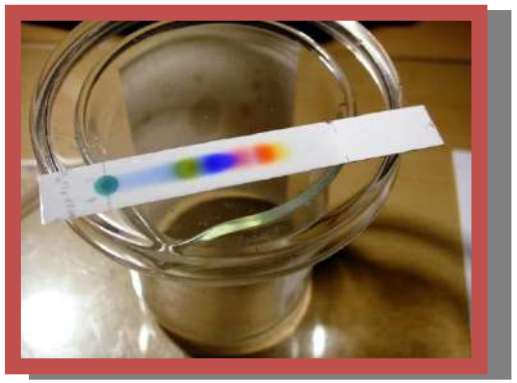
- Adsorption - for polar non-ionic compounds
- Ion Exchange - for ionic compounds
  - Anion - analyte is anion; bonded phase has positive charge
  - Cation – analyte is cation; bonded phase has negative charge
- Partition - based on the relative solubility of analyte in mobile and stationary phases
  - Normal – analyte is nonpolar organic; stationary phase MORE polar than the mobile phase
  - Reverse – analyte is polar organic; stationary phase LESS polar than the mobile phase
- Size Exclusion - stationary phase is a porous matrix; sieving

Types of Chromatography…



Paper





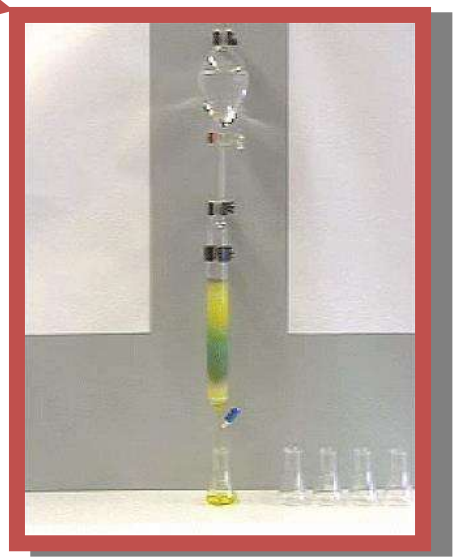
Thin layer



HPLC



Gas

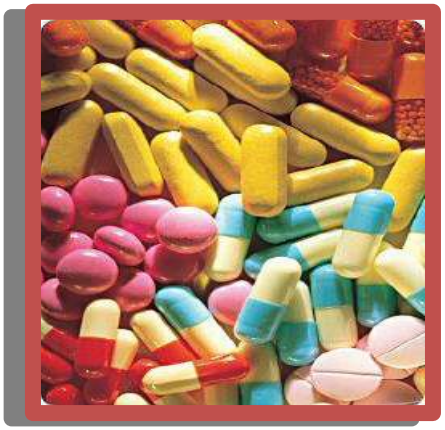


Column

Types of Chromatography

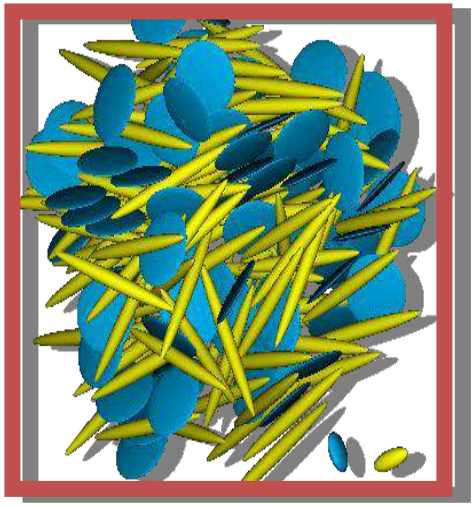
- Classification by mobile phase:
  - Gas - Gas chromatography (GC)
  - Liquid - Liquid chromatography (LC)
  - Supercritical fluid - Supercritical fluid chromatography (SFC)

Applications of Chromatography

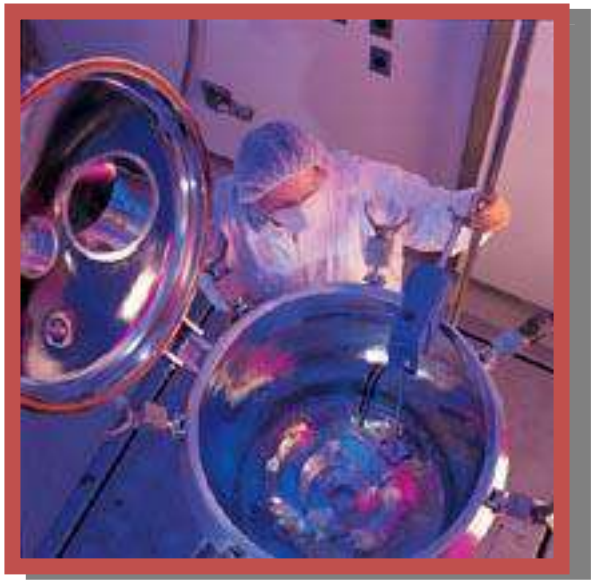




Forensics



Research



Pharmaceutical industry

Applications of Chromatography

Chromatography is widely used in:

- Research
- Pharmaceutical Industry
- Forensics (by forensic teams to analyse blood and urine samples for drugs)
- Paint analysis
- Testing for the presence of explosives.

Module # 079

Ion-Exchange Chromato-graphy

Ion-Exchange Chromatography

Ion-Exchange Chromatography

Ion exchange chromatography is a separation based on charge

Ion-Exchange Chromatography

Use of Ion-Exchange Chromatography

Used for almost any kind of charged molecules

Large proteins, small nucleotides & amino acids

Ion-Exchange Chromatography

Ion-exchange

chromatography preserves analyte molecules on the column based on ionic interactions

Ion-Exchange Chromatography

Two Type of Ion Exchangers

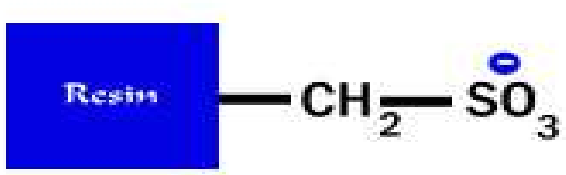
1. Cation Exchanger
2. Anion Exchanger

Ion-Exchange Chromatography

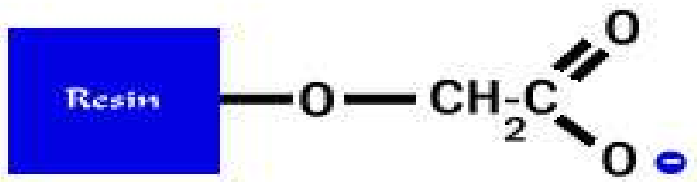
Cation Exchangers

Possess negatively charged groups and these will attract positively charged cations

Also called *acidic ion exchangers*



S-cation exchanger



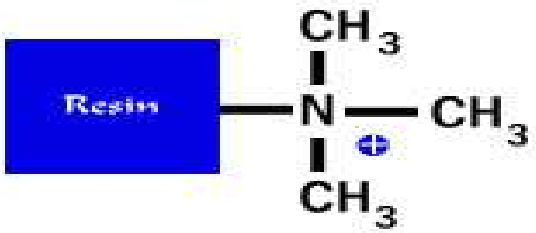
CM-cation exchanger

Ion-Exchange Chromatography

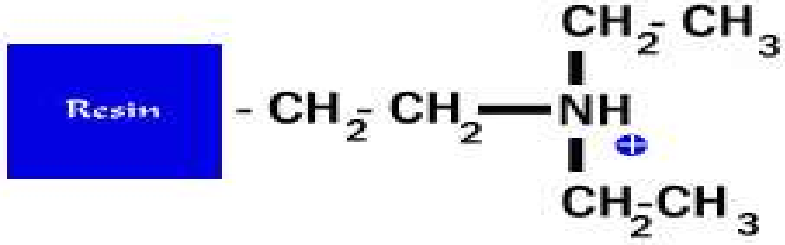
Anion Exchangers

Have positively charged groups that will attract negatively charged anions

Also called *basic ion exchangers*



Q-anion exchanger



DEAE-anion exchanger

Ion-Exchange Chromatography

Advantages

✓It is a non-denaturing technique. It can be used at all stages and scales of purification

✓An IEX separation can be controlled by changing pH, salt concentration and/or the ion exchange media

Ion-Exchange Chromatography

Advantages

✓ It can serve as a concentrating step. A large volume of dilute sample can be applied to a media

✓It offers high selectivity

Ion-Exchange Chromatography

Disadvantage

- ✓ Costly equipment and more expensive chemicals
- ✓ Turbidity should be below 10ppm

MB504T- METHODS IN MOLECULAR BIOLOGY

COMPLETE PPT

MID TERM PPT # 001 – 039

FINAL TERM PPT # 040 – 079

OLD CODE BIO- 203

MERGE BY BAQIR HUSSAIN BUZDAR

BS ZOOLOGY DATE 04-09-2024

03336361244 8TH SEMESTER

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